Electric Vehicle Subsidies: Cost-Effectiveness and Emission Reductions

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Abstract

I design a structural model of demand for electric vehicles and the supply of a public charging infrastructure by forward-looking local planners. Using Canadian data, I study the cost-effectiveness of electric vehicle incentives in this context. Subsidizing electric vehicle purchases doubled adoption in Quebec but had only a small impact on network provision. I conduct a rigorous cost-benefit analysis to study the environmental performance of Quebec's rebate program. I find that the marginal abatement cost of emissions was substantially higher than the social cost of carbon, suggesting that policymakers in Quebec overinvested in electric vehicle incentives.

Keywords: electric vehicles, charging stations, subsidies, emission abatement, cost-benefit analysis, indirect network effects.
JEL Codes: L91, H23, Q58.

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1 Introduction

Electric vehicles (EV) constitute one of the most promising innovations for lowering carbon emissions from the transportation sector, as long as clean energy production is available. Several barriers prevent the widespread adoption of this technology. The high initial purchase cost or the low availability of charging sites may lead potential buyers to select internal combustion engines over electric ones, even if they place a high value on reducing their carbon footprint, factor in future fuel cost savings, or the lower maintenance costs associated with driving an electric vehicle. At the same time, if few people own and use an electric vehicle, there is little incentive for network operators to invest in local charging infrastructures. This slows down the transition to electric vehicles.

Policymakers have introduced a wide range of incentives to convince consumers to adopt this new technology. Perhaps the most common intervention is to subsidize the purchase of new electric vehicles. This narrows the price gap between internal combustion and electric vehicles, and leads to increased adoption. At the same time, financial incentives contribute to establishing a market demand for charging services. This encourages network operators to invest in charging station infrastructures, which yields additional electric vehicle sales through indirect network effects.

While some jurisdictions leave the development of charging infrastructure to the private sector (often subsidizing new stations), others choose to provide charging services directly to electric vehicle owners. This is the case of the Canadian province of Quebec, where county-level governments are responsible for the provision of local charging infrastructures. I propose a structural model that reflects this reality. Its key innovation compared to previous works is that charging networks are provided by forward-looking local planners.[1] I find that ignoring the forward-looking behavior of network operators overestimates the importance of network effects in electric vehicle markets. This result is important for the design of electric vehicle incentive programs, as this would lead to overestimating their performance.

I study the introduction of the electric car in Canada and the cost-effectiveness of electric vehicle incentive programs. I gather a novel dataset of vehicle registrations and charging station installations in two Canadian provinces, Ontario and Quebec, for the period spanning 2012 to 2020. I start with a difference-in-differences analysis to study the direct impact of subsidies on sales, and their indirect impact on charging station provision. The identifying variation comes from changes to Ontario's rebate program that occurred in 2016 and 2018. Originally, Ontario and Quebec offered similar electric vehicle subsidies: $8,500 and $8,000 respectively, per electric vehicle sold. The government of Ontario improved subsidies to

[1] See for example Li et al. (2017), Pavan (2017), Springel (2021), Remmy (2025), or Li (2023).

$14,000 in February 2016 and phased out its incentive program in late 2018.

Subsidies are very effective at increasing electric vehicle adoption. The estimated intent-to-treat effect suggests that the improvements to Ontario's subsidies led to a 26.7% increase in electric vehicle adoption, while the abolition of the program reduced sales by 66.7% compared to baseline. I extend the analysis and estimate a continuous treatment effect model. I find that $1,000 in subsidies is associated with a 7.9% increase in electric vehicle sales. This is qualitatively similar to findings by Muehlegger and Rapson (2022) who study an electric vehicle incentive program in California using a quasi-experimental setup.

I also study the indirect effect of these subsidies on charging station deployment. The idea is that network supply might respond to shifts in the aggregate demand for charging services emanating from new electric vehicle owners.[2] I do not find evidence that the policy changed the configuration of local networks in the short-run. Furthermore, I cannot find evidence that local networks changed along other dimensions. For example, I see no change in the number of charging points at each site or the share of fast charging stations available. Together, these findings suggest that network provision is rigid in the short-run and cannot respond immediately to an unpredicted surge in demand from new electric vehicle owners. To the best of my knowledge, this is a new result in the literature.

I rely on a structural estimation to address the cost-effectiveness of electric vehicle subsidies and their overall environmental performance. I focus the analysis around the province of Quebec, where charging infrastructures are provided to users by local, county-level governments. This contrasts with other jurisdictions where network providers are private, profit maximizing firms. This setup is appealing for several reasons. First, it allows me to ignore the effects of price competition or product differentiation in the charging market. Charging and wholesale energy prices in Quebec are regulated, and the vast majority of chargers are homogeneous. Second, it avoids spatial competition concerns which can lead to multiple equilibria when networks are provided by more than one firm. Finally, it avoids issues related to platform competition, as all local networks are connected to the unified, government-provided platform.

I develop a structural model of consumers' demand for cars and the public provision of charging station infrastructures. In the model, consumers consider the current provision of charging stations in their surroundings, at the time of purchase, and are myopic about future states of the market. Public charging station infrastructures are provided by forward-looking local planners. Specifically, each local planner chooses the size of their charging infrastructure in each period, taking into account the lifetime valuation of the network by users and the

[2] Springel (2021) and Remmy (2025) have studied this question using a structural estimation. Both works focus on electric vehicle subsidies, charging station subsidies, and their interaction.

fixed cost of installing additional capacity. I show that ignoring the forward-looking behavior of local planners leads to overestimating their response to increases in the electric vehicle base. In the case of Quebec, this translates to overestimating the contribution of network effects to sales by 16.8%.

I conduct counterfactual simulations to validate the findings from the difference-in-differences analysis. I find that electric vehicle rebates doubled electric vehicle sales in Quebec between 2012 and 2020. This translates to a 11.5% increase per $1,000 in subsidies. Meanwhile network size increased by 10.5% over the same period. This suggests that the rigidities in network provision vanish in the long-run, leading to additional charging stations and electric vehicle sales. However, the indirect impact of electric vehicle incentives on charging station deployment remains small. To place these results in perspective, my results imply that charging stations are provided to electric vehicle owners at the rate of one station per 159 electric vehicles sold.

I construct a flexible framework to study the performance of non-marginal environmental policies, taking the decentralized equilibrium arising from the structural model as given. There is friction between the local planners' objective to provide public charging and the central government's objective to correct an environmental externality through electric vehicle subsidies. This friction is a central feature of Quebec's electric vehicle market.

I use this framework to conduct a rigorous cost-benefit analysis of the Canadian electric vehicle incentive programs. My findings suggest that the marginal abatement cost of emissions was $324 per ton of CO_2. This is above conventional measures of the social cost of carbon, which suggest an overinvestment in subsidies beyond what is efficient.

Related literature. This paper contributes to the literature on several fronts. First, I contribute the growing literature that studies electric vehicle markets. Notable contributions include Pavan (2017), Li et al. (2017), Springel (2021) on network effects in both alternative fuel and electric vehicle markets. I add to this literature by considering the case where network operators are forward-looking. As mentioned above, ignoring the forward-looking behavior of network operators inflates the contribution of network effects in counterfactual simulations. Li (2023) studies the impact of unifying competing standards for charging electric vehicles. She focuses on the case where car manufacturers both provide electric vehicles and the infrastructure to charge them. I instead focus on the case where charging stations are provided publicly by regional governments and compatibility is not an issue. Other works on electric vehicles include Remmy (2025) on driving range provision, Armitage and Pinter (2021) on electric vehicle mandates, Dorsey et al. (2022) on consumers' valuation of charging networks, and Johansen and Munk-Nielsen (2020) on the synergy between fuel and electric

vehicles within a multi-car household. Close to this research is the work of Xing et al. (2021) who show that recovering precise substitution patterns is crucial to estimating the environmental impact of electric vehicle incentives. My methodology allows for estimating very flexible elasticities of network supply which in turn enrich the substitution patterns on the demand side. In particular, this helps identify the substitution between internal combustion and electric vehicles which is important for environmental policy analysis.

This paper fits in the wider literature that studies the environmental regulation of the car market. Previous works have focused on the environmental performance of subsidies (Beresteanu and Li, 2011; D'Haultfoeuille et al., 2014; Huse and Lucinda, 2014; DeShazo et al., 2017; Azarafshar and Vermeulen, 2020; Sheldon and Dua, 2020), electric vehicle rebates passthrough (Beresteanu and Li, 2011; Sallee, 2011; Muehlegger and Rapson, 2022), gas taxes (Allcott and Wozny, 2014; Barla et al., 2016; Grigolon et al., 2018), emission standards (Durrmeyer and Samano, 2018; Reynaert, 2021), cash for clunker programs (Li et al., 2013; Grigolon et al., 2016; Li et al., 2022; Kitano, 2023), attribute-based regulation and taxation (Knittel, 2011; Ito and Sallee, 2018; Chaves, 2019), and comparing financial and non-financial incentives (Jenn et al., 2018). Advances in estimating the environmental impacts of these policies include Durrmeyer (2022) who studies the distributional impacts of the French rebate program, Tsanko (2023) on the environmental benefits of subsidizing plug-in hybrids when consumers do not recharge them optimally, and Holland et al. (2019) on air pollution patterns that occur upstream in the electricity production process. I provide a general framework for conducting cost-benefit analysis, based on the marginal cost of abatement rather than the average cost. I show that analyses based on the average abatement cost produce misleading policy recommendations (unless the policy change under study is marginal). My framework could be used to study a wide array of environmental regulations, including those described above.

Lastly, I contribute to the literature on estimating network effects and their role in the adoption of breakthrough innovations. Advances in this field touch a wide range of new products: green cars (Pavan, 2017; Li et al., 2017; Springel, 2021; Remmy, 2025; Li, 2023), compact discs (Gandal et al., 2000), video games (Clements and Ohashi, 2005; Corts and Lederman, 2009), software (Gandal, 1995), microcomputer chips (Gandal et al., 1999), and personal digital assistants (Nair et al., 2004). I extend these literatures to include the case where the complementary product (here, the charging station) is provided by forward-looking local planners. I find in general that network effects are weaker in this context.

The rest of the paper is organized as follows. Section 2 provides background information on the Canadian electric vehicle market. Section 3 studies the short-run effect of electric vehicle subsidies on sales and charging station deployment. I describe a structural model of

demand for cars and the supply of a public charging infrastructure in Section 4. Estimation and counterfactual results are presented in Section 5. I conduct a rigorous cost-benefit analysis in Section 6 to assess the environmental performance of subsidy programs. Section 7 provides concluding remarks.

2 The Canadian Market for Electric Vehicles

2.1 Background and data

The transportation of passengers and freight accounted for 22% of all Canadian greenhouse gas emissions in 2021, ranking second behind oil and gas production.[3] As such, the electrification of transportation has become a prime concern to both provincial- and federal-level policymakers in Canada.

I focus my analysis of electric vehicle incentives around the two largest Canadian provinces, Quebec and Ontario, which together account for two thirds of Canada's population. Both provinces offered generous rebates to new electric vehicle owners, starting as early as 2010. Moreover, electricity production in these provinces is almost exclusively emission-free.[4] This provides a clean setup to study emission reductions resulting from the electrification of transportation, as selling additional electric vehicles does not result in additional emissions upstream.

I begin with a description of the various Canadian policies that are relevant to the analysis. I focus on three financial incentive programs offered by the provincial government of Ontario, the provincial government of Quebec, and the federal government of Canada. To paint the broadest picture possible, I discuss the financial and the non-financial incentives that are offered in each jurisdiction. I also describe how charging station networks are developed, as there are significant differences between the two provinces. Ontario relies on a more traditional model which leaves the development of local networks to the private sector. Meanwhile, the government of Quebec develops public networks in partnership with county-level governments, with little contribution from private operators.

My analysis combines novel data from three sources. I obtain car registration data from the Société d'Assurance Automobile du Québec and the Ministry of Transportation of Ontario. The Quebec dataset is at the individual registration level while the Ontario dataset

[3]Source: Environment and Climate Change Canada.

[4]Quebec's electricity grid is almost entirely emission-free. In 2021, approximately 94% of generation came from hydropower and 5% from wind, with the remaining 1% from natural gas and other sources concentrated in remote communities not connected to the main grid. Ontario's grid is also predominantly low-carbon: 55% nuclear, 24% hydropower, 12% wind and solar, and natural gas making up the rest (mostly to answer peak demand). Source: Canada Energy Regulator, Provincial and Territorial Energy Profiles.

is aggregated at the product-county-quarter level. I combine the registration data with car characteristics obtained online from The Car Guide and the Auto Trader websites.[5] Both are leading source of information on passenger vehicles in Canada. Finally, I obtain the location, the operator's name, and the installation date of each charging station in Quebec and Ontario from Natural Resources Canada and Hydro-Québec. Additional details on the data are relegated to Appendix B.

2.2 Policy environment

Timeline. The transportation sector is one of the leading contributor to carbon emissions in Canada. Absent federal initiative, the provincial governments in Ontario and Quebec launched separate electric vehicle incentive programs in 2010 and 2012 respectively.[6] The stated goals of the policies were to support the transition to electric vehicles, reward first adopters, and create a market demand for this new technology. Increasing adoption also creates a market demand for charging services, which encourages investments into charging stations from private and public operators.

While Quebec's Roulez Vert Program was maintained over time, the government of Ontario modernized its Electric Vehicle Incentive Program in February 2016.[7] On that occasion, subsidies for battery electrics and long-range plug-in hybrids were significantly increased, from $8,500 to $14,000. For short-range plug-in hybrids, the rebate was slightly increased and made progressive in the capacity of the battery. Policymakers justified these improvements with Ontario's poor performance in terms of electric vehicle adoption, compared to the rest of Canada.

The election of a conservative government in June 2018 led to the abolition of Ontario's cap-and-trade carbon tax in early October 2018. Since this carbon tax was the principal source of funding for electric vehicle subsidies, the Electric Vehicle Incentive Program was terminated at the same time. There is some anecdotal evidence in news reports that consumers were taken by surprise by this sudden change, as the government did not campaign extensively on these subsidies during the 2018 election cycle. With only a few weeks' notice, there was not a lot of time to capitalize on the rebate before the program expired, since ordering an electric vehicle usually took between 6 to 12 months at the time.

In May 2019, the federal government of Canada stepped in with its Incentives for Light-Duty Zero-Emission Vehicles Program. The stated objectives were to make subsidies avail-

[5] See https://www.guideautoweb.com/en/ and https://www.autotrader.ca.

[6] Other Canadian provinces also offer subsidies. These include British Columbia (up to $4,000), Newfoundland and Labrador (up to $2,500), Prince Edward Island (up to $5,000), New Brunswick (up to $5,000), and Nova Scotia (up to $3,000).

[7] The implementation was made retroactive to November of 2015.

able to all Canadians and to ensure that electric vehicle sales targets were met nationwide.

Financial incentives. The detailed list of incentives is summarized in Table 1. Rebates are obtained automatically at the point of sale and are deducted from the transaction price.[8] To be eligible, consumers must either purchase the vehicle, or sign a long-term lease. Short-term leases are eligible for a fraction of the rebate, determined on a *pro rata* basis. Consumers also need to provide a proof of residence and the car needs to be purchased at a dealership located in the province that offers the subsidy.

Additional financial incentives are offered in Quebec. For example, the program includes subsidizing the purchased of a used electric vehicle (up to $4,000), the installation of a home-charger ($600), and the installation of large-scale charging capacity in multi-unit housing or in workplaces (up to 50% of installation costs). While these policies are interesting and could play a role in increasing adoption, I am forced to ignore their contribution due to data limitations.

I do not observe car ownership through time. As such, transactions on the secondary market are unobservable. To assess the relative size of the secondary market, I compare the total spending on used car subsidies to the total program expenditure. I find that 1.8% of the program's funds went to subsidizing used cars. To fix ideas, 94.1% of total spending went to subsidizing new cars. Back of the envelope calculations suggest that the primary market was around 30 times larger than the secondary market between 2012 and 2020.

I also ignore the effect of subsidizing home-chargers due to data limitations. The key problem is that I do not observe which consumer applied for, and received a home-charger subsidy. Furthermore, there is no requirement that new owners install a home-charger in the same year as they purchase an electric vehicle. Acquiring a home-charger can cost between a few hundred to a few thousand dollars, but is not absolutely necessary to charge at home. Government spending on home-chargers amounted to 4% of the total program expenditure in Quebec.

Non-financial incentives. Several non-financial incentives are offered to encourage electric vehicle adoption. They are normally tied to registering the car under a green license plate, which provides advantages all over Canada and in the United States. Registering an electric vehicle under a green license plate is mandatory for safety reasons. Both provinces offer similar non-financial incentives. They include a privileged access to dedicated lanes (e.g., carpool lanes), a free access to toll lanes or bridges, and dedicated free parking spaces.

[8] Tesla is an exception, as they did not have points of sale in Canada in that period. In that case, consumers must fill in additional paperwork and receive a mail-in refund a few weeks later.

Table 1: Canadian incentive programs

	Battery electric	Plug-in hybrid (long range)	Plug-in hybrid (short range)
Ontario program, phase 1 (2010 – 2015)			
MSRP below $150,000, batt. cap. 17 kWh or above	8,500	8,500	8,500
MSRP below $150,000, batt. cap. 4 kWh – 17 kWh	n/a	5,000	5,000
Non-financial incentives:			
• Privileged access to high occupancy vehicle lanes	✓	✓	✓
• Free access to high occupancy toll lanes	✓	✓	✓
• Free parking when charging	✓	✓	✓
Ontario program, phase 2 (2016 – 2018)			
MSRP below $75,000, batt. cap. 16 kWh or above	13,000	13,000	n/a
MSRP below $75,000, batt. cap. 5 kWh – 16 kWh	n/a	n/a	6,000 – 9,600
MSRP below $75,000, 5 seatbelts	+1,000	+1,000	+1,000
MSRP between $75,000 and $150,000	3,000	3,000	3,000
Non-financial incentives:			
• Privileged access to high occupancy vehicle lanes	✓	✓	✓
• Free access to high occupancy toll lanes	✓	✓	✓
• Free parking when charging	✓	✓	✓
Quebec program (2012 – pres.)			
MSRP below $75,000	8,000	8,000	4,000
MSRP between $75,000 and $125,000	3,000	0	0
Other financial incentives:			
• Used vehicle (original MSRP below $75,000)	4,000	0	0
• Installation of a home-charger	600	600	600
Non-financial incentives:			
• Privileged access to reserved lanes	✓	✓	✓
• Free access to toll bridges and toll lanes	✓	✓	✓
• Free parking (in some municipalities)	✓	✓	✓
• Free access to several ferries	✓	✓	✓
Federal program (2019 – pres.)			
Passenger car, base model MRSP below $55,000	5,000	5,000	2,500
SUV and minivan, base model MRSP below $60,000	5,000	5,000	2,500

Notes: All values are in current Canadian dollars. MSRP is the manufacturer's suggested retail price. The rebate for plug-in hybrids in Ontario, phase 2, increases from $6,000 to $9,600, in steps of $365 per kWh of battery capacity. The Chevrolet Volt is the only plug-in hybrid that qualifies as "long range" in Quebec. For the federal rebates, plug-in hybrids with a driving range above 50 km on electric mode qualify as "long range".

Additional details on non-financial incentives are available in Table 1.

2.3 Network deployment

Table 2 presents the distribution of all charging stations by province and operating network. There are striking differences between the two provinces. The first one is the sheer difference in the size of their networks. In per capita terms, there are more than four times more stations in Quebec than in Ontario. Second, Ontario's network is predominantly operated and developed by private firms (even though most stations are installed on the street and are considered as public). In contrast, Quebec's market is dominated by the government

Table 2: Network operators

	Ontario		Quebec	
	Number of stations	Share total	Number of stations	Share total
ChargeLab	18	0.02	1	4e-4
ChargePoint Network	219	0.20	70	0.02
Electric Circuit	29	0.03	1,960	0.70
Electrify Canada	4	4e-3	0	0
EV Connect	20	0.02	0	0
Flo	209	0.19	376	0.13
Ivy	23	0.02	0	0
Petro-Canada	19	0.02	7	2e-3
Shell Recharge	6	5e-3	0	0
SWTCH Energy	20	0.02	0	0
Tesla Destination	222	0.20	160	0.06
Non-networked	318	0.29	237	0.08
Total	1,107	1	2,811	1
Population, in 2020	14.22		8.44	
Nb. of counties	49		96	

Notes: The Electric Circuit is Quebec's public platform. All other networks are operated by private firms. Non-networked stations include small businesses and workplaces that allow for public charging on their private infrastructure. The network size is reported for year 2020. Population is in million.

provided platform, the Electric Circuit, with little competition from private firms.[9]

Network provision in Quebec does not follow a traditional model of demand and supply. Instead, the provincial government enters partnerships with regional governments, shopping malls, restaurant chains, and workplaces for the development of local charging station infrastructures. On the one hand, the provincial government provides the platform (including the software infrastructure, the phone app, and billing services) and coordinates maintenance. It also regulates both the the charging price paid by consumers and the wholesale energy price paid by the partner. On the other hand, the partner pays for the physical infrastructure (the actual station) and the installation cost. It then collects revenues from operating that station. Importantly, partners decide where and when to install stations, since they own property rights on the land.

The vast majority of partners are county-level governments. I assume throughout that they control the final decision about the size of local networks in Quebec. In practice,

[9]Tesla and Flo are the only other firms involved in Quebec. Tesla is involved in the development of its own network, which insure that Tesla owners can reach every destination in North America. Stations are typically located in strategic locations that facilitate long distance travels. Flo is the network developed by AddÉnergie, a company based in Quebec that manufactures and sells chargers.

they can forgo installing some stations if more private installations occur. I also maintain the assumption that they do not coordinate on a common deployment strategy. There are no unified political parties in Quebec that span both provincial and regional politics. County-level governments usually form around local political figures, and are insulated from provincial or federal politics. In that context, decisions are taken in isolation from other counties or higher levels of government.

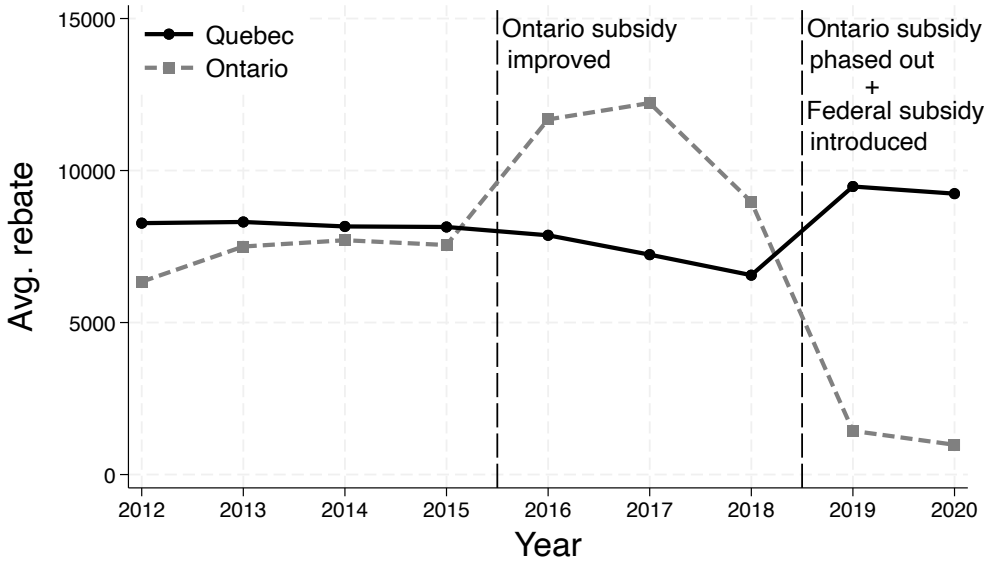
3 The Short-run Impact of Rebates

3.1 Setup

The Canadian market provides an ideal setup to study the short-run impact of electric vehicle subsidies using a difference-in-differences analysis. I leverage the fact that Ontario saw two changes in its electric vehicle rebate program to study the effect of rebates on electric vehicle adoption and their indirect effect on charging station installations. Electric vehicle subsidies have been more or less constant in Quebec, since their introduction in 2012, which provides me with an adequate control group.

Figure 1 depicts the average rebate offered to consumers of each province between 2012

Figure 1: Average rebate by province



Notes: This figure reports the average electric vehicle subsidy received by consumers, by province and year. The average rebate is weighted by sales and includes both battery electric and plug-in hybrid. All values are in 2018 CAD.

and 2020. All values are converted to 2018 Canadian dollars (CAD). Initially, both provinces had similar rebate programs. I will refer to the period from 2012 to 2015 as the pre-treatment period. I observe a first policy shock in 2016, when Ontario's rebate program was substantially increased, and a second policy shock at the end of 2018, when Ontario's program was phased out. The federal rebate program was introduced in early 2019. Since the phasing out of Ontario's program and the introduction of the federal program occurred in a short time window, I will consider them as a single policy shock. I will refer to these periods as the first and second post-treatment periods.

Similarly to the vast majority of studies that rely on a natural experiment for identification, it is important to discuss the potential endogeneity of these policy changes. Muehlegger and Rapson (2022) describe best the threat to the identification of a causal effect between subsidies and electric vehicle adoption: states are more likely to offer an incentive if the population they represent is predisposed to purchasing an electric vehicle. There is some anecdotal evidence that points in that direction. The government of Ontario significantly increased rebates because adoption of electric vehicles was low compared to other provinces. In this case, endogeneity would arise from a negative correlation between consumers' and the policymaker's preferences and would lead to underestimating the causal effect. The program was discontinued after Ontario abolished its cap-and-trade carbon tax which cut the main source of funding for subsidies. In this case, there is more chance that the change was exogenous.

It is very hard in practice to test the exogeneity assumption. I perform the analysis at the county level. This plays the double role of avoiding selection into treatment, but also, differences across counties make it less likely that the policy correlates with the outcomes, since it is defined at the provincial level. I include several county-level demographics and a rich set of fixed effects to control as best as possible for the potential unobserved factors that could bias my estimates. I am also very careful in my interpretation of the results.

A related threat to identification arises from anticipation. If electric vehicle owners or network providers anticipated subsidies to increase in the future, they could, for example, delay purchasing an electric vehicle, or preemptively install chargers to meet future demand for charging services. In both cases, these effects would lead to a violation of the parallel trends assumption, an overestimation of the true causal effect of subsidies on electric vehicle adoption, and an underestimation of the effect on network provision.

Figure A.1 and A.2 offer a visual representation of the pre-trends. The parallel trends assumption is slightly violated for all specifications concerning electric vehicle sales (the F-statistics are 5.21, 3.62, and 2.13) but not for the specifications that concern network provision (the F-statistics are below 1 for all specifications). Therefore we cannot completely

rule out anticipation in the case of electric vehicle adoption on the basis of parallel trends.

It seems unlikely, however, that consumers or network operators anticipated either changes to the policy environment. The announcement concerning the improvement to Ontario's rebate program was made on February 10, 2016, and was made retroactive to January 1 of that year. There is no evidence that the policy was disclosed to the press beforehand, leaving consumers (and network operators) no opportunity to act in anticipation. Regarding the 2018 phase-out, Ontario announced the cancellation of its rebate program mid-July 2018, with rebates remaining available until the end of September 2018. At the time, the typical lead time between placing an order and taking delivery of a new electric vehicle exceeded six months. A consumer placing an order after the announcement could not have received their vehicle before the program ended. Only consumers with an active order that preceded the announcement were in position to complete the transaction on time. These two factors together make it very difficult to attribute the mild violation in parallel trends to strategic anticipatory behavior.

Summary statistics concerning Ontario and Quebec are available in Table A.1. There are some significant differences between the two provinces, especially in terms of household income, which is much higher in Ontario. This is mitigated by the fact that housing costs are also higher in Ontario than in Quebec. Ontario residents are also, on average, more educated, more conservative, more likely to be homeowners, and more likely to belong to a visible minority group. Finally, Ontario's counties are on average three times as populous as Quebec's counties.

3.2 Effect on electric vehicle registrations

I first study the effect of rebates on electric vehicle adoption using a difference-in-differences analysis. I index counties by m and years by t. The difference-in-differences specification is

$$y_{mt} = \beta_1(\text{Ontario} \times \text{Post1})_{mt} + \beta_2(\text{Ontario} \times \text{Post2})_{mt} + X_{mt}\gamma + \mu_m + \lambda_t + \epsilon_{mt}, \quad (1)$$

where μ_m and λ_t are county and year fixed effects, and X_{mt} is a set of county-level demographics and controls. The treatment group contains counties located in Ontario and the two treatment periods are as defined above. The dependent variable, y_{mt}, is the logarithm of electric vehicle registrations, by county and year.

Results are presented in the first panel of Table 3. I estimate the model first using all electric vehicle registrations, then, battery electric vehicles and plug-in hybrids separately, to ensure that the overall effect is not carried by only one of the two segments. The effects all have the expected signs and are significant almost everywhere. The estimated intent-to-treat

Table 3: Difference-in-differences analysis

			No covariates		With covariates	
Dependent variable	Control mean	Observations	Treatment 1	Treatment 2	Treatment 1	Treatment 2
Log of sales						
(a) All electric vehicles	4.85	1,305	0.240	-0.710	0.263	-0.667
			(0.048)	(0.078)	(0.054)	(0.047)
(b) Battery electric only	4.61	1,305	0.163	-0.539	0.180	-0.530
			(0.056)	(0.095)	(0.061)	(0.051)
(c) Plug-in hybrid only	3.12	1,305	0.251	-1.363	0.196	-1.446
			(0.157)	(0.150)	(0.073)	(0.092)
Log of network						
(d) Nb. of locations	3.02	1,305	0.040	0.188	-0.008	0.142
			(0.198)	(0.250)	(0.126)	(0.153)
(e) New location openings	2.00	1,305	0.278	0.315	0.185	0.145
			(0.322)	(0.291)	(0.225)	(0.189)
(f) Nb. of chargers	3.48	1,305	0.039	0.359	-0.114	0.244
			(0.193)	(0.243)	(0.146)	(0.193)
(g) New charger installations	2.42	1,305	0.427	0.521	0.247	0.278
			(0.376)	(0.303)	(0.293)	(0.228)
Network characteristics						
(h) Share of Fast DC stations, Full network	0.05	1,041	0.022	0.026	0.049	0.100
			(0.037)	(0.041)	(0.031)	(0.027)
(i) Share of Fast DC stations, New locations	0.09	808	0.005	-0.037	0.005	0.014
			(0.052)	(0.047)	(0.057)	(0.048)
(j) Share of public stations, Full network	0.96	1,041	0.074	0.100	0.016	0.008
			(0.045)	(0.050)	(0.028)	(0.026)
(k) Share of public stations, New locations	0.97	808	0.056	0.092	-0.016	-0.023
			(0.048)	(0.044)	(0.035)	(0.038)
(l) Avg. chargers per location, Full network	1.76	1,041	-0.362	0.290	-0.700	0.008
			(0.407)	(0.338)	(0.275)	(0.288)
(m) Avg. chargers per location, New locations	1.86	808	0.399	0.382	0.219	0.321
			(0.313)	(0.276)	(0.497)	(0.363)
Log of price						
(n) All vehicles	3.91	362	0.006	0.019	-0.013	-0.030
			(0.062)	(0.067)	(0.045)	(0.037)

Notes: Regressions (a) to (m) include county and year fixed effects, and are weighted by population. The chosen covariates are the average household income, the average age, the average household size, the share of graduates, the share of conservative voters, and the share of homeowners. Standard errors in parenthesis are clustered at the county level. Regression (n) are unweighted and include product and year fixed effects. The included covariates are the horsepower, the curb weight, the fuel consumption or power consumption, and the two cost shifters described in Section 4.4. Each treated observation is matched to a single observation in the control group using Mahalanobis-metric matching based on the horsepower, the curb weight, the size, the brand, and the market segment (within year). Standard errors in parenthesis are clustered at the product level.

effects on electric vehicle registrations are 0.267 and -0.667 respectively, meaning that the improvement to subsidies increased sales by 26.7% over baseline, and the abolition of the program reduced sales by 66.7% compared to baseline. Interestingly, the effect of the two treatments is asymmetric. The abolition of Ontario's rebate program had a larger impact on registrations than its improvement (this is especially true for plug-in hybrids). This does not seem to be explained completely by the magnitude of the changes to subsidies in Ontario.

Instead, the overall smaller impact of the first treatment seems to be attributed to a smaller response to the policy in 2016. Since the increased rebates were announced and

introduced in February of 2016, imperfect information transmission about the new policy environment, or the delivery times of new electric vehicles, could have contributed to reducing the impact of the policy in the first year post-treatment. In contrast, information transmission or delivery times do not play a role when the program is phased out: consumers learn that rebates are no longer available at the point of sale and change their mind about purchasing an electric vehicle. In this case, the drop in demand would be immediate.

The corresponding event studies are available in Figure A.1. They corroborate the results from the difference-in-differences analysis.

3.3 Effect on networks

I next consider the effect of electric vehicle rebates on network deployment. I use four different definitions for network size: the total number of charging locations, the total number of chargers, new location openings, and new charger installations. Results are presented in the second panel of Table 3. The corresponding event studies are available in Figure A.2 and Figure A.3.

I do not find evidence that electric vehicle subsidies increased charging station deployment through network effects. Since installing stations requires planning (e.g., securing funding, finding adequate locations, etc.), network supply may react slowly to new market conditions. It is possible that I do not observe networks for long enough to capture an effect since there were two opposing policy changes in a short period of time. Therefore, these findings must be carefully interpreted as short-run effects. I rely on a structural estimation in the second half of the paper to address long-run effects.

The difference-in-differences specification raises a specific comparability concern when network size is the dependent variable, since charging stations are primarily operated by private firms in Ontario, whereas they are provided by municipal governments in Quebec (through the Electric Circuit platform). This institutional difference does not violate the identifying assumption. The parallel trends assumption requires only that, in the absence of treatment, both provinces would have followed the same trend in network provision, not that they share identical institutional structures.

Both private and public providers ultimately respond to the same demand fundamentals, namely, the local stock of electric vehicles. Even if institutional differences result in different baseline levels (as is the case here), the level of network provision should trend similarly in both provinces absent treatment, if the underlying stock of electric vehicles in each province also trends similarly. The event studies, available in Figure A.2 and A.3, show no differential trend between Ontario and Quebec before either policy change, providing direct empirical

evidence that the parallel trends assumption holds for network-related outcomes.

Finally, I verify if networks changed along other dimensions not captured by network size. For example, network operators could respond to the policy by installing more powerful chargers or stations that can accommodate more users simultaneously. This would not require finding additional sites. I re-estimate the model using network characteristics that are available in the data. Results are presented in the third panel of Table 3. The corresponding event studies are available in Figure A.4 and Figure A.5. Again, I do not find evidence that subsidies changed networks along other dimensions in the short-run.

3.4 Effect on prices

I consider the effect of subsidies on the price of electric vehicles. Since I am using list prices that are the same in both provinces, variation across counties cannot be used to construct a difference-in-differences estimator. Instead, I reorganize the data at the product level, and consider vehicles that were eligible to Ontario's rebate program as being part of the treatment group. This includes all electric vehicles with a battery capacity above 4 kWh and a list price below $150,000. Other vehicles (e.g., internal combustion engines and hybrids) are part of the control group.[10] I index products by j and years by t. The difference-in-differences specification is

$$y_{jt} = \beta_1(\text{Eligible} \times \text{Post1})_{jt} + \beta_2(\text{Eligible} \times \text{Post2})_{jt} + X_{jt}\gamma + \mu_j + \lambda_t + \epsilon_{jt}, \tag{2}$$

where μ_j and λ_t are product and year fixed effects, and X_{jt} is a set of product characteristics and cost shifters. The dependent variable, y_{jt}, is the logarithm of the list price.

Results are presented in the fourth panel of Table 3. The vast majority of electric vehicles were eligible for subsidies, so I construct the control group by picking the internal combustion vehicle or non-rechargeable hybrid with the closest characteristics to each treated unit. The pairing is done using Mahalanobis distance matching, based on the horsepower, the curb weight, the size, the brand, and the market segment. Each treated observation is matched to a single product from the control group.[11]

I do not find evidence that changes to the Ontario rebate schedule changed firms pricing decisions, at least in the short-run. The estimated effects are very small, -1.3% and -3%, and I can bound the price response to be below 10%. This aligns with anecdotal evidence that

[10] I ignore strategic effects that could lead car manufacturers to change the price of fuel vehicles when electric vehicles are subsidized. In practice, such strategic effects are possible. In this case, the estimated effect would be biased away from zero and I would overestimate the effect of rebates on electric vehicle prices.

[11] There are a few electric vehicles that were not eligible for subsidies in the data. These vehicles are not part of either the treatment or control group.

firms responded to both policy changes by moving inventory in and out of Ontario, toward other provinces, rather than changing prices. This means that consumers benefited almost exclusively from the rebates and car manufacturers benefited from increased sales.

To rule out the possibility that the effect is driven by a composition effect, I plot the evolution of prices for the top selling battery electric and plug-in hybrid vehicles, see Figure A.6 and Figure A.7. Prices are very stable at the product level over the period of study, with the exception of Tesla's Model S, see Figure A.8. For completeness, I also report the event study corresponding to the difference-in-differences specification in Figure A.9. Again, the price responses can be bound to values very close to zero.

3.5 Continuous treatment effect

I further the analysis and study the effect of rebates on sales and prices using a continuous treatment effect specification. Details are relayed to Appendix C.

I estimate that $1,000 in additional subsidies is associated with a 7.9% increase in electric vehicle registrations and I cannot reject the hypothesis of complete passthrough of the rebate to consumers. This is additional evidence that car manufacturers did not change list prices to take advantage of the changes in subsidies over time. Allcott et al. (2024) provide evidence of complete passthrough, in their study of the Inflation Reduction Act's electric vehicle incentives, while Muehlegger and Rapson (2022) finds evidence of incomplete passthrough, in their study of an electric vehicle incentive program in California. Both studies focus on transaction prices which are more likely to be manipulated by firms in the presence of subsidies than list prices.

The estimated parameters imply a market elasticity to price of -3.0 for electric vehicles. Muehlegger and Rapson (2022) obtain a slightly lower estimate using a similar methodology. They report an elasticity to price of -2.1. The difference between the two estimates can be explained partly by the fact that they use transaction prices which are typically lower after bargaining while I rely on list prices.

Due to data limitations, it is not possible to study the environmental performance of Canadian subsidies using the natural experiment setting. The difficulty arises from the fact that electric vehicles reduce emissions to the extent that they replace internal combustion engines. Extending the analysis to include emission abatement would require a survey of electric vehicle owners that informs me about their second choices (and, therefore, the exact composition of the counterfactual fleet of vehicles), or unreasonably strong assumptions about consumers' substitution patterns.

To circumvent these issues, I build on the findings presented in this section and estimate a

structural model of the demand for cars and the supply of a charging station infrastructure. I recover fundamental parameters which allow me to perform counterfactual experiments and evaluate the environmental performance of the Canadian subsidy programs. I present the model and the results in the following sections.

4 Model

I define a structural model to analyze the cost-effectiveness and the emission reduction potential of electric vehicle subsidies. Demand for cars is determined using the random coefficient logit model as in Berry et al. (1995). I augment the demand specification with county-level average demographics to capture differences in preferences among consumers, following Nevo (2001) and Gandhi and Houde (2019). Similarly to most works on the car market, I maintain the assumption that consumers are not forward-looking, and the demand for cars is static.[12]

I include a supply side for cars to account for potential price responses from car manufacturers to electric vehicle subsidies. However, I ignore the potential entry or exit of products that could result from subsidizing electric vehicles. The overwhelming majority of electric vehicles available in Canada are also available in the United States and vice versa. While it is possible that car manufacturers would develop new electric vehicles in response to the overall North-American level of electric vehicle subsidization, I find it unlikely that they would commit to large investment decisions in response to a local policy that affects a small part of the North-American market.

Finally, I define a model for charging station supply which fits the specific economic and political context in Quebec, where county-level governments are responsible for providing a public charging station infrastructure in their jurisdiction. The network supply model takes into account the forward-looking behavior of these local planners in a tractable, easy to implement way. Moreover, it solves the simultaneity issue between electric vehicle sales and station deployment by fully internalizing the demand response from potential electric

[12]There are several reasons that explain this modeling assumption. First, the overwhelming majority of works related to the car market also assume that the demand and supply for cars are static. A large share of consumers are renting a car on a fixed-term lease. For these consumers, the timing of their next purchase is determined by the end of their term, and a static model is appropriate in that context. Second, introducing dynamics to the random coefficient framework significantly increase the complexity of the estimation and the computation of counterfactuals, and is not crucial for estimating the environmental performance of subsidies. Finally, Springel (2021) provides some evidence to support the static demand assumption in electric vehicle markets. Ignoring dynamics on the demand side has potential implication for the decision to install chargers. For example, when electric vehicle adoption occurs, this mechanically reduces the pool of potential adopters in future periods and decreases future adoption. This in turns reduces the network operators' incentive to install chargers in the future versus today, changing the optimal evolution of networks over time.

vehicle owners in the supply equation. This allows for estimating the model without relying on instrumental variables. These advances improve on the currently available best practices.

4.1 Demand

Consider consumer i living in county m. Each period t, this consumer chooses to purchase one of the $j = 1, ..., J_{mt}$ car makes available, or to purchase nothing at all, denoted $j = 0$. In choosing which product to purchase, the consumer considers the net price of each product, $p_{jt} - \tau_{jt}$, where p_{jt} is the retail price and τ_{jt} a government subsidy on product j. It also considers observed product characteristics such as the horsepower, the driving cost, or the engine type. I denote the vector of observed product attributes by $\mathbf{x}_{jt}$. Furthermore, the consumer considers characteristics that are unobserved to the econometrician, summarized by the quality index ξ_{jmt}.

For all models with an electric engine, the consumer cares about the opportunity of charging at home or on the network. Let N_{mt} be the size of the charging network in county m at time t, and define the utility of charging as the deterministic function $v(N_{mt}, \theta_i)$, where θ_i is a consumer-specific preference parameter. For any product j, the associated utility of charging is

$$v_j(N_{mt}, \theta_i) = \begin{cases} v(N_{mt}, \theta_i), & \text{if } j \in EV \\ 0, & \text{if } j \notin EV \end{cases}. \tag{3}$$

The function $v(N_{mt}, \theta_i)$ represents the expected lifetime utility that a consumer derives from charging on the network. Since the model is static, and $v(N_{mt}, \theta_i)$ depends on current network provision, this means that consumers are myopic about future improvements to the charging infrastructure. Therefore, their purchasing decision is based on their perceived lifetime usage of the current network.

I select a flexible functional form for the utility of charging, namely,

$$v(N_{mt}, \theta_i) = \theta_i \frac{(1 + N_{mt})^{\zeta} - 1}{\zeta}, \tag{4}$$

where N_{mt} represents the stations available to consumer i in his county of residence, and the "1" represents the opportunity to charge at home. This means that, when no stations are available in a region, consumers are assumed to always have the option to charge at home. The parameter ζ controls the curvature of $v(N_{mt}, \theta_i)$.[13] The function $v(N_{mt}, \theta_i)$ is increasing

[13]The utility of charging follows the isoelastic (or constant relative risk aversion) functional form and is defined for $\zeta \leq 1$, $\zeta \neq 0$. It covers useful parametrizations as special cases: (1) linear ($\zeta = 1$), and (2)

at a decreasing rate in N_{mt} for $\theta_i > 0$ and $\zeta < 1$, such that each additional station is less valuable to the consumer than the previous one.

I allow consumers to have heterogeneous preferences in the observed product characteristics. Heterogeneity is introduced in two ways. First, the average taste for observed characteristics varies across regions through interactions with county-level average demographics, denoted $\mathbf{D}_{mt}$. Second, I allow for random coefficients to model the heterogeneity within county. Formally, the utility consumer i receives from purchasing product j is

$$u_{ijmt} = \beta_i^{\mathbf{p}}(p_{jt} - \tau_{jt}) + v_j(N_{mt}, \theta_i) + \mathbf{x}_{jt}\beta_i^{\mathbf{x}} + \xi_{jmt} + \epsilon_{ijmt}^{\mathbf{d}}, \tag{5}$$

where $\epsilon_{ijmt}^{\mathbf{d}}$ is a consumer-specific disturbance. Consumers' taste parameters take the following form,

$$\begin{aligned}
\beta_i^{\mathbf{p}} &= \beta^{\mathbf{p}} + \mathbf{D}_{mt}\Gamma^{\mathbf{p}} + \sigma^{\mathbf{p}}\nu_i^{\mathbf{p}}, \\
\beta_{ik}^{\mathbf{x}} &= \beta_k^{\mathbf{x}} + \mathbf{D}_{mt}\Gamma_k^{\mathbf{x}} + \sigma_k^{\mathbf{x}}\nu_{ik}^{\mathbf{x}}, \\
\theta_i &= \theta + \mathbf{D}_{mt}\Gamma^{\mathbf{N}} + \sigma^{\mathbf{N}}\nu_i^{\mathbf{N}},
\end{aligned} \tag{6}$$

where k indexes the different product characteristics in $\mathbf{x}_{jt}$ and the $\nu_i = \{\nu_i^{\mathbf{p}}, \nu_i^{\mathbf{x}}, \nu_i^{\mathbf{N}}\}$ are jointly distributed as independent standard normal.

The utility of the outside option is normalized to $u_{i0mt} = \epsilon_{i0mt}^{\mathbf{d}}$ in each market. The outside option includes purchasing a vehicle on the secondary market. For consumers purchasing an electric vehicle on the secondary market, subsidies are available, and they derive benefits from charging station networks, so, their utility should include $v(N_{mt}, \theta_i)$. In practice, the secondary market for electric vehicles in Quebec is very small over the period of study, about 30 times smaller than the primary market. Given its size, I choose to ignore the secondary market for electric vehicles to avoid needlessly complicating the model and its estimation.

I rewrite the utility function in terms of a mean utility and a consumer-specific deviation,

$$u_{ijmt} = \delta_{jmt} + \mu_{ijmt} + \epsilon_{ijmt}^{\mathbf{d}}, \tag{7}$$

logarithmic ($\zeta \to 0$).

with

$$\delta_{jmt} = \beta^{\mathbf{p}}(p_{jt} - \tau_{jt}) + \big((p_{jt} - \tau_{jt}) \otimes \mathbf{D}_{mt}\big)\Gamma^{\mathbf{p}} \tag{8}$$
$$+ \mathbf{x}_{jt}\beta^{\mathbf{x}} + \big(\mathbf{x}_{jt} \otimes \mathbf{D}_{mt}\big)\Gamma^{\mathbf{x}} + \xi_{jmt},$$

$$\mu_{ijmt} = \sigma^{\mathbf{p}}\nu_i^{\mathbf{p}}(p_{jt} - \tau_{jt}) + v_j(N_{mt}, \theta_i) + \sum_k x_{jt}^k \sigma_k^{\mathbf{x}} \nu_{ik}^{\mathbf{x}}. \tag{9}$$

Assuming that the taste shocks $\epsilon_{ijmt}^{\mathbf{d}}$ are independent and identically distributed as extreme value type I, the probability that consumer i purchases product j is given by

$$s_{ijmt}(\mathbf{p}_t, N_{mt}, \mathbf{x}_t, \mathbf{D}_{mt}, \nu_i) = \frac{\exp(\delta_{jmt} + \mu_{ijmt})}{1 + \sum_{j'=1}^{J_{mt}} \exp(\delta_{j'mt} + \mu_{ij'mt})}. \tag{10}$$

Taking expectation over all consumers yields the following aggregate demand for product j,

$$s_{jmt}(\mathbf{p}_t, N_{mt}, \mathbf{x}_t, \mathbf{D}_{mt}) = \int s_{ijmt}(\mathbf{p}_t, N_{mt}, \mathbf{x}_t, \mathbf{D}_{mt}, \nu_i)\, dF(\nu_i). \tag{11}$$

4.2 Supply

Consider car manufacturer f selling a subset of products $\mathcal{J}_f \subset \mathcal{J}$ in all counties, denoted by m. Each period t, the car manufacturer sets a uniform price for each product in all counties to maximize its total profits, that is,

$$\pi_{ft} = \max_{\{p_{jt}\}_{j \in \mathcal{J}_f}} \sum_{j \in \mathcal{J}_f} \sum_{m=1}^{M} q_{jmt}(\mathbf{p}_t, N_{mt}, \mathbf{x}_t, \mathbf{D}_{mt}) \cdot (p_{jt} - c_{jt}), \tag{12}$$

with $q_{jmt}(\mathbf{p}_t, N_{mt}, \mathbf{x}_t, \mathbf{D}_{mt}) = L_{mt} \cdot s_{jmt}(\mathbf{p}_t, N_{mt}, \mathbf{x}_t, \mathbf{D}_{mt})$ is the total quantity sold of car model j in market m and L_{mt} is the market size.

Firms compete in a multi-product oligopoly. The first-order conditions associated with profit maximization are

$$p_{jt} = c_{jt} + \left[\left(\sum_{m=1}^{M} L_{mt} \cdot \nabla_{mt}\right)^{-1}\right]_j \cdot \sum_{m=1}^{M} L_{mt} \cdot s_{jmt}, \quad j = 1, ..., J_t \tag{13}$$

where $[X]_j$ represents the j-th row of matrix X and the (j,k)-th element of matrix ∇_{mt} is[14]

$$\nabla_{mt}^{j,k} = \begin{cases} -\frac{\partial s_{kmt}}{\partial p_{jt}}, & \text{if } (j,k) \in \mathcal{J}_f \\ 0, & \text{otherwise} \end{cases}. \tag{14}$$

4.3 Network supply

I consider the case of county-level governments (henceforth "local planners") responsible for supplying a public network of charging stations in their respective jurisdiction.[15] Throughout, I maintain the assumption that these local planners are forward-looking: they install public chargers in their jurisdiction to maximize the current and future utility of consumers for the network, given installation costs. This objective is consistent with the mandate described by local policymakers, who view charging infrastructure as a public good whose primary beneficiaries are current and future electric vehicle owners.

Local governments collect revenues from operating their local network. However, anecdotal evidence suggests that they pursue non-profit motives in installing charging stations, with revenues mostly cover operational costs and maintenance.[16] I could not find evidence that local governments in Quebec held emission targets (related to electric vehicles) or electric vehicle adoption targets over the period of study, hence I leave these motives out of the local planners' problem. This introduces a friction between the local planners' objective and that of the central government, which uses subsidies to correct an environmental externality. I discuss and study alternative local planner objectives in Appendix F.

I assume throughout that the local planners do not coordinate on a joint deployment strategy, and that they control both the installation decision and the location of stations within their county, such that each local planner is a monopolist over network provision in its county. To keep the model tractable, I assume that local planners do not consider spillovers from drivers in other counties at the time of choosing network provision.[17] Finally, I consider local networks to be public goods, that is, they are non-excludable, non-rivalrous,

[14] See Appendix E.3 for details on the computation of the market share derivatives.

[15] In practice, network ownership in Quebec is split between the local planners and a number of fringe firms. Since local planners control around 70% of all stations, I assume throughout that they are the sole operator in each region. The remaining stations are installed by workplaces, shopping malls, or restaurant chains, and are typically not organized into what I consider as competing network operators.

[16] Non-profit objectives could include reducing local emissions, maximizing local adoption, maximizing political support from electric vehicle owners, or maximizing consumers' welfare, to give a few examples.

[17] This is reasonable in most counties, but not in or near large urban centers. For example, in the median county, 71% of household work within their county of residence (most of the charging occurs near home or near work). This number drops to 55% for the county at the 25th percentile, and 44% for the county at the 10th percentile. Studying these spillovers would require additional data on electric vehicle owners' charging patterns, hence, I leave the study of spillover effects for future work.

and publicly supplied by local planners. Since I define each station as a charging site that can accommodate two to four drivers simultaneously, I consider the non-rivalrous assumption to be satisfied in most cases.

Law of motion. Before I define the local planners' problem, I consider the law of motion of the electric vehicle base. In what follows, I consider a network of size $n \in \mathbb{N}$, where I leave the market indices out as n does not represent a particular observed outcome. I also do not restrict n to represent the optimal network provision in a given market at a given point in time. Let $Q^{ev}_{mt}(n)$ be the stock of electric vehicles in circulation (e.g., cumulative sales up to period t) and $q^{ev}_{mt}(n)$ be the flow of electric vehicles (e.g., sales) in county m and period t, given a network of size n. The law of motion of $Q^{ev}_{mt}(n)$ can be written as

$$Q^{ev}_{mt}(n) = (1-d)\, Q^{ev}_{m,t-1} + q^{ev}_{mt}(n), \tag{15}$$

where d is the fleet depreciation rate and $Q^{ev}_{m,t-1}$ includes only past sales, hence it is predetermined and does not depend on n. The term $q^{ev}_{mt}(n)$ can be recovered in each market by aggregating over market shares and multiplying by the market size L_{mt}, that is,

$$q^{ev}_{mt}(n) = L_{mt} \cdot \sum_{j \in EV} s_{jmt}(\mathbf{p}_t, n, \mathbf{x}_t, \mathbf{D}_{mt}), \tag{16}$$

where $\mathbf{p}_{jt}$, $\mathbf{x}_{jt}$, and $\mathbf{D}_{mt}$ are defined in the previous section.

Benefit function. I now turn to the local planners' problem. In what follows, the index m represents both a local planner and its associated county. Each local planner m installs charging stations in its jurisdiction to maximize the aggregate value of the network to electric vehicle owners, given the fixed cost of adding capacity.

I define Recall from the previous section that electric vehicle owner i derives a lifetime indirect utility of $v(n, \theta_i)$ from a network of size n, given preference parameter θ_i. For a given consumer discount rate of r and a scrapage rate of d, I can show that the per period indirect utility of charging is $\widetilde{r}v(n, \theta_i)$, where $\widetilde{r} = (r+d)/(1-d)$ is a composite discount rate.

I define the local planner's contemporaneous marginal benefits of increasing the network

size from $n-1$ to n as a function of consumers' network utilities, that is,

$$MB_{mt}(n) = \Psi_{mt} \cdot \underbrace{Q^{ev}_{mt}(n)}_{\substack{\text{Number of} \\ \text{EV owners}}} \cdot \left(\underbrace{\int \frac{\widetilde{r}v(n,\theta_i) - \widetilde{r}v(n-1,\theta_i)}{-\beta_i^{\mathbf{P}}} \, dF(\nu_i|EV)}_{\substack{\text{Average gain in utility per EV owner} \\ \text{from station } n}} \right)^{\gamma}, \tag{17}$$

where $Q^{ev}_{mt}(n)$ and $v(n,\theta_i)$ are defined above, $-\beta_i^{\mathbf{P}}$ is the marginal utility of income of consumer i ($\beta_i^{\mathbf{P}}$ is the price sensitivity), and $\gamma \in \mathbb{R}^+$ is a local planner preference parameter that controls the curvature of the marginal benefits function. Finally, $\Psi_{mt} \in \mathbb{R}^+$ is a time-varying, local planner-specific intercept that allows for heterogeneous preferences across local planners and over time. The integral is taken over the distribution of current and past consumers that have purchased an electric vehicle, denoted by $F(\nu_i|EV) \neq F(\nu_i)$.[18]

The term in parenthesis represents the monetary equivalent of the expected gain in utility that an electric vehicle owner receives when the network size increases from $n-1$ to n. The local planner's marginal benefit function, $MB_{mt}(n)$, can therefore be seen as the product between the total installed based of electric vehicles and the expected gain in utility from electric vehicle owners, accruing from the installation of station n, scaled by preference parameters Ψ_{mt} and γ.[19] The preference parameters are included to allow for the planner to value more or less charging on the network than electric vehicle owners themselves, and to avoid restricting the elasticity of network supply. To simplify the notation in what follows, I denote

$$\Delta v(n) = \int \frac{\widetilde{r}v(n,\theta_i) - \widetilde{r}v(n-1,\theta_i)}{-\beta_i^{\mathbf{P}}} \, dF(\nu_i), \tag{18}$$

and I can rewrite the benefit function as

$$MB_{mt}(n) = \Psi_{mt} \cdot Q^{ev}_{mt}(n) \cdot \Delta v(n)^{\gamma} \tag{19}$$

I impose two assumptions on the local planners' benefit function. First, I assume that local planners are price-takers in the charging market. This prevents local planners from affecting consumers' utility via driving costs. This is easily satisfied in this context, as Quebec's provincial government regulates both energy prices and charging prices. Second, I assume that $MB(n)$ is positive and decreasing in n. This condition is sufficient to have a unique equilibrium in network size for a given stock of electric vehicles. This is trivially

[18] Computational details on this conditional cumulative distribution function are relayed to Appendix E.2.
[19] See Appendix F for additional details on the construction of the marginal benefits function.

satisfied over some range of γ if $\partial v(n,\theta_i)/\partial n \geq 0$ and $\partial^2 v(n,\theta_i)/\partial n^2 < 0, \forall n \in \mathbb{N}$, and $\beta_i^{\mathbf{p}} < 0, \forall i$.

Installation decision. A local planner which chooses to install station n pays a one-time fixed cost FC_{mt}, unobserved to the econometrician, then reaps the lifetime benefits of operating that station. Therefore, the marginal value to the local planner of installing station n is

$$MV_{mt}(n) = -FC_{mt} + MB_{mt}(n) + \sum_{s=t+1}^{\infty} \rho^{s-t} \mathbb{E}_t MB_{ms}(n, \mathcal{I}_t), \tag{20}$$

$$= -FC_{mt} + MB_{mt}(n) + \rho \mathbb{E}_t \overline{MV}_{m,t+1}(n, \mathcal{I}_t), \tag{21}$$

where ρ is the planner's discount factor and $\overline{MV}_{mt} = MV_{mt} + FC_{mt}$.[20] Equation (20) introduces some new notation for the expected benefits. Let $\mathcal{I}_{t+k}$ indicate that station n was installed in period $t+k$. I define $\mathbb{E}_t MB_{ms}(n, \mathcal{I}_{t+k})$ as the expected benefit of station n in period $s > t+k$ given that station n was installed in period $t+k$. Keeping track of the installation date is important as the electric vehicle base accumulates differently before and after the installation of station n.

The local planner chooses to install station n in period t if it is more profitable than waiting. Its installation decision can be summarized as follows,

$$\iota_{mt}(n) = \begin{cases} \text{Install}, & \text{if} \quad MV_{mt}(n) \geq \max\limits_{k \geq 1} \left\{ \rho^k \mathbb{E}_t MV_{m,t+k}(n, \mathcal{I}_{t+k}) \right\} \\ \text{Not install}, & \text{otherwise} \end{cases}, \tag{22}$$

where the notation for $\mathbb{E}_t MV_{ms}(n, \mathcal{I}_{t+k})$ is similar to that of $\mathbb{E}_t MB_{ms}(n, \mathcal{I}_{t+k})$.

To simplify the expression in (22) and make the model tractable, I impose some additional assumptions on the planners' expectations. These assumptions are:

A1. $0 \leq \mathbb{E}_t F_{m,t+k} - \rho \mathbb{E}_t F_{m,t+k+1} \leq K_t(\rho), \quad \forall k \geq 1;$

A2. $\mathbb{E}_t q^{ev}_{m,t+k}(n) = (1 + g_{m,t+1}) \dots (1 + g_{m,t+k})\, q^{ev}_{mt}(n), \quad \forall n \in \mathbb{N}, \quad \forall k \geq 1;$

A3. $q^{ev}_{mt}(n) > q^{ev}_{mt}(n-1), \quad \forall n \in \mathbb{N}.$

[20] The function $MV(n)$ does not represent a "value at state" as in a traditional dynamic model. Instead, it captures the lifetime value of installing one more station and doesn't include, for example, the value of previously installed stations $(1, ..., n-1)$. The reader should interpret $MV(n)$ as a the marginal value of station n.

Assumption **A1** is the most restrictive and imposes limits on how the expected fixed costs vary over time, from the point of view of period t.[21] There is some anecdotal evidence that the costs of charging infrastructure decrease over time as a result of learning-by-doing and Assumption **A1** imposes that these costs do not decrease too fast. Assumption **A2** states that the local planners' best guess about future electric vehicle sales is current sales, multiplied by some exogenous growth rates $(1+g_{m,t+1}), ..., (1+g_{m,t+k})$. In some sense, local planners are uncertain about future market conditions, such that their expectations about future sales are based on current market conditions.[22] Finally, assumption **A3** holds trivially by strict monotonicity of consumer preferences.

Under these assumptions, I can show that

$$\max_{k\geq 1}\left\{\rho^k \mathbb{E}_t MV_{m,t+k}(n, \mathcal{I}_{t+k})\right\} = \rho \mathbb{E}_t MV_{m,t+1}(n, \mathcal{I}_{t+1}), \tag{23}$$

and the installation condition in (22) collapses to a comparison between period t and $t+1$. The full proof and an extended discussion on assumptions **A1** – **A3** are available in Appendix D, Lemma 1.

Optimal network provision. I denote the last station installed by $N \in \mathbb{N}$, also representing the optimal network provision. Again, I omit market indices, as N does not represent a particular observed market outcome. Since N represents the optimal network size, it must be that local planner m found it weakly beneficial to install station N, but not $N+1$. Hence, the optimal network size at any given point in time has to satisfy the following two inequality conditions,

$$MV_{mt}(N) \geq \rho \mathbb{E}_t MV_{m,t+1}(N, \mathcal{I}_{t+1}) \tag{24}$$

and

$$MV_{mt}(N+1) < \rho \mathbb{E}_t MV_{m,t+1}(N+1, \mathcal{I}_{t+1}). \tag{25}$$

I focus on equation (24) as solving (25) follows the same logic with the reversed inequality.

[21] The constant $K_t(\rho)$ is defined explicitly in Appendix D, Lemma 1.

[22] Assumption **A2** implies in practice that the expectation about future sales depends on the current level of sales and a series of exogenous growth rates $(1+g_{m,t+1}), ..., (1+g_{m,t+k})$. This means that the local planner's choice of network size can shift the level of future sales (through $q^{ev}_{mt}(n)$), but not the growth rate of future sales. This is potentially too restrictive. If growth rates were in fact endogenous, all of my results and the proof of Lemma 1 (see Appendix D) would hold, as long as $g_{m,t+k}(n) \geq g_{m,t+k}(n-1)$, for all n and k. That is, increasing network sizes increases sales in levels and in growth rates. This seems reasonable in the context of the electric vehicle market. To simplify the exposition and avoid burdening the notation, I maintain the exogeneity assumption throughout the paper.

Consider first the right-hand side of the inequality. By taking expectation over $MV_{m,t+1}$, I can show that

$$\begin{aligned}\rho\mathbb{E}_t MV_{m,t+1}(N,\mathcal{I}_{t+1}) &= -\rho\mathbb{E}_t FC_{m,t+1} + \sum_{s=t+1}^{\infty}\rho^{s-t}\mathbb{E}_t MB_{ms}(N,\mathcal{I}_{t+1}),\\ &= -\rho\mathbb{E}_t FC_{m,t+1} + \rho\mathbb{E}_t\overline{MV}_{m,t+1}(N,\mathcal{I}_{t+1}). \end{aligned} \tag{26}$$

I can then rewrite the inequality condition in (24) as

$$\underbrace{MB_{mt}(N) + \rho\mathbb{E}_t\overline{MV}_{m,t+1}(N,\mathcal{I}_t) - \rho\mathbb{E}_t\overline{MV}_{m,t+1}(N,\mathcal{I}_{t+1})}_{\substack{\text{Discounted future value of installing station N}\\ \text{in period } t \text{ versus period } t+1}} \geq FC_{mt} - \rho\mathbb{E}_t FC_{m,t+1}. \tag{27}$$

The bracketed term in equation (27) is the discounted future value of installing station N in period t, compared to installing it in period $t+1$. Very intuitively, installing station N will provide benefits from current electric vehicle owners and convince some additional consumers to buy an electric vehicle in period t, which permanently increases the stock of electric vehicles. Then, the local planners will reap benefits from these marginal consumers into the future. I can show that

$$\begin{aligned}&\rho\mathbb{E}_t\overline{MV}_{m,t+1}(N,\mathcal{I}_t) - \rho\mathbb{E}_t\overline{MV}_{m,t+1}(N,\mathcal{I}_{t+1})\\ &\qquad = \underbrace{\Psi_{mt}\cdot\frac{\widetilde{\rho}}{1-\widetilde{\rho}}\cdot\Delta v(N)^{\gamma}\cdot\left(q^{ev}_{mt}(N) - q^{ev}_{mt}(N-1)\right)}_{\substack{\text{Discounted lifetime benefits}\\ \text{of marginal consumers}}}, \end{aligned} \tag{28}$$

where $q^{ev}_{mt}(N) - q^{ev}_{mt}(N-1)$ are the marginal consumers that purchase an electric vehicle when station N is installed, $\Psi_{mt}\cdot\widetilde{\rho}/(1-\widetilde{\rho})\cdot\Delta v(N)^{\gamma}$ is the per driver discounted lifetime benefit of the local planner, and $\widetilde{\rho} = \rho\,(1-d)$ is a composite discount factor. The full proof is relayed to Appendix D, Lemma 2. To simplify the notation in what follows, I denote

$$\Delta q^{ev}_{mt}(N) = q^{ev}_{mt}(N) - q^{ev}_{mt}(N-1). \tag{29}$$

Replacing equation (28) in (27), we can rewrite the inequality conditions in (24) and (25)

as

$$\Psi_{mt} \cdot \left(Q^{ev}_{mt}(N) + \frac{\widetilde{\rho}}{1-\widetilde{\rho}} \Delta q^{ev}_{mt}(N) \right) \cdot \Delta v(N)^{\gamma} \geq FC_{mt} - \rho \mathbb{E}_t FC_{m,t+1} \tag{30}$$

and

$$\Psi_{mt} \cdot \left(Q^{ev}_{mt}(N+1) + \frac{\widetilde{\rho}}{1-\widetilde{\rho}} \Delta q^{ev}_{mt}(N+1) \right) \cdot \Delta v(N+1)^{\gamma} < FC_{mt} - \rho \mathbb{E}_t FC_{m,t+1}. \tag{31}$$

The term containing the fixed costs is unobserved to the econometrician. I impose the assumption that $FC_{mt} - \rho \mathbb{E}_t FC_{m,t+1}$ is independent and identically distributed as Lognormal, or equivalently,

$$\ln(FC_{mt} - \rho \mathbb{E}_t FC_{m,t+1}) = \eta_{mt} + \omega \epsilon^{\mathbf{n}}_{mt}, \tag{32}$$

with $\epsilon^{\mathbf{n}}_{mt} \sim \text{Normal}(0,1)$. Combining with equations (30) and (31) and taking logs yield an equilibrium condition of the form

$$\mathcal{G}_{mt}(N) \geq \omega \epsilon^{\mathbf{n}}_{mt} > \mathcal{G}_{mt}(N+1), \tag{33}$$

where N is the equilibrium network size and

$$\mathcal{G}_{mt}(N) = \ln \Psi_{mt} - \eta_{mt} + \gamma \ln(\Delta v(N)) + \ln \left(Q^{ev}_{mt}(N) + \frac{\widetilde{\rho}}{1-\widetilde{\rho}} \Delta q^{ev}_{mt}(N) \right). \tag{34}$$

Network supply function. From equation (34), we can write the observed network supply as

$$N_{mt} = \sum_{n=1}^{+\infty} n \cdot \mathbb{1} \left\{ \frac{1}{\omega} \mathcal{G}_{mt}(n) \geq \epsilon^{\mathbf{n}}_{mt} > \frac{1}{\omega} \mathcal{G}_{mt}(n+1) \right\}. \tag{35}$$

Essentially, the equilibrium condition in (35) is only valid at the observed equilibrium N_{mt}, so the sum will contain a single non-zero term at $n = N_{mt}$.

The network supply function in (35) emphasizes two important features of the model. First, the forward-looking behavior of local planners is explicit. For example, by setting ρ to zero, local planners stop valuing the future benefits of the network associated to marginal consumers, and the term $\widetilde{\rho}/(1-\widetilde{\rho})\, \Delta q^{ev}_{mt}(n)$ in $\mathcal{G}_{mt}(n)$ vanishes. In this case, we obtain a static model similar in spirit to Springel (2021).

Second, it clearly shows how local planners fully internalize the effect of station installations on future electric vehicle adoption. A careful inspection of the law of motion and the network supply equation reveals that network supply depends on $Q^{ev}_{m,t-1}$, which is predetermined, and on $q^{ev}_{mt}(n)$, which is constructed from the estimated demand system and evaluated at each potential network size $n \in \mathbb{N}$. Because the supply equation embeds the full demand system (with $q^{ev}_{mt}(n)$ evaluated at every potential network size rather than only at the observed equilibrium), local planners effectively choose both sides of the market simultaneously which internalizes the full equilibrium mechanism. In contrast, replacing $q^{ev}_{mt}(n)$ by its data counterpart q^{ev}_{mt} for all $n \in \mathbb{N}$ would treat market shares as invariant to network size, reintroducing simultaneity bias. Therefore, the specification presented in equation (35) suggests a path for estimating station supply without instrumental variables, provided that the estimated demand fundamentals are uncorrelated with the supply disturbances $\epsilon^{\mathbf{n}}_{mt}$. I discuss these potential endogeneity issues in Section 4.4.

4.4 Identification and estimation

Demand. I have to deal with several sources of endogeneity. First, prices depend not only on observed product characteristics, but also on unobserved product quality (to the econometrician), leading to the price endogeneity issue described in Berry et al. (1995). Second, my estimation routine relies on the inversion of the market shares to recover mean utilities $\delta(\mathbf{s}_t, \sigma)$. This implies that market shares are also endogenous since they are determined jointly with unobserved car attributes.[23] Concretely, this means that instrumental variables are needed for the prices and the market shares in the demand model. Finally, network deployment occurs simultaneously with electric vehicle sales, hence, network size is also endogenous in the demand model.

I solve the various endogeneity issues using instrumental variables. I use two separate cost shifters to instrument for prices. Similarly to D'Haultfoeuille et al. (2019), I construct a composite price index to capture variations in the production cost of the various car makes. I use four key input prices: steel, iron, plastics, and aluminum. I compute a weighted average cost per ton which I interact with each vehicle's curb weight to create the composite price index.[24] Next, I follow Grieco et al. (2023) and use the real exchange rate between Canada and the country each car was manufactured as an additional cost shifter.[25] The real exchange rate captures, among other things, variations in the cost of labor between Canada and the

[23] See Conlon and Gortmaker (2020) and Gandhi and Houde (2019).

[24] I assume cars are made of 56% steel, 8% iron, 8% plastics, 10% aluminum, and 18% of other materials not captured by the index.

[25] Real exchange rates are obtained from Penn World Tables, version 10.0, `pl_con`. See Grieco et al. (2023).

car's country of origin which affects the marginal production cost. Similarly to Grieco et al. (2023), I lag both cost shifters by one year to reflect planning horizons. I denote the set of price instruments by $\mathbf{z^p}$.

To solve for the endogeneity of the market shares, I follow the intuition in Gandhi and Houde (2019) to construct instruments based on product characteristic differences. I use the fact that the market segment is a strong dimension of differentiation, and interact it with other characteristics to construct basis functions. Formally, I construct the following instruments,

$$\mathbf{z}^{\mathbf{s}}_{jmt} = \begin{cases} \text{(a)} \ \sum_{j' \notin \mathcal{J}_f} \mathbb{1}(j' \text{ same segment as } j) \\ \text{(b)} \ \sum_{j' \notin \mathcal{J}_f} \mathbb{1}(j' \text{ same segment as } j) \times \mathbb{1}(j' \text{ same engine as } j) \\ \text{(c)} \ \sum_{j' \notin \mathcal{J}_f} \mathbb{1}(j' \text{ same segment as } j) \times d^{\widehat{p}}_{j',j} \\ \text{(d)} \ \sum_{j' \notin \mathcal{J}_f} \mathbb{1}(j' \text{ same segment as } j) \times \mathbf{d}^{\mathbf{x}}_{j',j} \\ \text{(e)} \ \sum_{j' \notin \mathcal{J}_f} \mathbb{1}(j' \text{ same segment as } j) \times (\mathbf{D}_{mt} \otimes d^{\widehat{p}}_{j',j}) \\ \text{(f)} \ \sum_{j' \notin \mathcal{J}_f} \mathbb{1}(j' \text{ same segment as } j) \times (\mathbf{D}_{mt} \otimes \mathbf{d}^{\mathbf{x}}_{j',j}) \end{cases}, \tag{36}$$

where $d^{x}_{j',j} = x_{j'} - x_j$ for some continuous characteristic $x \in \mathbf{x}$. To put it plainly, these instruments are (a) the number of competitors within segment, (b) the number of competitors within segment with the same engine type, (c) the sum of predicted price differences, and (d) the sum of exogenous characteristics differences between products of competitors in the same segment. Since prices are endogenous, they cannot be used to construct a differentiation instrument. They still contain a useful source of variation to identify consumers' heterogeneity in price sensitivity. I follow Reynaert and Verboven (2014) and Gandhi and Houde (2019) and use the projection of prices on exogenous characteristics and cost shifters, denoted $\widehat{p}_{jt} = \mathbb{E}(p_{jt} \mid \mathbf{x}_{jt}, \mathbf{z}^{\mathbf{p}}_{jt})$ to construct the instruments in (c). Finally, interactions with county-level average demographics in $\mathbf{D}_{mt}$ are used to construct instruments in (e) and (f).

I now address the endogeneity of charging stations in the demand equation. I follow the approach in Hausman (1996) and Nevo (2001), which use the panel structure of the data to construct instruments. Formally, the idea is to use networks in other regions to instrument for local charging stations. The installation of new stations depends on local consumption (e.g.., the installed base of electric vehicles in a given region) and a common cost component across regions that does not depend on consumption once I account for region fixed effects.[26]

[26]For example, this would be satisfied if the retail cost of acquiring a new charger is the same across regions, and the variations that can arise from ordering in bulk, delivery costs, and local variations in installation

Networks in other regions are valid instruments for local stations as long as the correlation between networks comes from sharing a common cost component and not from common shocks that affect several markets simultaneously or electric vehicle owners charging over region lines. This could be a concern for markets that are geographically close to each other.

Let $\text{dist}_{\ell,m}$ be the distance in kilometers between county ℓ and m. I impose a distance threshold, denoted by K, to select networks that are far enough to be valid instrument for local network size. I then construct a basis function, aggregating over these distant networks, that is,

$$z_{jmt}^{\mathbf{N}} = \begin{cases} \frac{\sum_{\ell \neq m} \mathbb{1}(\text{dist}_{\ell,m} > K) N_{\ell t}}{\sum_{\ell \neq m} \mathbb{1}(\text{dist}_{\ell,m} > K)} & \text{if } j \in EV \\ 0, & \text{if } j \notin EV \end{cases}. \tag{37}$$

To identify the curvature parameter ζ in the network utility function, I form additional instruments, based on $z_{jmt}^{\mathbf{N}}$, where I aggregate over functions of N_{mt} that map to particular realizations of ζ, see Birchall et al. (2024). The chosen functions are $\sqrt{N_{mt}}$ and $\ln(1+N_{mt})$, which correspond to curvature parameters $\zeta = 0.5$, and $\zeta \to 0$.

I use a radius of 300 kilometers from each county's centroid to determine which networks enter the basis function. The choice of a threshold is rather arbitrary. To document the robustness of my results to this assumption, I estimate a simple logit demand model, and vary the threshold in increments of 50 kilometers. The robustness analysis shows that we achieve very similar estimates with or without a distance threshold. The results are available in Table A.2, and the first-stage estimates in Table A.3 to A.5.

Several factors could break this instrumental variable strategy. Unobserved common shocks to network provision that affect all counties together would violate the exclusion restriction and invalidate instruments based on networks in other regions. A large scale advertisement campaign that raises awareness about environmental issues or a significant investment into charging stations from the provincial or federal governments that affects all regions together are examples. The network supply model includes demographics that capture county-level differences, and year fixed effects that should account for aggregate shocks to network supply, partially mitigating this concern. More importantly, and to the best of my knowledge, there was no change in the environment over the period of interest that would threaten identification. The exclusion restriction would also be violated if consumers compared local networks and exerted pressure on their local governments to increase network size. However, this is unlikely for counties that are distant to each other.

costs are captured by region fixed effects.

The full set of demand instruments is

$$\mathbf{Z}_{jmt} = \left\{ \mathbf{z}^{\mathbf{p}}_{jmt}, \mathbf{z}^{\mathbf{s}}_{jmt}, \mathbf{z}^{\mathbf{N}}_{jmt}, (\mathbf{z}^{\mathbf{p}}_{jmt} \otimes \mathbf{D}_{mt}), (\mathbf{z}^{\mathbf{N}}_{jmt} \otimes \mathbf{D}_{mt}) \right\}, \tag{38}$$

which includes interactions with county-level demographics in $\mathbf{D}_{mt}$.

Estimation is done using the Nested Fixed Point algorithm described in Berry et al. (1995). I perform the market share inversion to recover $\xi(\beta, \sigma)$, then minimize the following objective function,

$$(\beta^*, \sigma^*) = \underset{\beta,\sigma}{\operatorname{argmin}}\ \xi'(\beta, \sigma)\mathbf{Z}\mathbf{W}\mathbf{Z}'\xi(\beta, \sigma), \tag{39}$$

where $\mathbf{W}$ is some weighting matrix. As usual, the linear parameters $\beta = \{\beta^{\mathbf{p}}, \beta^{\mathbf{x}}, \theta, \Gamma^{\mathbf{p}}, \Gamma^{\mathbf{x}}, \Gamma^{\mathbf{N}}\}$ can be concentrated out, and the optimization is done over $\sigma = \{\sigma^{\mathbf{p}}, \sigma^{\mathbf{x}}, \sigma^{\mathbf{N}}, \zeta\}$. Additional details about the estimation routine can be found in Appendix E.1.

Station supply. I estimate the parameters of the station supply equation by maximum likelihood. For $\epsilon^{\mathbf{n}}_{mt}$ distributed as standard normal, the probability of observing a network of size N_{mt} is given by the following expression,

$$\Pr(N = N_{mt} \mid Q^{ev}_{m,t-1}, \mathbf{D}_{mt}) = \Phi\left(\frac{1}{\omega}\,\mathcal{G}(N_{mt})\right) - \Phi\left(\frac{1}{\omega}\,\mathcal{G}(N_{mt}+1)\right), \tag{40}$$

where

$$\begin{aligned}
\frac{1}{\omega}\,\mathcal{G}(N_{mt}) &= \frac{1}{\omega}\big(\ln \Psi_{mt} - \eta_{mt}\big) + \frac{\gamma}{\omega}\ln\big(\Delta v(N_{mt})\big) \\
&\quad + \frac{1}{\omega}\ln\left(Q^{ev}_{mt}(N_{mt}) + \frac{\widetilde{\rho}}{1-\widetilde{\rho}}\,\Delta q^{ev}_{mt}(N_{mt})\right), \\
&= \mathbf{D}_{mt}\lambda^{\mathbf{D}} + \lambda^{\mathbf{N}}\ln\big(\Delta v(N_{mt})\big) \\
&\quad + \lambda^{\mathbf{Q}}\ln\left(Q^{ev}_{mt}(N_{mt}) + \frac{\widetilde{\rho}}{1-\widetilde{\rho}}\,\Delta q^{ev}_{mt}(N_{mt})\right).
\end{aligned} \tag{41}$$

Since Ψ_{mt} and η_{mt} are both unobserved, I replace the function $(1/\omega)\,(\ln \Psi_{mt} - \eta_{mt})$ by a linear combination of county-level demographics, $\mathbf{D}_{mt}\lambda^{\mathbf{D}}$. The conditional log-likelihood is then

$$
\begin{aligned}
\ell(\lambda \mid Q^{ev}_{m,t-1}, \mathbf{D}_{mt}) &= \sum_m \sum_t \ln \Pr(N = N_{mt} \mid Q^{ev}_{m,t-1}, \mathbf{D}_{mt}) \\
&= \sum_m \sum_t \ln \Bigg[\Phi\Bigg(\mathbf{D}_{mt}\lambda^{\mathbf{D}} + \lambda^{\mathbf{N}} \ln\left(\Delta v(N_{mt})\right) \\
&\qquad + \lambda^{\mathbf{Q}} \ln\left(Q^{ev}_{mt}(N_{mt}) + \frac{\widetilde{\rho}}{1-\widetilde{\rho}} \Delta q^{ev}_{mt}(N_{mt}) \right) \Bigg) \\
&\qquad - \Phi\Bigg(\mathbf{D}_{mt}\lambda^{\mathbf{D}} + \lambda^{\mathbf{N}} \ln\left(\Delta v(N_{mt}+1)\right) \\
&\qquad + \lambda^{\mathbf{Q}} \ln\left(Q^{ev}_{mt}(N_{mt}+1) + \frac{\widetilde{\rho}}{1-\widetilde{\rho}} \Delta q^{ev}_{mt}(N_{mt}+1) \right) \Bigg) \Bigg].
\end{aligned} \tag{42}
$$

The discount factor $\widetilde{\rho} = \rho\,(1-d)$ is not identified without additional restrictions on the likelihood. The chosen approach to deal with this identification issue is to calibrate ρ and d to some known values, and perform the estimation on the remaining parameters $\lambda = \{\lambda^{\mathbf{N}}, \lambda^{\mathbf{Q}}, \lambda^{\mathbf{D}}\}$. The chosen values are $\rho = 0.95$ for the discount factor and $d = 0.0832$ for the depreciation rate (the expected lifetime of vehicles in the data is 12.02 years), so $\widetilde{\rho} = 0.871$. Estimation is done by maximizing the conditional log-likelihood function,

$$
\lambda^* = \underset{\lambda}{\operatorname{argmax}}\ \ell(\lambda \mid Q^{ev}_{m,t-1}, \mathbf{D}_{mt}). \tag{43}
$$

Computational details are relayed to Appendix E.2.

As discussed in Section 4.3, embedding the full demand system in the supply equation through $q^{ev}_{mt}(n)$ resolves the simultaneity between electric vehicle sales and station deployment. Identification of the supply-side parameters nonetheless relies on the additional assumption that prices and the unobserved product quality, ξ_{jmt}, are each uncorrelated with the network supply shocks $\epsilon^{\mathbf{n}}_{mt}$. I discuss each in turn, and assume throughout, as is standard in the car market literature, that the car characteristics in $\mathbf{x}_{jt}$ and the demographics in $\mathbf{D}_{mt}$ are predetermined and hence exogenous.

I first address the potential endogeneity of prices. The concern is that car manufacturers' pricing decisions could correlate with local shocks to network provision. This could be an issue if firms set prices at the local level or if the analysis relied on transaction prices (also set at the local level by dealers). In practice, I am relying on list prices (common to all counties) which are unlikely to correlate strongly with county-specific shocks, as the overwhelming majority of counties are atomistic in size compared to Quebec's population. More broadly, if prices did correlate with local network supply shocks through the equilibrium relationship, then re-optimizing prices inside the calculation of $q^{ev}_{mt}(n, \mathbf{p}_t(n))$ would internalize the change

in prices (alongside the demand-side response) and resolve the simultaneity issue between car manufacturer's pricing decisions and network provision.

I now turn to the potential endogeneity of ξ_{jmt}. Endogeneity can arise if local planners are more likely to install chargers where the population they represent is predisposed to purchase electric vehicles. In this case, the positive correlation between ξ_{jmt} and $\epsilon^{\mathbf{n}}_{mt}$ implies that I would overestimate $\lambda^{\mathbf{Q}}$. In practice, $q^{ev}_{mt}(n)$ is a complicated function of all the product-level ξ_{jmt} (including non-electric vehicles), so the impact of individual realizations of ξ_{jmt} is likely to be small since $\epsilon^{\mathbf{n}}_{mt}$ is defined at the county level. The network supply model also includes a rich set of demographics and controls that account for shifts in consumers' perception of green technology, making it even less likely that ξ_{jmt} and $\epsilon^{\mathbf{n}}_{mt}$ are correlated as described above.

The reduced-form evidence in Section 3.3 provides additional support for the exogeneity of ξ_{jmt}. I find no short-run response of network deployment to changes in electric vehicle subsidies, which suggests that local governments do not adjust infrastructure spending in response to contemporaneous demand conditions. If this slow response reflects the institutional nature of public infrastructure planning, where budgets are allocated well in advance and are not easily adjusted, then local governments are unlikely to react quickly to any source of demand variation, including fluctuations in ξ_{jmt}.

I gather some empirical evidence to assess whether the endogeneity concern raised in the previous paragraph is consequential in practice. The results are presented in Table A.8. I consider the static version of the model (with $\rho = 0$), which removes the influence of marginal electric vehicle owners and simplify the exposition. I estimate three specifications. The first is the proposed model, which handles the simultaneity by evaluating $q^{ev}_{mt}(n)$ at each potential network size n. The second treats the simultaneity as a standard endogeneity problem using a control function approach. This is done by replacing $Q^{ev}_{mt}(N_{mt})$ and $Q^{ev}_{mt}(N_{mt}+1)$ by their data counterpart Q^{ev}_{mt} in the likelihood equation (42) (e.g., using aggregated observed sales), and adding the residuals from a linear projection of Q^{ev}_{mt} on instruments and demographics as an additional control. The third specification ignores both issues. The first two specifications yield very similar estimates of $\lambda^{\mathbf{Q}}$, which confirms that the model correctly accounts for the equilibrium relationship and that no residual endogeneity remains. Both are substantially lower than the estimate from the third specification, which illustrates that ignoring the equilibrium relationship leads to upward bias in $\lambda^{\mathbf{Q}}$.

5 Estimation results and counterfactuals

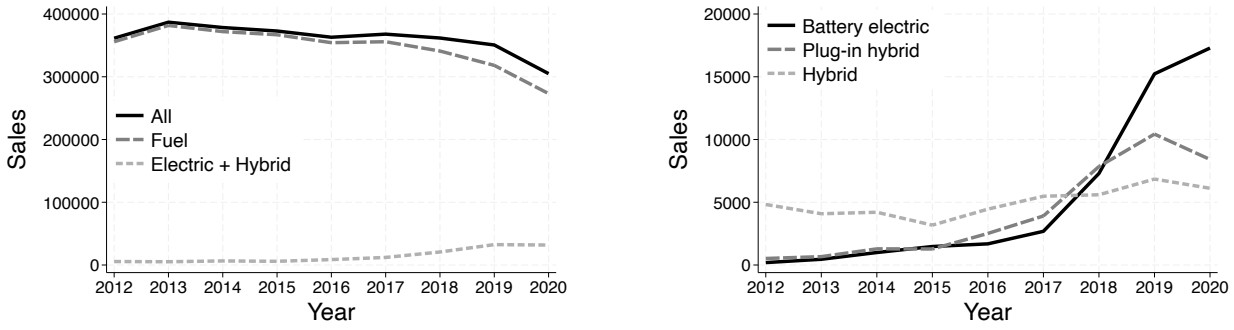
5.1 Demand

Summary statistics. I estimate both the demand for cars and supply of stations at the county-year level. I focus on the province of Quebec, where stations are provided publicly (charging stations are provided by private firms in Ontario). I define a product as a make-model-engine type combination and I set the market size to the number of households in each market. Table A.9 presents a summary of the characteristics of the available products. Battery electric vehicles are on average $18,000 more expensive than traditional combustion engines while plug-in hybrids are on average $6,000 more expensive. The combined rebates seem to cover fully the price difference for plug-in hybrids, but not for battery electric vehicles.

The evolution of vehicle sales in Quebec is depicted in Figure 2. Panel (a) shows that total sales are roughly constant until 2019, but then decrease due to supply chain disruptions and economic uncertainty caused by COVID-19. Panel (b) offers a breakdown by engine type for electric and hybrid vehicles. Sales of battery electrics and plug-in hybrids are rising steadily, with a sharp increase toward the end of the sample period. Sales of non-rechargeable hybrids are rising but slightly.

Several factors unrelated to the policy under study help explain the sustained growth in electric vehicle sales. One of them is the increasing electric vehicle offering, summarized in Table 4. The number of battery electric and plug-in hybrid alternatives is rising steadily from

Figure 2: Evolution of sales



(a) All vehicles

(b) Electric and hybrid vehicles

Notes: This figure presents the evolution of aggregate sales over time, by engine type. Panel (a) groups all electric and hybrid vehicles as these sales are marginal compared to fuel vehicles. Panel (b) offers a breakdown between battery electric, plug-in hybrid, and non-rechargeable hybrid vehicles.

Table 4: Evolution of choice set and charging infrastructure

Year	Number of products			Number of stations	Share of counties with			
	Fuel	Electric	Hybrid		0 station	1-10 stations	11-25 stations	25+ stations
2012	165	5	9	100	0.69	0.25	0.03	0.02
2013	176	8	9	192	0.48	0.46	0.04	0.02
2014	187	11	9	339	0.29	0.62	0.04	0.05
2015	188	11	10	623	0.13	0.72	0.09	0.05
2016	186	13	10	914	0.03	0.69	0.22	0.05
2017	184	23	13	1,266	0.02	0.61	0.31	0.06
2018	184	26	13	1,616	0	0.57	0.30	0.14
2019	177	29	13	2,371	0	0.39	0.39	0.22
2020	173	31	13	2,811	0	0.24	0.44	0.32

5 products available in 2012 to 31 in 2020. Meanwhile, the offering of internal combustion engines seems to decline slightly in 2019 and 2020, when sales of electric vehicles are highest.

The increasing availability of charging stations could also explain part of the growth in electric vehicle sales. The right-hand side of Table 4 shows the evolution of the charging station infrastructure over time. The number of stations available goes from 100 stations in 2012 to more than 2800 in 2020. Local networks are also densifying over time. This is especially important in predominantly rural counties which have low population density. A large share of counties initially had no charging station network. In this case, electric vehicle owners are constrained to charging at home which acts as a deterrent to the purchase of a battery electric vehicle. By 2020, 76% of counties had more than 10 stations available, 32% had more than 25 stations, and all counties had at least one open charging location.

Estimation. Results from the demand estimation are presented in Table 5. I include the horsepower (in 100 kW), the weight (in 100 kg), the driving cost (in CAD per 100 km),[27] and the engine type as observed characteristics. I also interact these car characteristics with average county-level demographics. The chosen demographics are the average income, the average age, the proportion of female, the population density (number of households per square kilometer), and a time trend.[28]

I include a large number of fixed effects: car makes (34 different makes), market segments (subcompact, compact, midsize, large/luxury, crossover utility, sport utility, and minivan), counties (96 counties), and years (9 years). These fixed effects capture unobservables such as

[27]For fuel and hybrid vehicles, driving cost is computed by multiplying fuel consumed for traveling 100 kilometers with the gas price in that county and year. For battery electric vehicles, driving cost is measured as power required for traveling 100 kilometers, times an average charging cost of 10.9 cents per kWh. For plug-in hybrid, I compute a weighted average of both measures based on the share of the total driving range that is achievable driving only on electric.

[28]All demographics are demeaned (including the time trend) such that they do not affect the coefficients on the observed characteristics they are interacted with.

Table 5: Demand estimation

		Demographic interactions						
	Estimate	Income	Age	Gender	Population density	Trend	ζ	σ
Price	-0.801	0.004						-0.154
	(0.035)	(0.003)						(0.021)
Network	0.214	0.068	-0.064				0.212	
	(0.069)	(0.034)	(0.022)				(0.124)	
Power	0.940		0.043	0.033				
	(0.021)		(0.017)	(0.004)				
Weight	0.203					0.093		
	(0.034)					(0.004)		
Driving cost	-0.037							
	(0.004)							
Battery electric	-2.114	-0.281		0.193	-0.617			
	(0.105)	(0.083)		(0.028)	(0.115)			
Plug-in hybrid	-2.055	-0.415		0.174	-0.662			
	(0.098)	(0.086)		(0.029)	(0.127)			
Hybrid	-1.720		0.329	0.152				
	(0.021)		(0.040)	(0.015)				
Constant								5.115
								(2.099)
Observations	126,397							
Nb. of markets	864							
Obj. Function	3,425.4							

Notes: Includes brand, market segment, county, and year fixed effects. Robust standard errors in parenthesis.

brand perception, or local unobserved consumer characteristics. Finally, I allow for heterogeneous preferences by including a random coefficient on the net price and on the constant.[29]

In practice, including a random coefficient on the price (or on one of the continuous characteristics) helps producing more diverse substitution patterns between products. In this case, it also allows for heterogeneous responses to the financial incentives. The random coefficient on the constant is useful to break the independence of irrelevant alternative between the inside and the outside good. Since this study aims at measuring the emission reduction potential of electric vehicle subsidies, it is crucial that I measure the substitution to the outside option accurately, or I am at risk of underevaluating the emission reduction potential of subsidies.

[29]I also estimate a specification with a random coefficient on θ, the network preference parameter. However, the random coefficient was estimated to be small with a large variance. Additionally, including this random coefficient produced insignificant estimates for the curvature parameter in the indirect utility of charging function. To avoid producing misleading results in counterfactual simulations, the final specification does not include a random coefficient on θ. For completeness, these results are available in Table A.10.

I estimate the price coefficient and its standard deviation to be -0.801 and 0.154 respectively. Both are highly significant. The interaction of the price coefficient with income is slightly positive but insignificant. Heterogeneity in price sensitivity is therefore primarily captured by the random coefficient rather than differences in average income.

The main coefficient on network size is 0.214 and significant, which means that consumers care about the availability of charging stations when considering the purchase of an electric vehicle. The curvature parameter on the network utility function is estimated to be 0.212. To fix ideas, this implies a curvature between the linear and logarithmic functions (close to the logarithmic function, but statistically significantly different).

I observe a positive interaction between the network size and the average income and a negative interaction with the average age, both significant. The average income is higher in dense urban areas where the potential for home-charging is lower. Consumers in high-income counties therefore rely more heavily on public charging, which raises their valuation of networks compared to other counties (and decrease their valuation of electric vehicles, for a given network size).

The remaining demographic interactions yield standard patterns. For example, the model suggests that the preference for powerful vehicle increases with age, and that women typically prefer more powerful vehicles compared to men. The consumer-level data also suggests that women (on average) purchase larger, hence more powerful cars than men. Men, on the other hand, tend to purchase cars with better acceleration (power-to-weight ratio). The model also predicts that women prefer electric and hybrid vehicles more than men. The estimates suggest that consumers in large cities dislike electric vehicles. One explanation is that the interactions between population density and the electric vehicle dummies capture partially the potential for home-charging which is lower in urban areas compared to suburban and rural areas. Finally, my estimates suggest that consumers prefer heavier vehicles (a proxy for security) and that this preference increases over time.

The interactions with average demographics are included to capture the heterogeneity in preferences for the observed car characteristics, across counties. The model suggest that the preference for powerful vehicle increases with age, and that women typically prefer more powerful vehicles compared to men. The consumer-level data also suggests that women (on average) purchase larger, hence more powerful cars than men. Men, on the other hand, tend to purchase cars with better acceleration (power-to-weight ratio). The model also predicts that women prefer electric and hybrid vehicles more than men. The estimates suggest that consumers in large cities dislike electric vehicles. The interactions between the electric vehicle dummies and the population density capture partially the potential for home-charging which is lower in urban areas compared to suburban and rural areas. Finally, my estimates suggest

Table 6: Market-level elasticities

	Fuel	Battery electric	Plug-in hybrid	Hybrid	Network
Fuel	-0.972	0.030	0.020	0.026	-0.010
Battery electric	1.781	-3.149	0.046	0.036	0.618
Plug-in hybrid	1.825	0.073	-2.684	0.034	0.562
Hybrid	1.842	0.042	0.026	-2.748	-0.015
Outside option	6.16e-04	8.80e-06	6.85e-06	7.72e-06	-2.38e-06

Notes: The market-level elasticities are computed by increasing the prices of all products by 1%, by engine type, and computing the resulting change on sales (including the outside option) keeping everything else constant. The market-level elasticity to network size is computed similarly by increasing all local network sizes by 1%.

that consumers prefer heavier vehicles (a proxy for security) and that this preference increases over time.

The market-level elasticities implied by these estimates are presented in Table 6. The market-level own-price elasticities for battery electric and plug-in hybrid vehicles are -3.15 and -2.68, respectively, which leads to a combined elasticity of -2.78. This is close to what I estimate in the reduced form analysis (-3.00). I also compute the market-level passthrough of rebates to prices to be 1.03 (reduced form estimate is 1.03), meaning that the subsidies did not meaningfully change firms' pricing decisions, and consumers benefited from the full subsidy amount.

Table 6 reveals other interesting patterns. Increasing the price of all fuel vehicles (together) induces substitution toward electric and hybrid engines. The cross-price elasticities range between 1.78 and 1.84, suggesting that taxing internal combustion engines may be an effective way to promote electric vehicle adoption. The large magnitude of these cross-price elasticities is due to the fact that fuel vehicles represent more than 95% of all sales and small transfers from fuel to electric generate large increases in the electric vehicle base. Finally, I report the market-level elasticity to network provision. I find large elasticities, in the range of 0.56 to 0.62, suggesting that network effects are strong on the consumer side. This is in line with previous findings by Springel (2021) and Remmy (2025).

5.2 Network supply

Results from the network supply estimation are presented in Table 7. The results from the alternative local planner objectives are relayed to Appendix F. I include several demographics that capture regional differences in consumer characteristics which may induce local planners to install chargers. I use the share of residents that have an undergraduate degree as a proxy for consumers' environmental awareness and their aggregate taste for green

Table 7: Station supply estimation

	Forward-looking		Static	
	Estimate	Standard error	Estimate	Standard error
$\lambda^{\mathbf{N}}$ $(=\gamma/\omega)$	2.383	(0.132)	2.445	(0.139)
$\lambda^{\mathbf{Q}}$ $(=1/\omega)$	0.375	(0.110)	0.417	(0.111)
Avg. income	-0.580	(0.240)	-0.591	(0.243)
Avg. age	1.824	(0.556)	1.856	(0.557)
Avg. household size	-0.092	(1.497)	-0.103	(1.517)
Share graduates	8.405	(2.461)	8.323	(2.489)
Share homeowners	-6.024	(1.989)	-5.911	(1.990)
Urban	0.435	(0.328)	0.390	(0.327)
Elasticity to EV	0.1084		0.1444	
EV for one additional station	75.3		57.0	
Observations	830		830	
Log-likelihood	-2,123.0		-2,116.9	

Notes: Includes year fixed effects. Markets where electric vehicles are not in the choice set and which have no electric vehicle in circulation are excluded. The forward-looking model is estimated using a discount factor of $\widetilde{\rho} = 0.871$. Standard errors in parenthesis are clustered at the county level and are computed using 500 bootstrap replications.

technologies. Additionally, I measure the potential for home-charging by the share of homeowners and an indicator for urban counties. Demand for charging services should be higher if some electric vehicle owners cannot install and use a home-charger. This in turn should lead to more station installations.

Because of the highly nonlinear nature of the model, I cannot include county fixed effects, as these would not be identified with only nine years of data. Instead, I include the average income, the average age, and the average household size to account for any remaining regional differences.

I report the coefficients of both the model with forward-looking planners and the static model. The coefficients on the share of graduates, the share of homeowners, and the urban indicator are significant and have the correct signs. This reinforces the idea that environmental awareness and the potential for home-charging are two important drivers of charging station supply.

The coefficients $\lambda^{\mathbf{N}}$ and $\lambda^{\mathbf{Q}}$ are difficult to interpret directly. I use them to recover the underlying local-planner preference parameter, γ, estimated to be 6.35 and highly significant. This means that local planners value charging less than the underlying electric vehicle owners (since the terms $\Delta v(n) < 1$ everywhere). I also recover the elasticity of network supply to assess the magnitude of the network effects on the station supply side. The elasticity implied

by the forward-looking model is 0.108, meaning that a 10% increase in the stock of electric vehicles is associated with a 1.08% increase in network size. This is much smaller than the demand elasticity to network size (0.562–0.618), meaning that network effects are much stronger on the consumer side than on the network supply side. To put things in perspective, the estimates from the forward-looking model suggest that one additional station is installed for every 75 electric vehicles sold.

Alternatively, the estimates from the static model imply an elasticity of supply of 0.146 and that one additional station is installed for every 57 electric vehicle sold. Ignoring the forward-looking behavior of the social planner therefore leads to overestimating the magnitude of the elasticity of supply and the marginal effect by about 33% and 32% respectively. This happens because ignoring the future gains from marginal consumers at the estimation stage means that local planners' decisions are explained entirely by users current valuation of networks, which leads to a larger estimated coefficient. At the same time, the term $\widetilde{\rho}/(1-\widetilde{\rho})\,\Delta q^{ev}_{mt}(n)$ representing the marginal users decreases mechanically the marginal effect, see Appendix E.3. Both factors contribute to overestimating the elasticity of network supply and the marginal effect when ignoring the forward-looking behavior of network suppliers.

The magnitude of network effects on both sides of the market is crucial for policy design, as they can create a feedback loop that can enhance the effect of subsidization. For example, with strong network effects on both sides of the market, subsidizing electric vehicles increases adoption directly through prices, and the larger pool of electric vehicle drivers would induce local planners to increase the roll-out of chargers through network effects. These additional chargers would then generate additional sales since consumers care about the opportunity for charging, amplifying the impact of the policy.

When network effects on one side of the market are weak, as is the case in the Canadian context, this limits the gains that accrue from the feedback loop described above, and most of the policy impact from subsidization comes from the direct effect of subsidies on adoption.

5.3 Robustness to spillovers

The demand and supply estimates account only for within county effects. For example, consumers care about the availability of charging in their county of residence, and local planners care about the local installed base of electric vehicles. In practice, consumers travel between neighboring regions for work or other daily activities which could lead to spillover effect. To assess the extent of these effects, I estimate a simplified version of the demand and supply models presented in Section 4, that include these spillover effects (e.g., networks and installed base in neighboring counties).

The results are relayed to Table A.6 and A.7. On the one hand, I find some evidence of spillover effects on the demand side. I estimate a positive and significant coefficient for networks located in counties within 50 km. The coefficient is small, slightly above 10% of the main estimated effect (0.027 versus 0.243). There is no effect beyond 50 km. On the other hand, I cannot find evidence that local planners care about electric vehicle adoption in other regions (the coefficient is small, insignificant, and has the wrong sign).

Since the estimated effects are small and including spillover effects considerably complicates the computation of counterfactual experiments, I ignore their contribution in what follows.

5.4 Counterfactual analysis

I conduct several counterfactual simulations. Computational details are relayed to Appendix E.4. I want first to disentangle the direct effect of subsidies from network effects. To that end, I compare the outcomes of a counterfactual experiment where I remove all subsidies, keeping networks at their original levels, to another experiment where network is updated using the forward-looking supply model. Second, I want to understand the impact of ignoring the forward-looking behavior of local planners in the station supply model. To achieve this, I compare a counterfactual experiment where the local planner is forward-looking to the case where the re-optimization of networks is done using the static model.

The results of the counterfactual experiments are reported in Table 8. I set the baseline to be the observed outcomes from the data (with rebates). I first consider the direct effect of the rebate program, when networks are fixed at their observed values. The rebates contributed to increasing sales of electric vehicles by 39,226 units, representing 46.6% of all registrations. Around two thirds of these additional electric vehicles replaced internal combustion engines. Registrations of non-rechargeable hybrids increase slightly, by 777 units.[30] This led to a reduction in total carbon emissions in the range of 1.014 million metric tons over the lifetime of these vehicles, or 0.7% of the total emissions attributed to the vehicles sold between 2012 and 2020. The reduction is small compared to total fleet emissions, since electric vehicle sales represent a small fraction of total sales.

I next focus on the contribution of network effects when the local planners are forward-looking. I find that allowing for the re-optimization of local networks leads to an additional 2.740 electric vehicle sales and 267 additional charging station installations. These additional sales attributed to the network expansion represent around 7% of the direct effect of subsidies and the additional stations represent around 9.5% of the observed networks. Out of the 2,740

[30]Some non-rechargeable hybrids are eligible to a small subsidy of around $500, resulting in a small market expansion effect.

Table 8: Counterfactual simulation

		Counterfactuals: No Subsidies					
	(1) Baseline	(2) Fixed network		(3) Forward-looking		(4) Static	
Key outcomes							
Δ Total sales	3,248,085	-13,628	(-0.42%)	-14,567	(-0.45%)	-14,724	(-0.45%)
Δ Sales (fuel)	3,119,123	+26,375	(+0.85%)	+28,140	(+0.90%)	+28,438	(+0.91%)
Δ Sales (battery electric)	47,305	-24,230	(-51.22%)	-25,664	(-54.25%)	-25,909	(-54.77%)
Δ Sales (plug-in hybrid)	36,869	-14,996	(-40.67%)	-16,302	(-44.22%)	-16,519	(-44.80%)
Δ Sales (hybrid)	44,788	-777	(-1.73%)	-741	(-1.65%)	-735	(-1.64%)
Δ Charging stations	2,811	0	(0.00%)	-267	(-9.50%)	-307	(-10.92%)
Δ CO_2 emissions	141.46	+1.014	(+0.72%)	+1.080	(+0.76%)	+1.092	(+0.77%)
Δ Consumer surplus	0	-548.7	–	-589.4	–	-596.2	–
Δ Total cost	737.8	0	(-100%)	0	(-100%)	0	(-100%)
Implied abatement costs							
Avg. cost per ton CO_2	–	-727	–	-683	–	-676	–
Avg. cost per electric vehicle	–	-18,809	–	-17,581	–	-17,389	–

Notes: Column (1) reports the baseline values from the data, where rebates are available. Columns (2) to (4) report the change from baseline for various counterfactuals where rebates are removed. Sales of electric vehicles includes both battery electric and plug-in hybrid vehicles. CO_2 emissions are calculated over the lifetime of vehicles, in million tons, based on a 22,053 average mileage per year, an average lifetime of 12.02 years, and a discount factor of 0.95. Consumer surplus in the baseline case is normalized to zero. Consumer surplus and the total cost of the program are in million 2018 CAD. The average cost per ton of CO_2 and the average cost per electric vehicle are in 2018 CAD.

new electric vehicle sales, 1,800 replaced internal combustion engines or non-rechargeable hybrids, leading to further emission reductions. The abated emissions reach 1.081 million tons, or 0.76% of total emissions, and network effects account for roughly 6.2% of these abated emissions.

We can compare these estimates with previous works. For example, Springel (2021) and Remmy (2025) find that electric vehicle adoptions generate additional station installations at a rate of 1:38 and 1:11 respectively.[31] They define charging stations as separate chargers, while I define charging stations as sites that can host more than one charger. In the data, I observe that each charging site holds on average 2.27 chargers. Using back of the envelope calculations, my results imply that new chargers are installed at a rate of 1:70 electric vehicle sold (new charging sites are opened at a rate of 1:159 electric vehicle sold). Using the estimates from the static model, I find that new chargers are installed at a rate of 1:61. In both cases, the results indicate weaker network effects than Springel (2021) or Remmy (2025).

[31] These figures are computed from Table 5 in each paper.

I compare the estimated impact of the subsidy from the structural estimation to the reduced form results of Section 3. The counterfactual simulation in column (3) of Table 8 suggests that electric vehicle sales increased from 42,208 to 84,174 due to subsidies over the full period. This is an increase of around 99%. Back of the envelope calculations suggest that this is equivalent to a 11.5% increase in sales per $1,000 in average subsidy. Meanwhile, the reduced form estimates suggests an increase of 7.9% per $1,000 in subsidies.

The structural model predicts that total network size increases from 2,544 to 2,811. That is, the total number of charging locations increased by 10.5%. In contrast, the reduced form estimates revealed no significant change in network size due to the policy. These findings shed light on the rigidities affecting network supply which ultimately affect electric vehicle adoption in the short-run. For example, the opening of new location requires planning to find appropriate sites, approving budgets, ordering the required material, and allocating human resources for installation. This makes it difficult for network operators to change supply in the short-run following a sudden surge in demand for charging services. However, network supply eventually adapts to the new market condition in the long-run, as evidenced from the counterfactual simulations.

I next quantify the effect of ignoring the forward-looking behavior of local planners in the station supply model. I find that doing so leads to overestimating the impact of network effects. For example, the simulation predicts that network size increases by 307 instead of 267 and the network effect contribution to sales is 3,201 instead of 2,740. That is, the counterfactual simulation which rely on the static supply model overestimates the contribution of network effects to key outcomes by 16.8% and 15% respectively. These differences are moderately small, and given that the contribution of the network effects is small compared to the direct effect of the policy, they do not translate into meaningful differences in terms of electric vehicle adoption, emission abatement, or abatement costs. This could, however, be problematic in other contexts where network effects are more important.[32]

The total spending on subsidies by both levels of government reached $721.5 million. I

[32]The magnitude of the bias to aggregate outcomes introduced by ignoring dynamics depends crucially on two factors: the elasticity of demand to network size (how react when you add charging capacity) and the elasticity of supply to the installed base of electric vehicles (how many chargers are installed when more electric vehicle adoption occur). The elasticity of demand depends on the preference parameters in the utility of charging function, θ_i, and the curvature parameter ζ. The elasticity of supply depends on the curvature parameter γ, which is inversely related to the elasticity of supply. The demand side parameters also matter for the elasticity of supply. Our estimates imply a moderately high elasticity of demand, and a low elasticity of supply. In this context, local planners do not react much to electric vehicle adoption, and the impact of subsidies attributed to network effects is an order of magnitude lower than their direct effect. Ignoring the dynamics therefore does not introduce to a large bias. In other context, the bias could be significant. In particular, with a much lower γ, the elasticity of supply would be higher, resulting in a much stronger feedback loop, and the bias introduced by ignoring dynamics would have a larger impact on the aggregate outcomes.

use this figure to compute some preliminary cost measures, which are useful to compare my results to previous literature that use simple measures of abatement cost. I estimate the average cost of reducing emissions to be $683 per ton of CO_2. This is close to the results from Xing et al. (2021), who estimate an average abatement cost between $581 and $662 (484–552 USD) per ton for a similar rebate program in the United States. Other studies of similar financial incentives typically find lower estimated costs. Examples include Huse and Lucinda (2014) on the Swedish green car rebate ($131–158), Beresteanu and Li (2011) on tax incentives on hybrids in the United States ($212), and Azarafshar and Vermeulen (2020) on the Canadian electric vehicle market ($480).

These cost measures, while informative, are inadequate to study the cost-effectiveness of environmental policies. The reason is that the policies under study are non-marginal, so the average abatement cost measured in these works is potentially far from the marginal abatement cost. They also ignore the fiscal externalities to consumers, firms, and the policymaker, and the marginal cost of providing public funds. These factors could drastically change how we evaluate different programs. I propose a unifying framework for studying non-marginal environmental policies in what follows.

6 Cost-benefit analysis

6.1 Setup

I propose a unified framework to study the cost-effectiveness of the Canadian electric vehicle rebate programs. My framework is sufficiently general to be used to study other types of environmental regulations, for example, gas taxes or emission standards. In what follows, I consider a social planner, distinct from the local planners, who determine the optimal level of electric vehicle subsidies to maximize social welfare given the environmental externality from car emissions and the total cost of the policy. In choosing an optimal level of subsidization, the social planner takes as given the decentralized equilibrium arising from consumer preferences, car manufacturers' pricing decisions, and network provision by local planners.[33]

[33]Local planners maximize the utility of electric vehicle owners from charging networks, given the cost of infrastructure, while the social planner maximizes social welfare, accounting for the environmental externality from car emissions. There is a friction between the two objectives. As a result, decisions about network size may not maximize social welfare. If both objectives aligned, or equivalently, if the social planner could control both the level of subsidies and the provision of charging networks, then the network supply model would be redundant and network size would become an additional policy variable. The social planner would then solve for an optimal combination of subsidies and network investments to maximize social welfare, subject to budget constraints. In this setting, the optimal policy would involve lower subsidies on electric vehicles and a greater direct investment in charging infrastructures.

Consider the following social planner objective, where τ is the targeted policy variable,

$$\tau^* = \underset{\tau \geq 0}{\text{argmax}} \quad \underbrace{\mathcal{W}(\tau) - \text{Cost}(\tau) + \text{Tax}(\tau)}_{\text{Value to society}} - \underbrace{\text{E}(\tau) \cdot P^E}_{\substack{\text{Value of} \\ \text{environmental} \\ \text{damage}}}. \tag{44}$$

The social planner is looking to pick the policy τ^* that maximizes the value to society and minimizes the value of the environmental damage that arises from the policy. For now, I take the carbon price P^E as given.

The government objective function has four key inputs: a social welfare function $\mathcal{W}(\tau)$ which accounts for the fiscal externalities of the policy on consumers and firms, a cost function $\text{Cost}(\tau)$ which summarizes government spending on the policy, a tax function $\text{Tax}(\tau)$ which accounts for revenues from the provincial gas tax (or other taxes associated with the purchase and driving of vehicles), and an emission function $\text{E}(\tau)$ which accounts for the lifetime environmental impact of vehicles.

Provided all functions are continuously differentiable, the cost-effective policy rule $\tau^*(P^E)$ can be obtained by inverting the following first-order condition,[34]

$$\underbrace{\frac{\partial \mathcal{W}(\tau^*)}{\partial \text{E}(\tau^*)} - \frac{\partial \text{Cost}(\tau^*)}{\partial \text{E}(\tau^*)} + \frac{\partial \text{Tax}(\tau^*)}{\partial \text{E}(\tau^*)}}_{\substack{\text{Marginal} \\ \text{Abatement} \\ \text{Cost}}} = \underbrace{P^E}_{\substack{\text{Cost of} \\ \text{Carbon}}}. \tag{45}$$

I now provide more details on each component of the objective function.

Social welfare function. Let Θ be the fundamental parameters governing consumers' preferences and network provision. I define the social welfare function as a weighted sum of the firms' variable profits and consumers' welfare (defined by consumer surplus). Let ψ_1 and ψ_2 be welfare weights and $q_{jmt}(\tau)$ be the quantity sold of model j in county m and period t, given policy τ. Furthermore, let c_{jt} be the marginal cost of product j in period t.[35] The

[34] The social planner problem could in principle be more complex. For example, the policymaker could take into account congestion issues, mileage decisions of users (i.e. through modal choices), fairness concerns, political constraints, or other types of externalities. Therefore the solution to the problem is the "cost-effective policy" (for lack of a better terminology) only in terms of equating the marginal abatement cost to the social cost of carbon.

[35] Marginal costs can be estimated from demand side parameters and car manufacturers' first-order conditions, see Berry et al. (1995).

social welfare function is

$$\mathcal{W}(\tau) = \mathcal{W}(\tau, \Theta) = \psi_1 \, \pi(\tau, \Theta) + \psi_2 \, \mathcal{CS}(\tau, \Theta), \tag{46}$$

where

$$\pi(\tau, \Theta) = \sum_t \sum_m \sum_j q_{jmt}(\tau, \Theta) \cdot (p_{jt} - c_{jt}) \tag{47}$$

are the firms' aggregate variable profits, and

$$\mathcal{CS}(\tau, \Theta) = \sum_t \sum_m L_{mt} \int \frac{1}{-\beta_i^{\mathbf{p}}} \ln \left(1 + \sum_{j=1}^{J} \exp \Big(\delta_{jmt}(\tau, \Theta) + \mu_{ijmt}(\tau, \Theta, \nu_i) \Big) \right) dF(\nu_i) + C \tag{48}$$

is the aggregate expected consumers' surplus (identified up to a constant C).

Cost function. The cost function accounts for all government expenditures on the policy. Recall that $\tau > 0$ represents a subsidy in the utilty specification and $\tau < 0$ a tax. The cost function can be computed as the sum of all government subsidies,

$$\text{Cost}(\tau) = \text{Cost}(\tau, \Theta) = \phi \cdot \sum_t \sum_m \sum_j q_{jmt}(\tau, \Theta) \cdot \tau_{jt}, \tag{49}$$

weighted by the marginal cost of providing public funds ϕ.

Tax function. The tax function accounts for the revenues from the 19.2 cents per liter provincial gas tax, computed over the lifetime consumption of vehicle sold under subsidy τ. It depends on each car's fuel consumption f_{jt}, its expected lifetime T_j, and the average mileage by year traveled by a typical owner m_{js}. I assume that the policymaker discounts future revenues at some discount factor ρ. Under these assumptions, the present-value of the gas tax revenues is

$$\text{Tax}(\tau) = \text{Tax}(\tau, \Theta) = 0.192 \cdot \sum_t \sum_m \sum_j \left(q_{jmt}(\tau, \Theta) \cdot \sum_{s=t}^{t+T_j} \rho^{t-s} m_{js} f_{jt} \right). \tag{50}$$

Emission function. Lifetime emissions depend on several parameters, including the car's level of emission per kilometer e_{jt}, its expected lifetime T_j, and the average mileage by year

that a typical owner travels m_{js}.[36] I assume that the policymaker discounts future emissions according to the same discount factor ρ. The present-value of the aggregated emissions can be computed as

$$\mathrm{E}(\tau) = \mathrm{E}(\tau, \Theta) = \sum_t \sum_m \sum_j \left(q_{jmt}(\tau, \Theta) \cdot \sum_{s=t}^{t+T_j} \rho^{t-s} m_{js} e_{jt} \right). \tag{51}$$

6.2 Calibration

I calibrate the various parameters of the social planner's objective function. The parameters are the welfare weights (ψ_1, ψ_2), the marginal cost of public funds (ϕ), the discount factor (ρ), the average mileage per year (m_{js}), and the expected lifetime of vehicles (T_j).

I assume that the social planner cares about consumer surplus and the taxable part of firms' profits, to account for the fiscal externality of firms on the social planner's finances. The corporate tax rate in Quebec is around 27% for large firms, so I set the welfare weight on profits to 0.27 and the welfare weight on consumer surplus to one. The discount factor is set to 0.95 as previously. I use data on fuel spending from the Canadian Survey of Household Spending and local fuel costs to compute the average mileage of a representative Quebec household in 2017. Unfortunately, the data doesn't distinguish between households that own one versus two cars, so I assume that all mileage is done on one vehicle. The average mileage is set to 22,053 kilometers for all j and s. I compute the expected lifetime of vehicles using the micro-level car registration data. I have access to the full fleet of vehicles in ten successive years which I use to track vehicles of various ages and estimate their expected lifetime. The expected lifetime of a new vehicle estimated to be 12.02 years.[37] Finally, I assume that the government can provide subsidies without friction at no additional cost, hence I set the marginal cost of public funds to one. I consider alternative calibrations in which the government incurs an administrative fee for providing subsidies in Section 6.5.

I restrict the policy space to rebate programs that are proportional to the currently implemented scheme. To fix ideas, let τ_0 be the currently available rebate program. The set of policies that are available to the policymaker satisfies

$$\tau = \kappa \cdot \tau_0, \quad \kappa \in \mathbb{R}^+, \quad \tau_0 \in \mathbb{R}^J. \tag{52}$$

[36] Since Quebec's grid is near emission-free, I assume that battery electric vehicle do not generate emissions.

[37] Combining the expected lifetime with the average mileage by year implies that cars have an expected total mileage of around 265,000 kilometers.

With this restriction, the marginal abatement cost can be computed as

$$\text{MAC}(\kappa) = \frac{\partial \mathcal{W}(\kappa)}{\partial \text{E}(\kappa)} - \frac{\partial \text{Cost}(\kappa)}{\partial \text{E}(\kappa)} + \frac{\partial \text{Tax}(\kappa)}{\partial \text{E}(\kappa)}. \tag{53}$$

Restricting the policy space serves two purposes. First, it reduces the computational burden associated with evaluating all possible policies. With J different electric vehicle models available, solving for the cost-effective rebate program entails solving a problem in $\mathbb{R}^J$ which is impractical or infeasible. More importantly, there are strong incentives for policymakers to subsidize all models equally to avoid picking winners and losers among firms.

6.3 Cost-effective policy

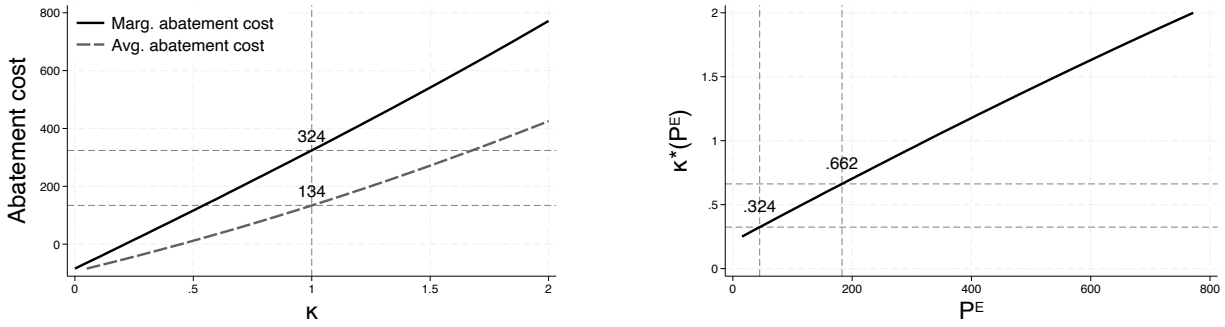
I study the cost-effectiveness of rebates by considering the social planner's optimality condition. I compute counterfactuals on a grid $\{\kappa_1, ..., \kappa_{\mathcal{N}}\}$ using the forward-looking supply model, then I estimate the marginal abatement cost as

$$\begin{aligned}\text{MAC}(\kappa_n) = &\frac{\mathcal{W}(\kappa_{n+1}) - \mathcal{W}(\kappa_n)}{\text{E}(\kappa_{n+1}) - \text{E}(\kappa_n)} - \frac{\text{Cost}(\kappa_{n+1}) - \text{Cost}(\kappa_n)}{\text{E}(\kappa_{n+1}) - \text{E}(\kappa_n)} \\ &+ \frac{\text{Tax}(\kappa_{n+1}) - \text{Tax}(\kappa_n)}{\text{E}(\kappa_{n+1}) - \text{E}(\kappa_n)}.\end{aligned} \tag{54}$$

I collect the results to construct the marginal abatement cost curve as a function of κ. There are two interpretations to the social planner's first-order condition. On the one hand, I can assume that it holds at the current rebates. In this case, equation (45) provides an estimate for the cost of carbon, $P^E = \text{MAC}(\kappa = 1)$. On the other hand, I can calibrate the cost of carbon to known estimates and recover the cost-effective policy $\kappa^*(P^E)$. In what follows, I focus on the second interpretation.

Figure 3, panel (a), depicts the marginal abatement cost curve and the cost-effective policy curve for the chosen calibrated parameters. I observe that the marginal abatement cost is strictly increasing in the subsidy, which ensures that a stable solution to the planner's problem exists and is unique, for a given P^E. In practice, I expect rebates to exhibit decreasing returns in term of emission abatement since the number of infra-marginal consumers increases with the rebate but emissions abated per new owner does not. I evaluate the marginal abatement cost at current rebates to be \$324 per ton of carbon emissions. This is larger than current measures of the social cost of carbon.

Figure 3: Abatement cost and cost-effective policy curves



(a) Abatement cost

(b) Cost-effective policy

Notes: This figure presents the average and marginal abatement costs as a function of the government's policy in panel (a) and the optimal government policy as a function of the price of carbon in panel (b). Both graphs represent the same relationship between the policy and the marginal abatement cost (which is equal to the price of carbon at the optimum).

Figure 3, panel (a), also reports the corresponding average abatement costs.[38] A key observation is that the average abatement cost sits below the marginal abatement cost over the full policy support. This has important implications for policy design: determining the cost-effective policy based on the average abatement cost systematically leads to an overinvestment from the social planner.

I invert the marginal abatement cost curve to recover the cost-effective policy curve. I evaluate the cost-effective policy for two separate estimates of the social cost of carbon. The chosen values are \$45 and \$183, which correspond to the average social cost of carbon and the 95th percentile of the distribution in 2018, according to the Government of Canada. For the lowest estimate, the cost-effective policy corresponds to 32.4% of the current rebate programs. For the highest estimate, the cost-effective policy corresponds instead to 66.2% of current rebates. In both cases, my analysis suggests that policymakers are overinvesting in rebates.

6.4 Accounting for future environmental gains

One reason why governments may be ready to overinvest in subsidies is that quickly increasing the stock of electric vehicles and charging stations may lead to future environmental benefits through network effects, once subsidies are phased out. In that spirit, I consider a

[38] I construct the average abatement curve as

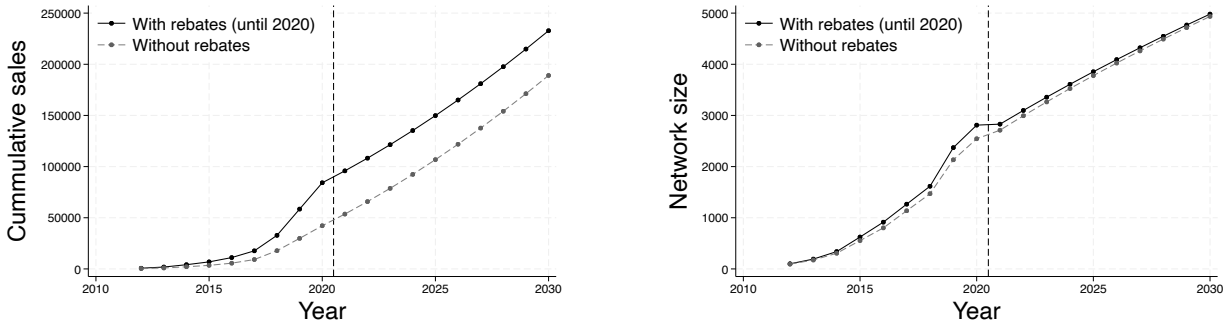
$$AAC(\kappa_n) = \frac{\mathcal{W}(\kappa_n) - \mathcal{W}(0)}{\mathrm{E}(\kappa_n) - \mathrm{E}(0)} - \frac{\mathrm{Cost}(\kappa_n) - \mathrm{Cost}(0)}{\mathrm{E}(\kappa_n) - \mathrm{E}(0)} + \frac{\mathrm{Tax}(\kappa_n) - \mathrm{Tax}(0)}{\mathrm{E}(\kappa_n) - \mathrm{E}(0)}.$$

counterfactual experiment where the policymaker subsidizes electric vehicles from 2012 to 2020, phases out rebates in 2021, and I measure the environmental benefits until 2030. I compare the results to another counterfactual experiment where electric vehicles are never subsidized between 2012 and 2030. Computational details and limitations are discussed in Appendix G.

Figure 4 depicts the evolution of the stock of electric vehicles and charging stations under both counterfactual experiments described above. The first observation concerns electric vehicle adoption, see panel (a) of Figure 4. Providing subsidies for a period of time leads to a permanently higher stock of electric vehicles, as electric vehicle sales accumulate faster when subsidies are offered. However, this improved growth in electric vehicle sales does not continue once subsidies are phased out and electric vehicle adoption is essentially the same between 2021 and 2030, whether or not subsidies were initially offered. This occurs because I estimate the contribution of network effects to be relatively small and the direct effect of subsidies explains most of the additional electric vehicle sales.

The second observation concerns network provision, see panel (b) of Figure 4. An interesting pattern emerges. I observe that networks grow slightly faster while subsidies are offered. However, once rebates are phased out, network operators forgo installing some stations, as they expect significantly fewer future sales, and network size is almost the same in both counterfactual experiments between 2021 and 2030. This limits the potential future gains from the policy that could occur if network provision was kept at a permanently higher level once rebates are phased out.

Figure 4: Evolution of the stock of electric vehicles and charging stations, from 2012 to 2030

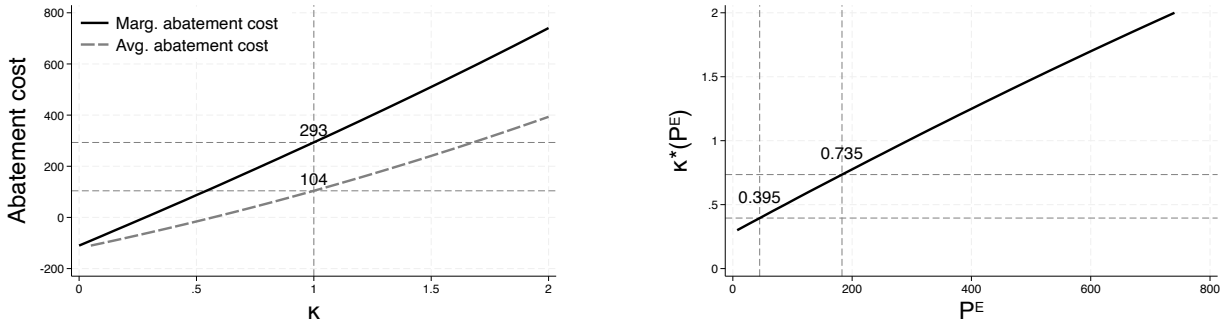


(a) Cumulative EV sales

(b) Network size

Notes: This figure presents the evolution of the stock of electric vehicles and the stock of charging stations, from 2012 to 2030, under two alternative counterfactual experiments. In the first case, subsidies are offered from 2012 to 2020, and I forecast the outcomes from 2021 to 2030 using the 2020 data as reference. In the second counterfactual, subsidies are never available, and I forecast the outcomes from 2021 to 2030 using the 2020 data as reference.

Figure 5: Abatement cost and policy curves, accounting for future environmental gains



(a) Abatement cost

(b) Cost-effective policy

Notes: This figure presents the average and marginal abatement costs as a function of the government's policy in panel (a) and the optimal government policy as a function of the price of carbon in panel (b). The policymaker phases out subsidies in 2021, and sales and network size are forecasted until 2030. Both graphs represent the same relationship between the policy and the marginal abatement cost (which is equal to the price of carbon at the optimum).

I re-evaluate the marginal abatement cost curve and the cost-effective policy curve, accounting for the future environmental benefits of the policy. The results are available in Figure 5. I find that the marginal abatement cost is slightly lower, \$293 instead of \$324 per ton of carbon, and the cost effective policy associated with a \$183 social cost of carbon correspond to 73.5% of the currently implemented rebates instead of 66.2%.

6.5 Alternative parametrizations

To paint the broadest picture possible, I redo the cost-benefit analysis using alternative sets of calibrated parameters. The results are available in Table 9. I start from the extreme case in which the policymaker cares only about government spending and not welfare. This calibration is used widely in the literature related to the car market. In this case, the cost-effective policy is a corner solution: the marginal abatement cost is well above any conventional measures for the social cost of carbon for any level of the subsidy, hence the social planner chooses not to subsidize electric vehicles.

I repeat the exercise for various combinations of the welfare weights and the marginal cost of providing public funds. I consider three cases for the welfare weights. In the first case, the policymaker takes into account consumer surplus but not profits. In the second case, it takes into account consumer surplus and the taxable part of firms' profits (the corporate tax rate is around 27% in Quebec). Finally, I consider the case where the policymaker cares fully about consumer surplus and profits. I interact these welfare weights with two distinct values for the marginal cost of public funds and I focus on the highest value for the social cost of

Table 9: Alternative calibration results

Description	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Parameters							
• Profit weight (ψ_1)	0	0	0	0.27	0.27	1	1
• Consumer surplus weight (ψ_2)	0	1	1	1	1	1	1
• Marginal cost of public funds (ϕ)	1	1.3	1	1.3	1	1.3	1
Cost-benefit analysis, without future env. gains							
• Marginal abatement cost	912	629	374	579	324	444	189
• Average abatement cost	737	387	184	337	134	200	-3
• Cost-effective policy (κ)	0	0.142	0.539	0.243	0.662	0.510	0.986
• Cost-effective provincial rebate (τ^*_{prov})	0	1,136	4,312	1,944	5,296	4,080	7,888
• Cost-effective federal rebate (τ^*_{fed})	0	710	2,695	1,215	3,310	2,550	4,930
Cost-benefit analysis, with future env. gains							
• Marginal abatement cost	885	590	344	540	293	405	158
• Average abatement cost	712	350	155	299	104	162	-34
• Cost-effective policy (κ)	0	0.210	0.612	0.313	0.735	0.584	1.06
• Cost-effective provincial rebate (τ^*_{prov})	0	1,680	4,896	2,504	5,880	4,672	8,480
• Cost-effective federal rebate (τ^*_{fed})	0	1,050	3,060	1,565	3,675	2,920	5,300

Notes: In all parametrization, we have $\rho = 0.95$, $T_j = 12.02$, and $\boldsymbol{m}_{js} = 22,053$. The social cost of carbon (SCC) is set to \$183 per ton of carbon. Column (5) is the main specification. It is reproduced for comparability. The marginal abatement cost and average abatement cost are in CAD per metric ton of carbon. The provincial rebate is computed by multiplying the cost-effective policy κ^* by \$8,000. The federal rebate is computed by multiplying the cost-effective policy κ^* by \$5,000. The federal rebate is available starting in 2019.

carbon, that is, \$183 per metric ton of carbon. Finally, I provide two sets of results, one in which the government ignores the future environmental gains associated with the policy and one where it takes into account these future benefits.

In almost all parameterizations, I find that the social planner offers excessive subsidies. The cost-effective policy varies between 14.2% of the currently implemented rebate schemes for the least favorable specification (excluding cases with a corner solution) and 106% for the most favorable specification. The estimated marginal costs of abatement are between \$158 and \$912.

The only parameterization that approximately rationalizes the observed level of subsidization is column (7) of Table 9. In this case, the policymaker assigns full weight to both consumer and producer surplus, and faces no additional cost for providing public funds. Under a social cost of carbon of \$183, the cost-effective policy reaches 98.6% of the currently implemented rebate and 106% after we account for future environmental benefits. These experiments, however, imply that the social planner values producer surplus beyond the 27%

fiscal share captured through corporate taxation.

The Canadian electric vehicle subsidy programs were designed around sales targets rather than a rigorous cost-benefit assessment. For example, the Quebec program explicitly targeted 100,000 registered electric vehicles by the end of 2020, a goal that was (barely) achieved when corporate fleet purchases are included.

Relying on sales targets does not necessarily align with an optimal policy design, as evidenced here, highlighting the importance of the type of structural work presented in this paper. Understanding the cost-effectiveness of environmental policies is crucial. With limited resources, policymakers need to choose where to allocate public funds to maximize the impact of their interventions on environmental outcomes.

7 Conclusion

The Canadian electric car market presents a unique opportunity to study the cost-effectiveness of subsidizing electric vehicle adoption. Evaluating the environmental performance of such policies is important. With limited financial resources, policymakers should strive to reduce emissions at the lowest cost possible.

My findings suggest that electric vehicle subsidies are an effective way to diffuse the technology and increase adoption. I find no evidence that these additional sales generate additional charging station installations in the short-run. In the long-run, I find that the program led to a small increase in network provision. This study provides a rigorous cost-benefit analysis to evaluate the cost-effectiveness of rebate programs. I find in general that the marginal cost of abatement remains high compared to traditional measures of the social cost of carbon.

The generalizability of these results depends on the institutional context governing network provision. In Quebec, charging infrastructures are primarily deployed by local governments with non-profit motives. Once financial resources are allocated through planning and budgeting processes, network expansion is less sensitive to realized electric vehicle adoption, which explains the low estimated network elasticity. In markets where private firms compete to provide charging, the responsiveness of network operators to electric vehicle demand is likely larger, as suggested by the larger network effects estimated by Springel (2021) and Remmy (2025). This does not necessarily imply more stations overall. Public provision in Quebec has resulted in approximately four times as many chargers per capita as in Ontario, where private firms dominate. Researchers and policymakers should account for these institutional differences when applying these findings to other contexts.

Whether or not governments should subsidize electric vehicle adoption should be carefully

considered against a broader set of alternative environmental policies. One such alternative policy is to subsidize charging station installations directly. In the Canadian context, because network effects are strong on the demand side and the cost of infrastructure is relatively low, subsidizing local planners to provide charging could potentially abate emissions at a much lower cost to the central government. Previous works such as Springel (2021) find that subsidizing charging stations in Norway was more impactful than subsidizing electric vehicle adoption directly on a per dollar basis.

Finally, the results of this study need to be put in perspective against alternative forms of investment into clean air policies. Investments into cleaner electricity production, reforestation, or the modernization of particularly polluting industries could abate emissions at a lower marginal cost than subsidizing electric vehicle purchases, especially in contexts where the upstream electricity production relies on fossil fuels. This study contributes to creating a unified framework to study and compare environmental policies and help policymakers make these crucial decisions.

References

Allcott, H., Kane, R., Maydanchik, M. S., Shapiro, J. S., and Tintelnot, F. (2024). The effects of "buy american": Electric vehicles and the inflation reduction act. Technical report, National Bureau of Economic Research.

Allcott, H. and Wozny, N. (2014). Gasoline prices, fuel economy, and the energy paradox. *Review of Economics and Statistics*, 96(5):779–795.

Armitage, S. and Pinter, F. (2021). Regulatory mandates and electric vehicle product variety. *Working Paper*.

Azarafshar, R. and Vermeulen, W. N. (2020). Electric vehicle incentive policies in Canadian provinces. *Energy Economics*, 91:104902.

Barla, P., Couture, E., and Samano, M. (2016). Gasoline prices and fuel economy of new vehicles in Quebec. *Canadian Public Policy*, 42(2):181–193.

Beresteanu, A. and Li, S. (2011). Gasoline prices, government support, and the demand for hybrid vehicles in the United States. *International Economic Review*, 52(1):161–182.

Berry, S., Levinsohn, J., and Pakes, A. (1995). Automobile prices in market equilibrium. *Econometrica: Journal of the Econometric Society*, pages 841–890.

Birchall, C., Mohapatra, D., and Verboven, F. (2024). Estimating substitution patterns and demand curvature in discrete-choice models of product differentiation. *Review of Economics and Statistics*, pages 1–40.

Blundell, R. W. and Powell, J. L. (2004). Endogeneity in semiparametric binary response models. *The Review of Economic Studies*, 71(3):655–679.

Chaves, D. (2019). Market power, taxation and product variety in the Brazilian automobile industry. *Working paper*.

Clements, M. T. and Ohashi, H. (2005). Indirect network effects and the product cycle: Video games in the U.S., 1994–2002. *The Journal of Industrial Economics*, 53(4):515–542.

Conlon, C. and Gortmaker, J. (2020). Best practices for differentiated products demand estimation with PyBLP. *The RAND Journal of Economics*, 51(4):1108–1161.

Corts, K. S. and Lederman, M. (2009). Software exclusivity and the scope of indirect network effects in the U.S. home video game market. *International Journal of Industrial Organization*, 27(2):121–136.

DeShazo, J., Sheldon, T. L., and Carson, R. T. (2017). Designing policy incentives for cleaner technologies: Lessons from California's plug-in electric vehicle rebate program. *Journal of Environmental Economics and Management*, 84:18–43.

D'Haultfoeuille, X., Durrmeyer, I., and Février, P. (2019). Automobile prices in market equilibrium with unobserved price discrimination. *The Review of Economic Studies*, 86(5):1973–1998.

D'Haultfoeuille, X., Givord, P., and Boutin, X. (2014). The environmental effect of green taxation: The case of the French bonus/malus. *The Economic Journal*, 124(578):F444–F480.

Dorsey, J., Langer, A., and McRae, S. (2022). Fueling alternatives: Gas station choice and the implications for electric charging. Technical report, National Bureau of Economic Research.

Durrmeyer, I. (2022). Winners and losers: The distributional effects of the French feebate on the automobile market. *The Economic Journal*, 132(644):1414–1448.

Durrmeyer, I. and Samano, M. (2018). To rebate or not to rebate: Fuel economy standards versus feebates. *The Economic Journal*, 128(616):3076–3116.

Gandal, N. (1995). Competing compatibility standards and network externalities in the pc software market. *The Review of Economics and Statistics*, pages 599–608.

Gandal, N., Greenstein, S., and Salant, D. (1999). Adoptions and orphans in the early microcomputer market. *The Journal of Industrial Economics*, 47(1):87–105.

Gandal, N., Kende, M., and Rob, R. (2000). The dynamics of technological adoption in hardware/software systems: The case of compact disc players. *The RAND Journal of Economics*, pages 43–61.

Gandhi, A. and Houde, J.-F. (2019). Measuring substitution patterns in differentiated products industries. Technical report, National Bureau of Economic Research.

Grieco, P. L., Murry, C., and Yurukoglu, A. (2023). The evolution of market power in the U.S. automobile industry. *The Quarterly Journal of Economics*, page qjad047.

Grigolon, L., Leheyda, N., and Verboven, F. (2016). Scrapping subsidies during the financial crisis – Evidence from Europe. *International Journal of Industrial Organization*, 44:41–59.

Grigolon, L., Reynaert, M., and Verboven, F. (2018). Consumer valuation of fuel costs and tax policy: Evidence from the European car market. *American Economic Journal: Economic Policy*, 10(3):193–225.

Hausman, J. A. (1996). Valuation of new goods under perfect and imperfect competition. In *The economics of new goods*, pages 207–248. University of Chicago Press.

Holland, S. P., Mansur, E. T., Muller, N. Z., and Yates, A. J. (2019). Distributional effects of air pollution from electric vehicle adoption. *Journal of the Association of Environmental and Resource Economists*, 6(S1):S65–S94.

Huse, C. and Lucinda, C. (2014). The market impact and the cost of environmental policy: Evidence from the Swedish green car rebate. *The Economic Journal*, 124(578):F393–F419.

Ito, K. and Sallee, J. M. (2018). The economics of attribute-based regulation: Theory and evidence from fuel economy standards. *Review of Economics and Statistics*, 100(2):319–336.

Jenn, A., Springel, K., and Gopal, A. R. (2018). Effectiveness of electric vehicle incentives in the United States. *Energy Policy*, 119:349–356.

Johansen, B. G. and Munk-Nielsen, A. (2020). Portfolio complementarities and electric vehicle adoption. *Working paper*.

Kitano, T. (2023). Greening vehicle fleets: A structural analysis of scrappage programs during the financial crisis. Technical report, Research Institute of Economy, Trade and Industry (RIETI).

Knittel, C. R. (2011). Automobiles on steroids: Product attribute trade-offs and technological progress in the automobile sector. *American Economic Review*, 101(7):3368–99.

Li, J. (2023). Compatibility and investment in the U.S. electric vehicle market. *Working paper*.

Li, S., Linn, J., and Spiller, E. (2013). Evaluating "cash-for-clunkers": Program effects on auto sales and the environment. *Journal of Environmental Economics and management*, 65(2):175–193.

Li, S., Liu, Y., and Wei, C. (2022). The cost of greening stimulus: A dynamic analysis of vehicle scrappage programs. *International Economic Review*, 63(4):1561–1594.

Li, S., Tong, L., Xing, J., and Zhou, Y. (2017). The market for electric vehicles: indirect network effects and policy design. *Journal of the Association of Environmental and Resource Economists*, 4(1):89–133.

Muehlegger, E. and Rapson, D. S. (2022). Subsidizing low-and middle-income adoption of electric vehicles: Quasi-experimental evidence from California. *Journal of Public Economics*, 216:104752.

Nair, H., Chintagunta, P., and Dubé, J.-P. (2004). Empirical analysis of indirect network effects in the market for personal digital assistants. *Quantitative Marketing and Economics*, 2(1):23–58.

Nevo, A. (2001). Measuring market power in the ready-to-eat cereal industry. *Econometrica*, 69(2):307–342.

Pavan, G. (2017). Green car adoption and the supply of alternative fuels.

Remmy, K. (2025). Adjustable product attributes, indirect network effects, and subsidy design: The case of electric vehicles. *Conditionally accepted, AEJ: Economic Policy*.

Reynaert, M. (2021). Abatement strategies and the cost of environmental regulation: Emission standards on the European car market. *The Review of Economic Studies*, 88(1):454–488.

Reynaert, M. and Verboven, F. (2014). Improving the performance of random coefficients demand models: The role of optimal instruments. *Journal of Econometrics*, 179(1):83–98.

Sallee, J. M. (2011). The surprising incidence of tax credits for the Toyota Prius. *American Economic Journal: Economic Policy*, 3(2):189–219.

Sheldon, T. L. and Dua, R. (2020). Effectiveness of China's plug-in electric vehicle subsidy. *Energy Economics*, 88:104773.

Springel, K. (2021). Network externality and subsidy structure in two-sided markets: Evidence from electric vehicle incentives. *American Economic Journal: Economic Policy*, 13(4):393–432.

Tsanko, I. (2023). Would you like to super-size your car? The effect of environmental subsidies on emissions. *The Effect of Environmental Subsidies on Emissions (August 3, 2023)*.

Varadhan, R. and Roland, C. (2008). Simple and globally convergent methods for accelerating the convergence of any em algorithm. *Scandinavian Journal of Statistics*, 35(2):335–353.

Xing, J., Leard, B., and Li, S. (2021). What does an electric vehicle replace? *Journal of Environmental Economics and Management*, 107:102432.

Online Appendix

Electric Vehicle Subsidies: Cost-Effectiveness and Emission Reductions

Jean-François Fournel

A Additional Tables and Figures

Table A.1: County-level demographics

	Ontario			Quebec		
	Pre	Post1	Post2	Pre	Post1	Post2
Avg. household income	100,436	104,507	112,740	77,641	82,209	89,108
	(15,592)	(15,809)	(15,638)	(10,995)	(10,935)	(10,914)
Avg. after-tax household income	83,363	85,754	92,651	64,339	67,373	72,834
	(11,275)	(11,121)	(10,944)	(7,986)	(7,866)	(7,806)
Unemployment rate	0.081	0.074	0.12	0.073	0.072	0.075
	(0.012)	(0.010)	(0.018)	(0.023)	(0.021)	(0.020)
Avg. household size	2.65	2.63	2.62	2.33	2.32	2.28
	(0.28)	(0.29)	(0.28)	(0.16)	(0.16)	(0.16)
Avg. age	40.6	41.0	41.8	42.0	42.0	42.8
	(2.66)	(2.02)	(2.03)	(3.22)	(2.41)	(2.53)
Share of graduates	0.25	0.32	0.36	0.20	0.25	0.29
	(0.10)	(0.11)	(0.12)	(0.09)	(0.11)	(0.12)
Share of conservatives	0.42	0.35	0.33	0.17	0.16	0.16
	(0.10)	(0.08)	(0.08)	(0.10)	(0.11)	(0.11)
Work location within 30 min drive	–	0.57	0.61	–	0.61	0.66
		(0.14)	(0.11)		(0.14)	(0.11)
Work location within county of residence	–	0.74	0.76	–	0.67	0.69
		(0.15)	(0.12)		(0.24)	(0.21)
Median commuting time	23.4	–	–	22.1	–	–
	(6.64)			(6.57)		
Share of homeowners	0.72	0.70	0.69	0.62	0.62	0.60
	(0.11)	(0.11)	(0.10)	(0.15)	(0.14)	(0.14)
Share of visible minority	0.27	0.29	0.34	0.12	0.13	0.16
	(0.21)	(0.21)	(0.22)	(0.12)	(0.13)	(0.15)
Population, in million	13.07	13.45	14.22	7.95	8.11	8.44
Nb. of counties	49	49	49	96	96	96

Notes: All values are averaged over counties, weighted by population. Standard deviations are in parenthesis. "Pre" is based on the 2011 Canadian Census Survey. "Post1" is based on the 2016 Canadian Census Survey. "Post2" is based on the 2021 Canadian Census Survey. Income variables are deflated to 2018 Canadian dollars.

Table A.2: Robustness to distance threshold

	(1) No instr	(2) 0 km	(3) 50 km	(4) 100 km	(5) 150 km	(6) 200 km	(7) 250 km	(8) 300 km	(9) 350 km	(10) 400 km	(11) 450 km	(12) 500 km
Price	-0.662	-0.677	-0.675	-0.671	-0.669	-0.674	-0.672	-0.677	-0.676	-0.677	-0.676	-0.675
	(0.031)	(0.031)	(0.031)	(0.031)	(0.031)	(0.031)	(0.031)	(0.031)	(0.031)	(0.031)	(0.031)	(0.031)
Price × Income	0.007	0.007	0.007	0.007	0.007	0.007	0.007	0.007	0.007	0.007	0.007	0.007
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Network	0.134	0.248	0.242	0.229	0.246	0.235	0.234	0.243	0.244	0.239	0.237	0.238
	(0.016)	(0.020)	(0.020)	(0.020)	(0.020)	(0.020)	(0.020)	(0.020)	(0.020)	(0.020)	(0.019)	(0.019)
Network × Income	0.037	0.041	0.049	0.057	0.053	0.056	0.062	0.058	0.055	0.056	0.055	0.052
	(0.009)	(0.013)	(0.012)	(0.012)	(0.012)	(0.013)	(0.013)	(0.012)	(0.012)	(0.013)	(0.013)	(0.013)
Network × Age	-0.004	-0.081	-0.080	-0.073	-0.077	-0.076	-0.074	-0.088	-0.086	-0.086	-0.086	-0.080
	(0.027)	(0.029)	(0.029)	(0.029)	(0.030)	(0.029)	(0.029)	(0.029)	(0.029)	(0.029)	(0.029)	(0.029)
Observations	126,397	126,397	126,397	126,397	126,397	126,397	126,397	126,397	126,397	126,397	126,397	126,397
R-squared	0.121	0.112	0.113	0.115	0.116	0.114	0.115	0.112	0.112	0.112	0.112	0.113

Notes: This table highlights how the coefficients on network utility and its interactions with demographics change as I vary the distance threshold used to construct the network instrument $\mathbf{z^N}$. The model is estimated with an IV logit specification, without the random coefficients, and I fix the curvature parameter to $\zeta = 0.2115$ to avoid performing a nonlinear optimization. Distance thresholds are in kilometers from centroid to centroid for each region pair. Column (1) does not instrument for the network (i.e., excludes $\mathbf{z^N}$) and networks are taken as exogenous. Column (2) uses all stations that are located outside of a given county without filtering for distance. Column (8) is the chosen specification. Comparing columns (1) and (2) shows that controlling for endogeneity increases the coefficient on network size, indicating a downward bias. This result may seem surprising, as intuition suggests that network deployment should be positively correlated with unobserved electric vehicle demand. The intuition breaks down because network size is a function of the unobserved qualities of all vehicles, not just electric ones. While network provision increases with the unobserved quality of electric vehicles, it decreases with the unobserved quality of internal combustion vehicles through a substitution effect. Since internal combustion vehicles vastly outnumber electric vehicles in every market, this negative effect tends to dominate, which can generate a downward bias. All regressions include car characteristics and their interaction with county-level average demographics. All regressions include brand, market segment, county, and year fixed effects. All regressions include the instruments described in Section 4.4. The associated first stage regressions are available in Table A.3, Table A.4, and Table A.5. Standard errors in parenthesis are clustered at the product × county level.

Table A.3: Robustness to distance threshold, first stage (1/3)

	Dependent variable: Network												
	(1) 0km	(2) 50km	(3) 100km	(4) 150km	(5) 200km	(6) 250km	(7) 300km	(8) 350km	(9) 400km	(10) 450km	(11) 500km	(12) 550km	(13) 600km
$z_1^{\mathbf{N}}$	-0.096	1.322	2.647	-0.286	2.965	0.990	3.045	3.593	3.877	4.626	4.969	6.509	5.119
	(1.212)	(0.335)	(0.309)	(0.237)	(0.265)	(0.219)	(0.254)	(0.212)	(0.235)	(0.204)	(0.201)	(0.231)	(0.253)
$z_2^{\mathbf{N}}$	-5.580	-0.552	-1.108	-0.330	0.685	0.191	-0.297	-0.522	-0.262	-0.492	-0.357	-0.297	-1.163
	(0.523)	(0.199)	(0.213)	(0.197)	(0.207)	(0.166)	(0.191)	(0.188)	(0.197)	(0.177)	(0.212)	(0.201)	(0.234)
$z_3^{\mathbf{N}}$	-0.207	-0.430	0.319	0.253	1.009	0.070	0.425	0.617	0.444	0.547	0.437	0.342	-0.926
	(0.483)	(0.513)	(0.485)	(0.459)	(0.417)	(0.392)	(0.393)	(0.384)	(0.377)	(0.360)	(0.360)	(0.337)	(0.357)
$z_1^{\mathbf{N}}$ × Income	2.401	0.453	-1.186	3.170	-0.904	1.768	-1.121	-1.934	-2.308	-3.281	-3.784	-5.915	-3.807
	(1.722)	(0.464)	(0.406)	(0.303)	(0.338)	(0.272)	(0.326)	(0.265)	(0.303)	(0.264)	(0.260)	(0.298)	(0.314)
$z_2^{\mathbf{N}}$ × Income	7.711	0.489	1.440	0.272	-0.922	-0.547	0.177	0.593	0.192	0.508	0.315	0.295	1.295
	(0.741)	(0.222)	(0.238)	(0.219)	(0.235)	(0.180)	(0.217)	(0.216)	(0.235)	(0.205)	(0.260)	(0.245)	(0.282)
$z_3^{\mathbf{N}}$ × Income	0.436	0.640	0.312	0.408	-0.075	0.313	0.232	0.185	0.337	0.298	0.321	0.400	0.900
	(0.309)	(0.351)	(0.331)	(0.325)	(0.300)	(0.288)	(0.283)	(0.281)	(0.277)	(0.264)	(0.261)	(0.247)	(0.256)
$z_1^{\mathbf{N}}$ × Age	-0.148	-0.038	0.043	-0.238	-0.031	-0.197	-0.022	0.034	0.051	0.097	0.128	0.238	0.058
	(0.101)	(0.028)	(0.023)	(0.017)	(0.019)	(0.014)	(0.018)	(0.014)	(0.017)	(0.015)	(0.016)	(0.017)	(0.017)
$z_2^{\mathbf{N}}$ × Age	-0.442	-0.013	-0.082	-0.006	0.060	0.063	0.016	-0.020	0.011	-0.005	0.008	0.005	-0.034
	(0.044)	(0.012)	(0.013)	(0.012)	(0.012)	(0.008)	(0.012)	(0.012)	(0.014)	(0.012)	(0.015)	(0.014)	(0.014)
$z_3^{\mathbf{N}}$ × Age	-0.042	-0.054	-0.088	-0.100	-0.092	-0.032	-0.076	-0.107	-0.117	-0.124	-0.123	-0.124	0.004
	(0.014)	(0.014)	(0.014)	(0.015)	(0.014)	(0.013)	(0.016)	(0.016)	(0.015)	(0.015)	(0.015)	(0.018)	(0.017)
Observations	126,397	126,397	126,397	126,397	126,397	126,397	126,397	126,397	126,397	126,397	126,397	126,397	126,397
R-squared	0.945	0.944	0.943	0.946	0.943	0.946	0.942	0.942	0.943	0.944	0.944	0.944	0.942

Notes: This table provides the first stage regressions associated to each specification in Table A.2. The dependent variable is the network utility function, $v_j(N_{mt}) = \mathbb{1}(j \in EV)\frac{(1+N_{mt})^{\zeta}-1}{\zeta}$, where $\zeta = 0.2115$. The related instruments are $z_{1mt}^{\mathbf{N}} = \frac{\sum_{\ell \neq m} \mathbb{1}(\text{dist}_{\ell,m}>K)N_{\ell t}}{\sum_{\ell \neq m} \mathbb{1}(\text{dist}_{\ell,m}>K)}$, $z_{2mt}^{\mathbf{N}} = \frac{\sum_{\ell \neq m} \mathbb{1}(\text{dist}_{\ell,m}>K)\sqrt{N_{\ell t}}}{\sum_{\ell \neq m} \mathbb{1}(\text{dist}_{\ell,m}>K)}$, $z_{3mt}^{\mathbf{N}} = \frac{\sum_{\ell \neq m} \mathbb{1}(\text{dist}_{\ell,m}>K)\ln(1+N_{\ell t})}{\sum_{\ell \neq m} \mathbb{1}(\text{dist}_{\ell,m}>K)}$, and their interaction with income and age. Distance thresholds are in kilometers from centroid to centroid for each region pair. Column (1) uses all stations that are located outside of a given county without filtering for distance. Column (7) is the chosen specification. All regressions include car characteristics and their interaction with county-level average demographics. All regressions include brand, market segment, county, and year fixed effects. All regressions include the instruments described in Section 4.4. Standard errors in parenthesis are clustered at the product × county level.

Table A.4: Robustness to distance threshold, first stage (2/3)

	Dependent variable: Network × Income												
	(1) 0km	(2) 50km	(3) 100km	(4) 150km	(5) 200km	(6) 250km	(7) 300km	(8) 350km	(9) 400km	(10) 450km	(11) 500km	(12) 550km	(13) 600km
$z_1^{\mathbf{N}}$	-13.831	-0.183	-1.468	1.667	1.023	0.556	-1.457	-1.194	-1.267	-2.279	-0.836	-0.256	0.048
	(2.217)	(0.655)	(0.486)	(0.410)	(0.382)	(0.382)	(0.380)	(0.332)	(0.380)	(0.331)	(0.314)	(0.330)	(0.342)
$z_2^{\mathbf{N}}$	1.269	-4.185	-4.510	-3.104	5.061	4.987	-0.891	-1.167	-0.736	0.276	3.076	6.988	7.174
	(1.909)	(0.440)	(0.455)	(0.452)	(0.706)	(0.475)	(0.451)	(0.432)	(0.459)	(0.470)	(0.504)	(0.523)	(0.564)
$z_3^{\mathbf{N}}$	-1.021	-0.732	-2.544	-1.283	-0.361	-0.331	-1.610	-1.338	-2.557	-2.499	-2.122	-1.014	-0.810
	(0.908)	(1.001)	(1.009)	(0.951)	(0.883)	(0.830)	(0.820)	(0.773)	(0.766)	(0.750)	(0.737)	(0.654)	(0.694)
$z_1^{\mathbf{N}}$ × Income	19.772	0.413	1.915	-2.305	-1.059	-0.484	1.831	1.504	1.698	3.035	1.213	0.435	-0.047
	(3.115)	(0.885)	(0.607)	(0.506)	(0.461)	(0.472)	(0.482)	(0.413)	(0.486)	(0.417)	(0.378)	(0.407)	(0.417)
$z_2^{\mathbf{N}}$ × Income	-0.213	8.207	8.557	7.499	-3.194	-2.955	4.492	4.819	3.676	2.380	-1.202	-6.062	-6.161
	(2.657)	(0.489)	(0.498)	(0.518)	(0.850)	(0.553)	(0.509)	(0.496)	(0.578)	(0.586)	(0.630)	(0.661)	(0.696)
$z_3^{\mathbf{N}}$ × Income	1.758	1.995	2.719	2.349	1.517	1.503	2.194	2.075	2.608	2.530	2.354	1.740	1.726
	(0.615)	(0.720)	(0.719)	(0.694)	(0.651)	(0.612)	(0.601)	(0.574)	(0.572)	(0.560)	(0.551)	(0.496)	(0.499)
$z_1^{\mathbf{N}}$ × Age	-1.158	-0.035	-0.102	0.155	0.073	0.043	-0.064	-0.045	-0.070	-0.134	-0.035	0.022	0.061
	(0.181)	(0.050)	(0.031)	(0.026)	(0.022)	(0.025)	(0.025)	(0.021)	(0.025)	(0.022)	(0.019)	(0.018)	(0.023)
$z_2^{\mathbf{N}}$ × Age	0.034	-0.543	-0.562	-0.564	0.033	0.000	-0.436	-0.447	-0.321	-0.255	-0.065	0.156	0.122
	(0.154)	(0.025)	(0.026)	(0.028)	(0.042)	(0.026)	(0.025)	(0.026)	(0.033)	(0.032)	(0.033)	(0.034)	(0.033)
$z_3^{\mathbf{N}}$ × Age	-0.174	-0.230	-0.188	-0.298	-0.241	-0.257	-0.276	-0.292	-0.205	-0.201	-0.216	-0.269	-0.331
	(0.029)	(0.036)	(0.032)	(0.032)	(0.027)	(0.025)	(0.028)	(0.028)	(0.026)	(0.025)	(0.025)	(0.028)	(0.029)
Observations	126,397	126,397	126,397	126,397	126,397	126,397	126,397	126,397	126,397	126,397	126,397	126,397	126,397
R-squared	0.914	0.928	0.924	0.921	0.905	0.910	0.917	0.918	0.912	0.911	0.908	0.911	0.914

Notes: This table provides the first stage regressions associated to each specification in Table A.2. The dependent variable is the network utility function, $v_j(N_{mt}) = \mathbb{1}(j \in EV)\,\frac{(1+N_{mt})^{\zeta}-1}{\zeta}$, where $\zeta = 0.2115$, multiplied by income. The related instruments are $z^{\mathbf{N}}_{1mt} = \frac{\sum_{\ell \neq m} \mathbb{1}(\text{dist}_{\ell,m}>K) N_{\ell t}}{\sum_{\ell \neq m} \mathbb{1}(\text{dist}_{\ell,m}>K)}$, $z^{\mathbf{N}}_{2mt} = \frac{\sum_{\ell \neq m} \mathbb{1}(\text{dist}_{\ell,m}>K) \sqrt{N_{\ell t}}}{\sum_{\ell \neq m} \mathbb{1}(\text{dist}_{\ell,m}>K)}$, $z^{\mathbf{N}}_{3mt} = \frac{\sum_{\ell \neq m} \mathbb{1}(\text{dist}_{\ell,m}>K) \ln(1+N_{\ell t})}{\sum_{\ell \neq m} \mathbb{1}(\text{dist}_{\ell,m}>K)}$, and their interaction with income and age. Distance thresholds are in kilometers from centroid to centroid for each region pair. Column (1) uses all stations that are located outside of a given county without filtering for distance. Column (7) is the chosen specification. All regressions include car characteristics and their interaction with county-level average demographics. All regressions include brand, market segment, county, and year fixed effects. All regressions include the instruments described in Section 4.4. Standard errors in parenthesis are clustered at the product × county level.

Table A.5: Robustness to distance threshold, first stage (3/3)

	Dependent variable: Network × Age												
	(1) 0km	(2) 50km	(3) 100km	(4) 150km	(5) 200km	(6) 250km	(7) 300km	(8) 350km	(9) 400km	(10) 450km	(11) 500km	(12) 550km	(13) 600km
$z_1^{\mathbf{N}}$	4.878	0.341	0.321	0.088	0.735	0.369	0.838	0.896	0.882	1.098	1.038	0.993	1.100
	(0.264)	(0.076)	(0.096)	(0.080)	(0.075)	(0.075)	(0.080)	(0.074)	(0.076)	(0.070)	(0.073)	(0.068)	(0.073)
$z_2^{\mathbf{N}}$	-1.659	0.461	0.588	0.639	-0.274	-0.428	0.262	0.159	0.117	0.084	-0.192	-0.772	-0.760
	(0.190)	(0.060)	(0.072)	(0.078)	(0.062)	(0.057)	(0.065)	(0.067)	(0.073)	(0.064)	(0.068)	(0.062)	(0.065)
$z_3^{\mathbf{N}}$	1.880	1.883	2.584	3.210	3.056	2.584	2.602	2.634	2.691	2.724	2.576	2.345	2.258
	(0.131)	(0.131)	(0.122)	(0.123)	(0.120)	(0.111)	(0.119)	(0.121)	(0.124)	(0.115)	(0.115)	(0.111)	(0.110)
$z_1^{\mathbf{N}}$ × Income	-6.773	-0.294	-0.217	0.104	-0.796	-0.294	-0.894	-1.016	-1.033	-1.321	-1.282	-1.262	-1.387
	(0.372)	(0.098)	(0.121)	(0.101)	(0.091)	(0.093)	(0.100)	(0.091)	(0.095)	(0.086)	(0.088)	(0.082)	(0.088)
$z_2^{\mathbf{N}}$ × Income	2.430	-0.646	-0.728	-0.747	0.427	0.556	-0.364	-0.202	-0.101	-0.040	0.312	1.055	1.052
	(0.267)	(0.066)	(0.083)	(0.093)	(0.072)	(0.064)	(0.077)	(0.081)	(0.089)	(0.075)	(0.080)	(0.076)	(0.079)
$z_3^{\mathbf{N}}$ × Income	-0.386	-0.415	-0.626	-0.764	-0.647	-0.399	-0.406	-0.395	-0.351	-0.325	-0.225	-0.018	0.072
	(0.094)	(0.093)	(0.087)	(0.090)	(0.093)	(0.089)	(0.092)	(0.093)	(0.096)	(0.090)	(0.089)	(0.087)	(0.085)
$z_1^{\mathbf{N}}$ × Age	0.390	0.008	-0.000	-0.020	0.030	-0.002	0.032	0.042	0.046	0.061	0.062	0.061	0.064
	(0.022)	(0.006)	(0.007)	(0.006)	(0.005)	(0.005)	(0.006)	(0.005)	(0.005)	(0.005)	(0.005)	(0.004)	(0.005)
$z_2^{\mathbf{N}}$ × Age	-0.144	0.045	0.043	0.039	-0.024	-0.025	0.033	0.020	0.009	0.005	-0.013	-0.051	-0.050
	(0.016)	(0.003)	(0.004)	(0.005)	(0.004)	(0.003)	(0.005)	(0.005)	(0.005)	(0.004)	(0.004)	(0.004)	(0.004)
$z_3^{\mathbf{N}}$ × Age	0.014	0.019	-0.021	-0.075	-0.079	-0.061	-0.062	-0.070	-0.091	-0.101	-0.103	-0.112	-0.113
	(0.007)	(0.007)	(0.007)	(0.007)	(0.006)	(0.006)	(0.007)	(0.006)	(0.006)	(0.006)	(0.005)	(0.005)	(0.006)
Observations	126,397	126,397	126,397	126,397	126,397	126,397	126,397	126,397	126,397	126,397	126,397	126,397	126,397
R-squared	0.933	0.932	0.930	0.934	0.931	0.933	0.935	0.935	0.935	0.938	0.934	0.935	0.936

Notes: This table provides the first stage regressions associated to each specification in Table A.2. The dependent variable is the network utility function, $v_j(N_{mt}) = \mathbb{1}(j \in EV)\,\frac{(1+N_{mt})^{\zeta}-1}{\zeta}$, where $\zeta = 0.2115$, multiplied by age. The related instruments are $z_{1mt}^{\mathbf{N}} = \frac{\sum_{\ell \neq m} \mathbb{1}(\text{dist}_{\ell,m}>K) N_{\ell t}}{\sum_{\ell \neq m} \mathbb{1}(\text{dist}_{\ell,m}>K)}$, $z_{2mt}^{\mathbf{N}} = \frac{\sum_{\ell \neq m} \mathbb{1}(\text{dist}_{\ell,m}>K) \sqrt{N_{\ell t}}}{\sum_{\ell \neq m} \mathbb{1}(\text{dist}_{\ell,m}>K)}$, $z_{3mt}^{\mathbf{N}} = \frac{\sum_{\ell \neq m} \mathbb{1}(\text{dist}_{\ell,m}>K) \ln(1+N_{\ell t})}{\sum_{\ell \neq m} \mathbb{1}(\text{dist}_{\ell,m}>K)}$, and their interaction with income and age. Distance thresholds are in kilometers from centroid to centroid for each region pair. Column (1) uses all stations that are located outside of a given county without filtering for distance. Column (7) is the chosen specification. All regressions include car characteristics and their interaction with county-level average demographics. All regressions include brand, market segment, county, and year fixed effects. All regressions include the instruments described in Section 4.4. Standard errors in parenthesis are clustered at the product × county level.

Table A.6: Robustness to spillovers, demand-side

	(1) Baseline (IV Logit)	(2) Spillovers (50km)	(3) Spillovers (100km)
Price	-0.677	-0.679	-0.681
	(0.031)	(0.031)	(0.031)
Price × Income	0.007	0.007	0.007
	(0.004)	(0.004)	(0.004)
Network	0.243	0.206	0.197
	(0.020)	(0.022)	(0.024)
Network × Income	0.058	0.077	0.073
	(0.012)	(0.029)	(0.029)
Network × Age	-0.088	0.016	-0.034
	(0.029)	(0.057)	(0.078)
Networks within 50 km		0.031	0.027
		(0.007)	(0.011)
Networks within 50 km × Income		-0.015	-0.009
		(0.009)	(0.009)
Networks within 50 km × Age		-0.027	-0.033
		(0.028)	(0.032)
Networks within 100 km			0.006
			(0.010)
Networks within 100 km × Income			-0.010
			(0.008)
Networks within 100 km × Age			0.024
			(0.033)
Observations	126,397	126,397	126,397
R-squared	0.112	0.112	0.111

Notes: This table presents the results from an IV logit demand specification, without random coefficients, to study the contribution of spillover effects in the demand estimation. Variables Network, Networks within 50 km, and Networks within 100 km keep the estimated functional form as in the main specification, with curvature parameter $\zeta = 0.2115$. I assume that networks in other regions are uncorrelated to local shocks, hence I leave them as exogenous. Column (1) presents the result from the baseline specification, equivalent to Table 5. Column (2) introduces networks in other counties, within 50 km of the home county. Column (3) adds networks beyond 50 km and up to 100 km. All regressions include car characteristics and their interaction with county-level average demographics. All regressions include brand, market segment, county, and year fixed effects. All regressions include the instruments described in Section 4.4. Standard errors in parenthesis are clustered at the product × county level.

Table A.7: Robustness to spillovers, network supply-side

	(1) Baseline (OLS)	(2) Baseline (IV)	(3) Spillovers (50km)	(4) Spillovers (100km)
Log of EV stock	0.275	0.171	0.194	0.134
	(0.040)	(0.071)	(0.132)	(0.162)
Log of EV stock within 50km			-0.008	-0.027
			(0.028)	(0.026)
Log of EV stock within 100km				0.035
				(0.033)
Avg. income	0.068	0.069	0.060	0.082
	(0.091)	(0.090)	(0.093)	(0.097)
Avg. age	0.494	0.385	0.413	0.379
	(0.182)	(0.197)	(0.239)	(0.251)
Avg. household size	-0.011	-0.032	0.061	0.160
	(0.624)	(0.616)	(0.725)	(0.689)
Share graduates	3.849	4.607	4.539	4.939
	(1.005)	(1.039)	(1.090)	(1.175)
Share homeowners	-3.193	-3.498	-3.441	-3.807
	(0.779)	(0.870)	(0.923)	(1.010)
Urban	0.066	0.174	0.172	0.199
	(0.140)	(0.172)	(0.175)	(0.191)
Observations	864	864	864	864
R-squared	0.832	0.651	0.655	0.648

Notes: This table presents the results from a linear regression of the log of network size on the log of the installed base of electric vehicles to study the contribution of spillover effects in the supply estimation. Column (1) estimates the model without instrumental variables. Column (2) introduces three instruments for the installed base of electric vehicles, namely, the gas station density (number of gas stations per 5,000 inhabitant), a fuel price index, and an interaction between the two, similarly to Springel (2021). The identifying assumptions are that competition in the fuel market affects electric vehicle sales through substitution between fuel and electric and that network operators and refueling stations do not compete with each other once sales are realized. Column (3) and (4) add the installed base of electric vehicles in counties within 50 km and between 50 and 100 km of the origin. The installed base in other counties is assumed to be uncorrelated to local shocks, hence is left as exogenous. All regression include year fixed effects. Standard errors in parenthesis are clustered at the county level.

Table A.8: Simultaneity, endogeneity, and network supply

		Alternative specifications	
	(1) Static model	(2) Control function	(3) No control function
$\lambda^{\mathbf{N}} = \gamma/\omega$	2.445 (0.139)	2.484 (0.139)	2.473 (0.140)
$\lambda^{\mathbf{Q}} = 1/\omega$	0.417 (0.111)	0.397 (0.208)	0.573 (0.105)
Avg. income	-0.591 (0.243)	-0.602 (0.252)	-0.604 (0.250)
Avg. age	1.856 (0.557)	1.849 (0.604)	2.009 (0.548)
Avg. household size	-0.103 (1.517)	-0.125 (1.556)	-0.151 (1.568)
Share graduates	8.323 (2.489)	8.675 (2.748)	7.392 (2.516)
Share homeowner	-5.911 (1.990)	-6.035 (2.266)	-5.412 (1.967)
Urban	0.390 (0.327)	0.419 (0.402)	0.234 (0.326)
Control function		0.243 (0.246)	
Observations	830	830	830
Likelihood	-2,116.9	-2,082.0	-2,084.5

Notes: This table compares alternative methods for dealing with the simultaneity and the endogeneity issues in the network supply model. Column (1) reports the estimates from the static version of the model presented in Section 4.3. Column (2) reports the estimates from a static model that does not internalize the demand response in electric vehicle sales from additional stations. This is done by replacing the demand system $Q^{ev}_{mt}(n)$ by the observed market outcomes Q^{ev}_{mt} in the likelihood equation. The simultaneity issue is tackled using a control function approach. The set of instrumental variables includes the county-level demographics, as well as the gas station density (number of gas stations per 5,000 inhabitant), a fuel price index, and an interaction between the two, similarly to Springel (2021). The identifying assumptions are that competition in the fuel market affects electric vehicle sales through substitution between fuel and electric and that network operators and refueling stations do not compete with each other once sales are realized. Column (3) reports estimates for the same model as in (2), without the control function approach such that the simultaneity issue is not addressed. Standard errors in parenthesis are clustered at the county level and are computed using 500 bootstrap replications.

Table A.9: Average characteristics, by engine type

	Fuel	Battery electric	Plug-in hybrid	Hybrid
Characteristics				
List price, in CAD	36,720	54,373	42,755	36,768
Net price, in CAD	36,720	44,564	35,771	36,405
Rebate, in CAD	0	9,809	6,984	363
Power, in kW	149.6	168.9	154.9	163.5
Length, in m	4.55	4.41	4.63	4.65
Width, in m	1.83	1.85	1.82	1.83
Height, in m	1.57	1.52	1.51	1.58
Weight, in 100kg	15.8	16.9	16.9	16.0
Driving range, in km	661	349	809	873
Fuel consumption, in L/100km	8.85	0	5.94	5.89
Electricity consumption, in kWh/100km	0	16.1	24.8	0
Cost of driving 100km	11.37	1.74	6.78	7.33
CO_2 emissions, in g/km	205.6	0	60.1	137.5
Transmission				
Manual	0.10	0	0	0
Automatic/single-speed	0.90	1	1	1
Fuel type				
Regular	0.82	0	0.88	1
Premium	0.15	0	0.12	0
Diesel	0.03	0	0	0
Market segment				
Subcompact	0.11	0.19	0.01	0
Compact	0.33	0.59	0.66	0.11
Midsize	0.05	0	0.09	0.19
Luxury/executive	0.02	0.03	0	0
Crossover utility (CUV)	0.18	0.18	0.06	0.64
Sport utility (SUV)	0.27	0.01	0.15	0.05
Minivan	0.03	0	0.02	0

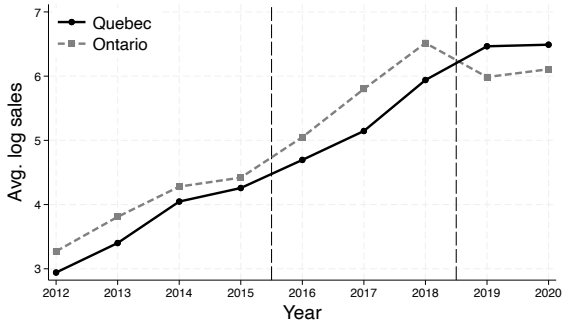
Notes: All averages are weighted by sales. All dollars values are in 2018 CAD.

Table A.10: Alternative demand specification

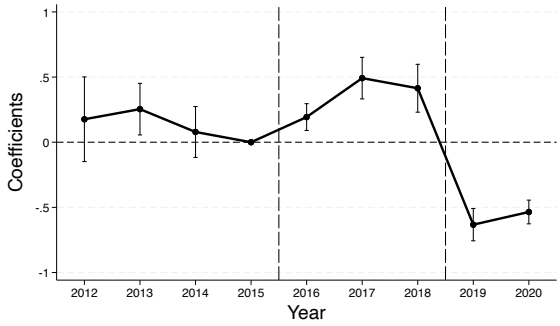
		Demographic interactions						
	Estimate	Income	Age	Gender	Population density	Trend	ζ	σ
Price	-0.844	0.006						0.167
	(0.032)	(0.003)						(0.014)
Network	0.206	0.066	-0.063				0.211	-0.038
	(0.067)	(0.037)	(0.024)				(0.174)	(0.395)
Power	0.947		0.043	0.034				
	(0.021)		(0.017)	(0.004)				
Weight	0.233					0.093		
	(0.034)					(0.004)		
Driving cost	-0.033							
	(0.004)							
Battery electric	-2.064	-0.274		0.193	-0.615			
	(0.104)	(0.084)		(0.028)	(0.128)			
Plug-in hybrid	-2.024	-0.405		0.173	-0.658			
	(0.096)	(0.086)		(0.029)	(0.145)			
Hybrid	-1.708		0.330	0.153				
	(0.021)		(0.041)	(0.016)				
Constant								5.536
								(1.987)
Observations	126,397							
Nb. of markets	864							
Obj. Function	3,397.3							

Notes: The table presents the results from an alternative demand specification with a random coefficient on the network utility. The additional random coefficient is insignificant. The other parameters are robust to including this additional random coefficient, but we lose significance on the curvature parameter, ζ. Includes brand, market segment, county, and year fixed effects. Robust standard errors in parenthesis.

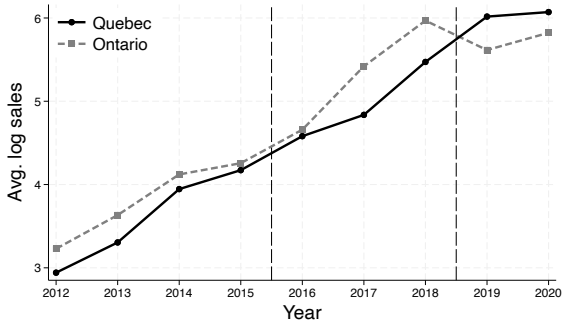
Figure A.1: Event-study, log of sales



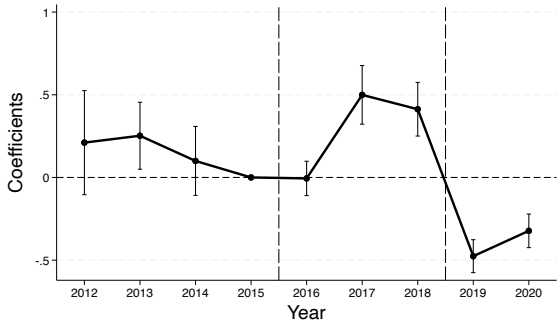
(a) All electric vehicles



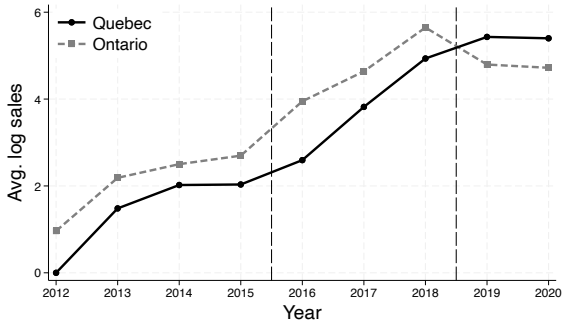
(b) Event study estimates



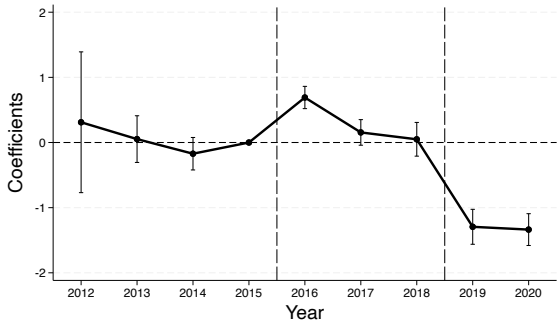
(c) Battery electric only



(d) Event study estimates



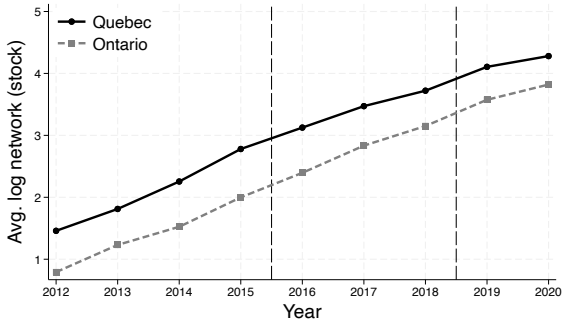
(e) Plug-in hybrid only



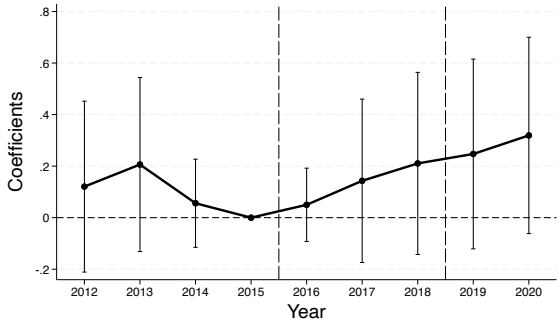
(f) Event study estimates

Notes: All regressions include county-level demographics and are weighted by population. Standard errors are clustered at the county level.

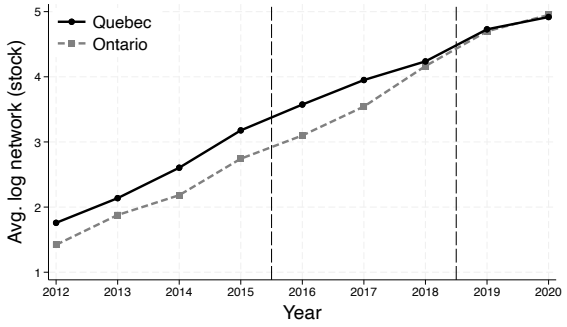
Figure A.2: Event-study, log of network size (stock)



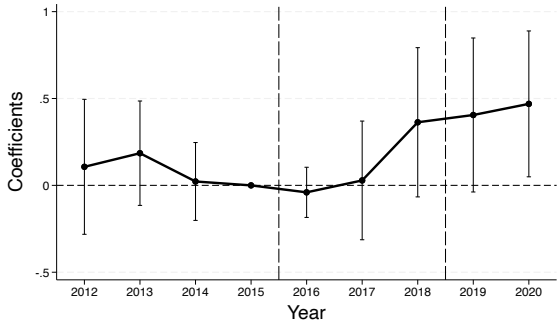
(a) Nb. of locations



(b) Event study estimates



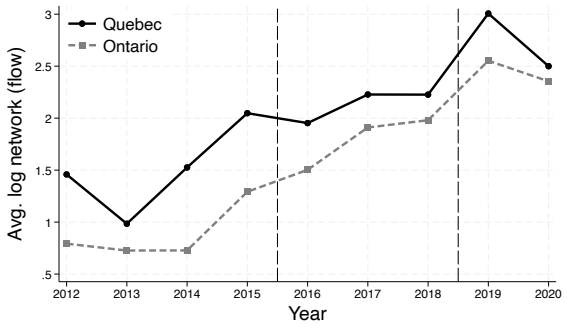
(c) Nb. of chargers



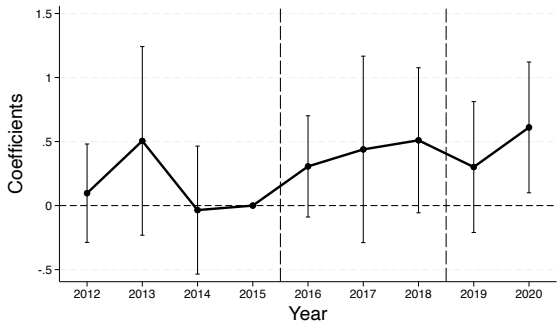
(d) Event study estimates

Notes: All regressions include county-level demographics and are weighted by population. Standard errors are clustered at the county level.

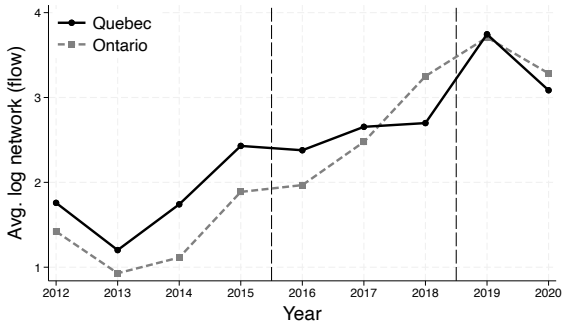
Figure A.3: Event-study, log of network size (flow)



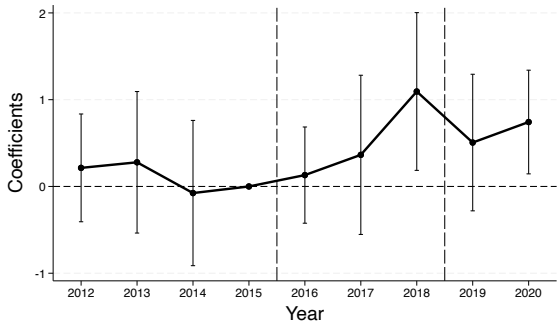
(a) New location openings



(b) Event study estimates



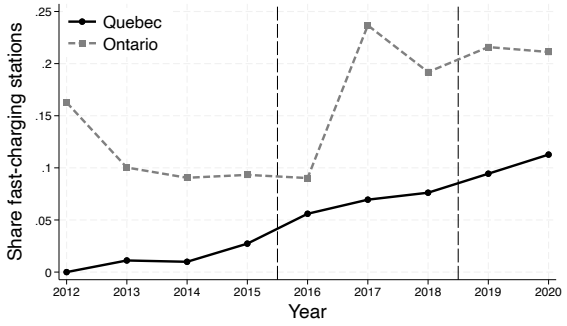
(c) New charger installations



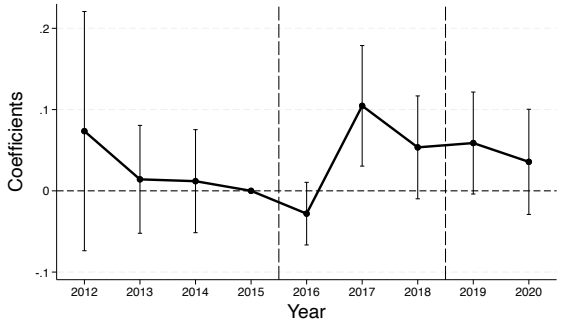
(d) Event study estimates

Notes: All regressions include county-level demographics and are weighted by population. Standard errors are clustered at the county level.

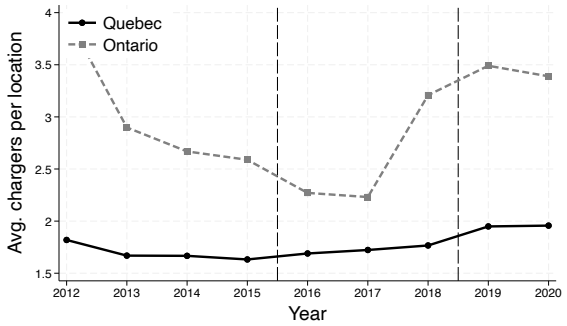
Figure A.4: Event-study, network characteristics (stock)



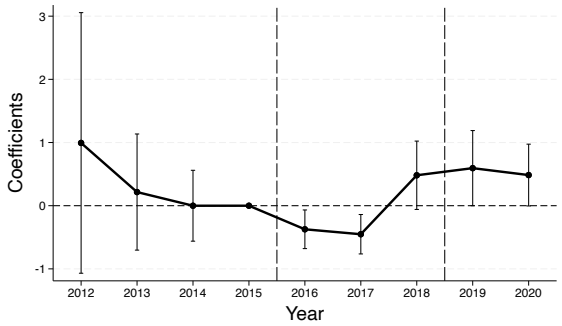
(a) Ratio fast-charging stations (stock)



(b) Event study estimates



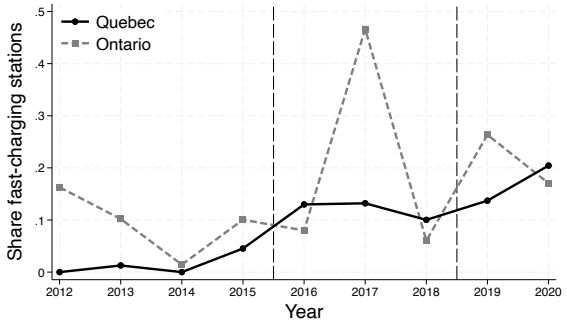
(c) Avg. chargers per location (stock)



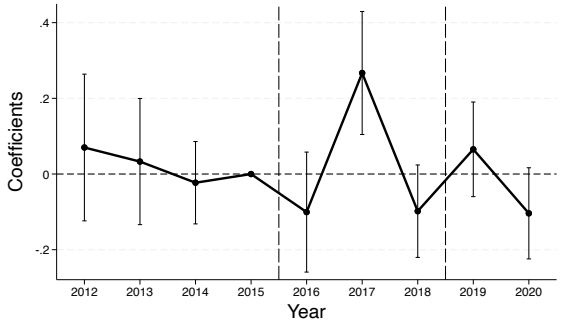
(d) Event study estimates

Notes: All regressions include county-level demographics and are weighted by population. Standard errors are clustered at the county level.

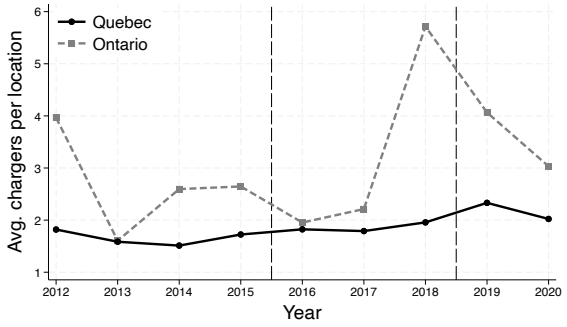
Figure A.5: Event-study, network characteristics (flow)



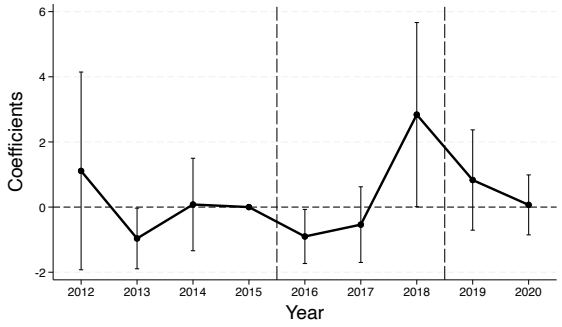
(a) Ratio fast-charging stations (flow)



(b) Event study estimates



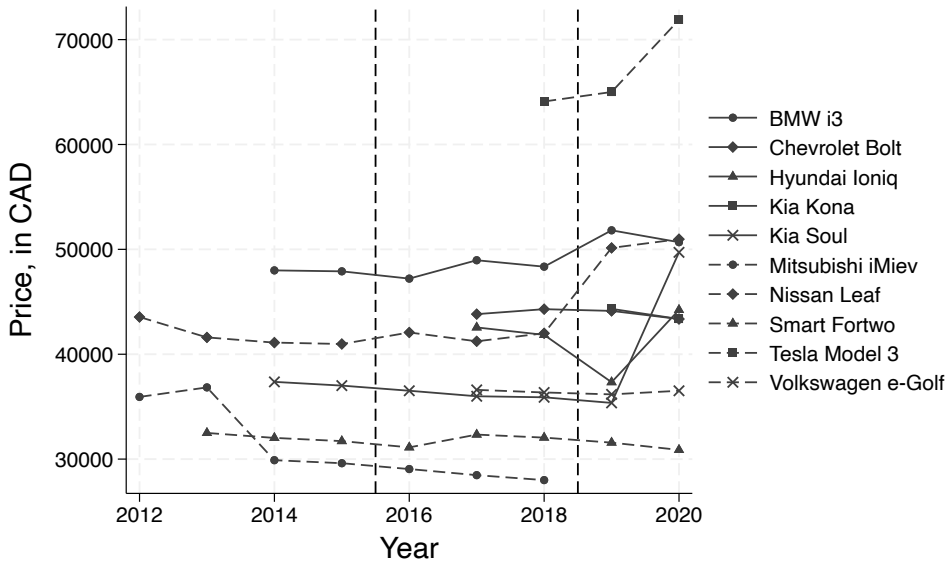
(c) Avg. chargers per location (flow)



(d) Event study estimates

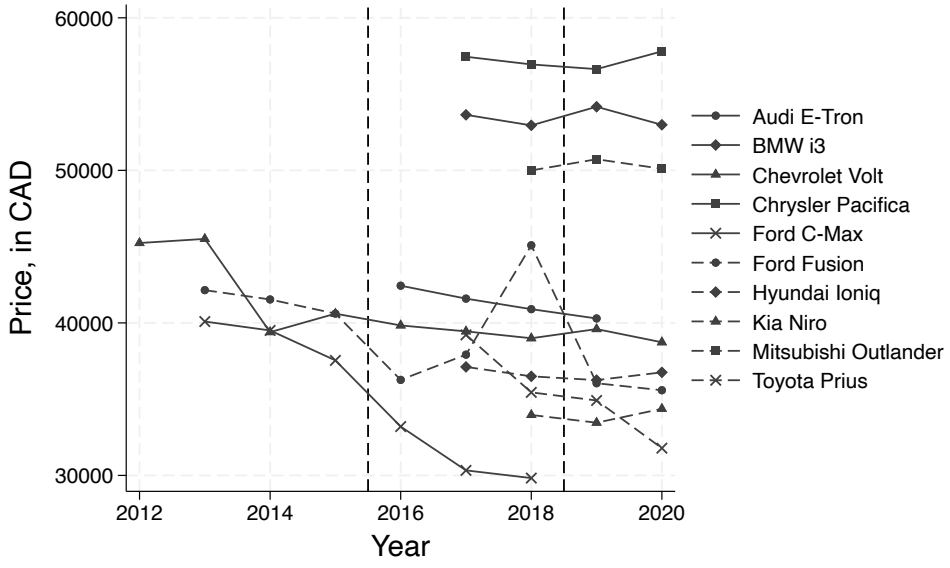
Notes: All regressions include county-level demographics and are weighted by population. Standard errors are clustered at the county level.

Figure A.6: Evolution of prices for selected battery electric vehicles



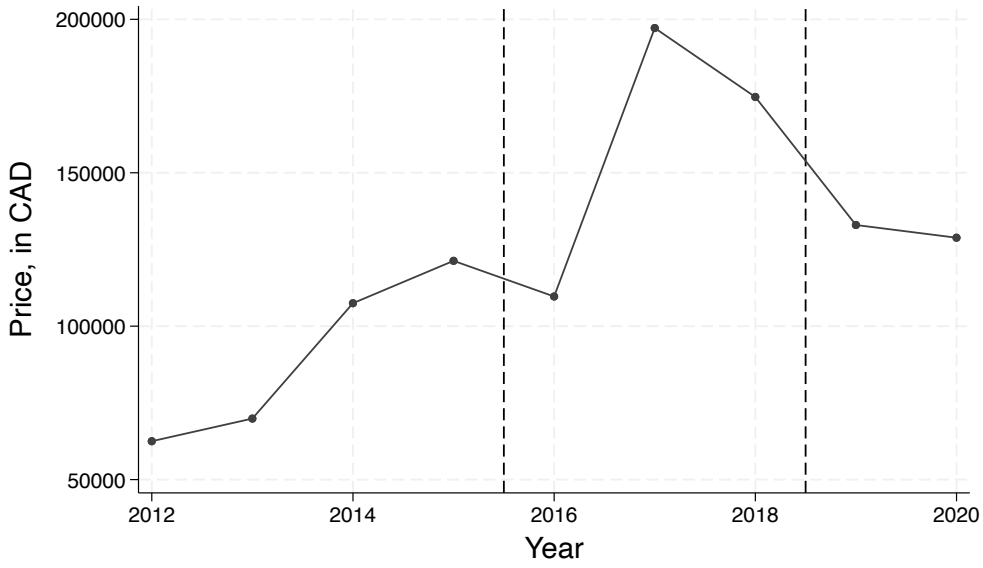
Notes: This figure depicts the evolution of prices for the top 10 selling battery electric vehicles, excluding Tesla Model S (see Figure A.8).

Figure A.7: Evolution of prices for selected battery electric vehicles



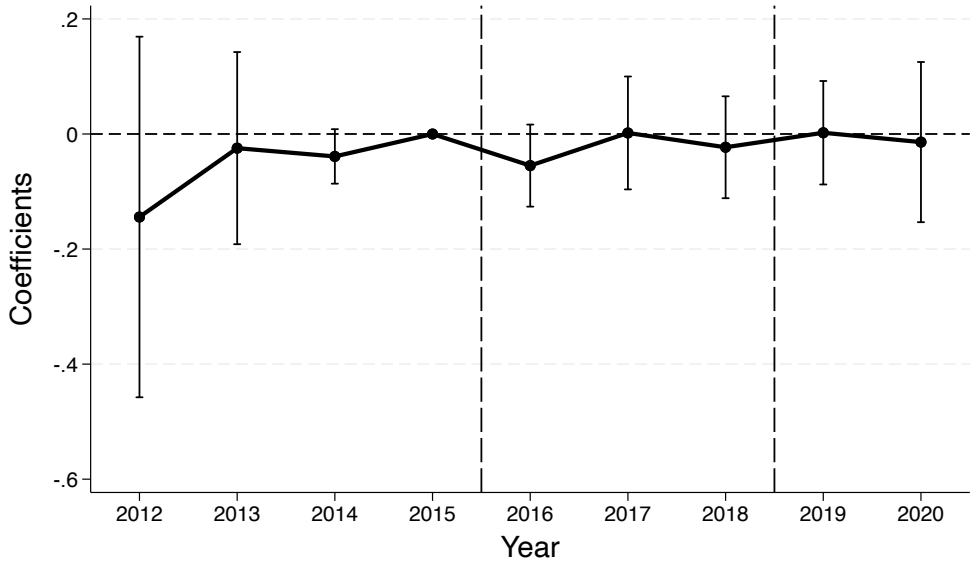
Notes: This figure depicts the evolution of prices for the top 10 selling plug-in hybrid vehicles.

Figure A.8: Evolution of prices for Tesla Model S



Notes: This figure depicts the evolution of prices for Tesla Model S.

Figure A.9: Event-study, log of price



Notes: Event-study for the effect of rebates on prices. The treatment group includes all electric vehicles eligible under Ontario's rebate program. The control group is constructed by matching each treated observation with a single ineligible car model, including internal combustion and non-rechargeable hybrid vehicles, using Mahalanobis-metric matching based on the horsepower, the curb weight, the size, the brand, and the market segment (within year). The regression is unweighted and includes product and year fixed effects. Standard errors are clustered at the product level.

B Details on the Data

I use data from several sources, described below. The data is aggregated at the county level, following Statistics Canada's Census Divisions. Markets are defined as county-year combinations. I choose this level of aggregation for two reasons. First, counties capture relatively well the day-to-day commuting area of car owners: about 72% of them work within their county of residence. Also, county-level governments are the largest contributors to charging networks, which reinforces the idea that network provision is decided at the level of the county.

Car registration. The data on car registration comes from two main sources: the Ministry of Transportation of Ontario and the Société d'Assurance Automobile du Québec. The Ontario dataset includes quarterly car registrations aggregated at the product-county level for the years 2011-2021. The data includes the make (e.g., Ford), the model (e.g., Focus), the engine type (e.g., battery electric), and the quantity sold.

The Quebec dataset comprises 10 yearly datasets that detail the full fleet of vehicles in circulation at the end of each year, from 2011 to 2020. The data includes the make, the model, the model year, some demographics of the owner (age, gender, county of residence), as well as additional vehicle characteristics (color, number of cylinders, cylinder capacity, and curb weight). The engine type is available from the 2017 dataset onward. I impute the engine type in the 2011-2016 datasets using the information available in the 2017 dataset. Since battery packs are relatively heavy, I find that the make, the model, the model year, and the curb weight of the vehicles allow me to identify battery electric, plug-in hybrids, and non-rechargeable hybrids reliably. In some edge cases, I also leverage information in the other variables (number of cylinder, cylinder capacity, and the consumer demographics) to assign an engine type to all vehicles. Vehicles with a model year prior to 2011 are assumed to be internal combustion engines, unless the model is exclusively electric or exclusively hybrid with no other engine type variant.

I use the following algorithm to reconstruct sales in Quebec in each year between 2012 and 2020.

1. Take dataset t;
2. Keep model years that could have been sold as new in year t (e.g., $t-1, t, t+1$);
3. Remove vehicles that also appear in dataset $t-1$, by comparing the make, the model, the model year, the color of the vehicle, the age of the owner, the gender of the owner, and the county of residence of the owner;

4. Repeat for dataset $t+1$.

Since vehicles could have been resold in the first year of ownership, or owners could have moved to a different county, I risk overestimating sales. After a careful verification against governmental statistics, I find that this is not a serious issue.

Car characteristics. The car characteristics were scraped from The Car Guide[39] which publishes on their website comprehensive information on all makes and models available in Canada. This website has been one of the go-to references for information about the different car makes since the mid-90s and has widespread public recognition in Canada. The car characteristics dataset includes the manufacturer's suggested retail price and various characteristics such as the engine type, horsepower, size, fuel consumption, and carbon emissions, all recorded at the brand-model-year-specification level (e.g., Ford Focus 2017 S-Sedan). The data has a non-negligible number of missing values in key variables. Specifications with a missing price or a missing curb weight are removed entirely.[40] Missing values in other variables are filled in using the data from other specifications that share the same make-model-year. If information is missing for all specifications for a make-model-year combination, I use information from other vehicles with the same make-model but a different model year. Remaining missing values are imputed using data collected on the Auto Trader's website.[41]

Product definition. I define a product to be a combination of the make, the model, and the engine type. The final dataset is aggregated at the product-county-year level. The car characteristics dataset is at a more disaggregated level than the registration data. I select the characteristics of the most sold specification over all sales to define product attributes. To find this most popular specification, I first recover the exact specification for each entry in the registration data by matching on the make, the model, the engine type, and then picking the specification with the closest curb weight.[42] I then aggregate the data over counties and keep the specification with the most sales. Once the specification is chosen, I assign these characteristics to all products.

To avoid the proliferation of products in the structural estimation, I remove all products with fewer than 1,000 sales over all counties and years (100 sales for battery electric and

[39] See https://www.guideautoweb.com/en/.

[40] Curb weight is particularly important in this context since I use it to match the make-model-year registration data to the make-model-year-specification characteristics data.

[41] See https://www.autotrader.ca.

[42] In case two specifications have the same weight in the characteristics data, I keep the specification that is closest to the base model.

plug-in hybrid vehicles). I also remove exotic makes, and all vehicles with a retail price above $150,000. Finally, I remove pickup trucks which are poor substitutes for electric vehicles and are not relevant to this study.

Other data sources. I complement these datasets with data from various other sources. The data on charging stations comes from Natural Resources Canada and Hydro-Quebec. They contain the exact geographical location of each station, the entry date, the operator's name, pricing, and relevant physical attributes such as the type of station, the number of chargers, and whether a station is publicly or privately owned. I obtain detailed data on government expenditure on rebates, which include the exact rebate that was given to each model in each year. Consumer demographics are taken from the Canadian Census Survey, the Institut de la Statistique du Québec, and Election Canada. Information on gas prices and gas stations are obtained from the Régie de l'Énergie du Québec.

C Continuous treatment effect

Effect on electric vehicle registrations. I study the effect of rebates at the intensive margin using a continuous treatment effect specification. This allows me to identify the underlying market elasticity of demand for electric vehicles using a similar approach to Muehlegger and Rapson (2022). I start by constructing a continuous measure of the treatment variable, $\overline{\tau}_{mt}$, the average rebate received in county m and year t. I then estimate the following continuous treatment effect specification,

$$y_{mt} = \alpha\overline{\tau}_{mt} + X_{mt}\gamma + \mu_m + \lambda_t + \epsilon_{mt}, \tag{A1}$$

where μ_m and λ_t are fixed effects, and X_{mt} is a vector of county-level demographics as previously. The dependent variable is the log of electric vehicle registrations by county and year. The parameter of interest is α, the semi-elasticity to the rebate.

The average rebate is constructed by aggregating over individual-level rebates within a county and year. Therefore, it depends on the composition of the underlying fleet of electric vehicles and is endogenous by construction. If the proportion of plug-in hybrids is higher in a given county, the average rebate would decrease mechanically as plug-in hybrids are usually not eligible for the same subsidy as battery electric vehicles. Endogeneity arises if unobserved shocks to consumer preferences shift both the total quantity of electric vehicles sold and the proportion of battery electrics to plug-in hybrids.

I propose two different instrumental variables to address this issue. First, I consider

Table C.1: Effect of rebates on electric vehicle registrations

	Log of EV sales		
	(1) OLS	(2) IV	(3) IV
Avg. rebate	0.120	0.079	0.078
	(0.007)	(0.003)	(0.003)
First stage:			
Ontario × Post1		4.949	
		(0.171)	
Ontario × Post2		-6.791	
		(0.154)	
Avg. rebate in other counties			0.981
			(0.012)
Covariates	Y	Y	Y
County FE	Y	Y	Y
Year FE	Y	Y	Y
Observations	1,232	1,232	1,232

Notes: The dependent variable is the log of electric vehicle sales. The average rebate is computed at the county-year level, in thousand 2018 CAD. Covariates include the average household income, the average age, the average household size, the share of conservative voters, the share of graduates, and the share of homeowners by county and year. All regressions are weighted by county-level populations. Standard errors in parenthesis are clustered at the county level.

using the discrete version of the treatment variable as instruments. These instruments are naturally highly correlated to the average rebate. The exclusion restriction would be satisfied if the timing of the policy changes in Ontario were uncorrelated with local shocks. This assumption is difficult to test in practice. I construct a second instrument in the spirit of Hausman (1996) and Nevo (2001). The idea is to use the cross-sectional variation in the data to construct a valid instrument for the average rebate. In this context, this means using the average rebate in other counties within a province. The instrument's validity rests on the assumption that the proportion of battery electrics to plug-in hybrids in other counties is uncorrelated with local preference shocks. This assumption would be violated if preference shocks not accounted for by fixed effects affected the ratio of battery electrics to plug-in hybrids in many counties simultaneously.

The results are presented in Table C.1. I report the main coefficient of interest for a specification estimated by least squares and two specifications estimated using the instrumental

variables described above. Both instrumental variable regressions give a very similar result: a \$1,000 increase in rebates is associated with a 7.9% increase in sales of electric vehicles. This is significantly lower than the specification estimated without instruments.

Effect on prices. I am also interested in the effect of rebates on prices. Since I am using national prices that do not vary across provinces, aggregating at the county level does not allow me to identify the effect of rebates on prices. I rely instead on data aggregated at the product level, including both electric vehicles (treated), fuel vehicles, and hybrid vehicles (not treated). The continuous measure of the treatment variable, $\overline{\tau}_{jt}$, is the average rebate offered to consumers purchasing product j in year t. I then estimate the following continuous treatment effect specification,

$$y_{jt} = \psi \overline{\tau}_{jt} + X_{jt}\gamma + \mu_j + \lambda_t + \epsilon_{jt}, \tag{A2}$$

where μ_j and λ_t are fixed effects, and X_{jt} is a vector of product characteristics and cost shifters. The dependent variable is the price of product j after rebates, and the parameter of interest is ψ, the rebate passthrough.

The results are presented in Table C.2. I consider the net price both in levels and in logs. The average rebate is constructed by aggregating over counties at the product level, hence it depends on how many sales of each product occurred in each province. This could lead to endogeneity if preference shocks that change the composition of the fleet at the province level correlate with prices. To instrument for the average rebate, I use the discrete version of the treatment variable, that is, I interact a rebate eligibility indicator variable with post-treatment period dummies.

I find that the estimation leads to very similar results with or without instrumental variables, indicating that endogeneity is not a serious issue in this case. Columns (1) and (2) of Table C.2 report the estimated effect of rebates on the net price in levels. I find that increasing subsidies by one dollar changed the net price by \$0.95–0.98 and I cannot reject the hypothesis of perfect passthrough (although the coefficients are not very precisely estimated and have a large standard error). This provides some additional evidence that firms did not respond to changes in the Ontario rebate program by increasing the prices of electric vehicles nationwide. Column (3) and (4) report the impact of rebates on the net price in logs. These coefficients are hard to interpret on their own, however, they are useful to recover the elasticity to price for electric vehicle.

Market elasticity and passthrough. Combining both sets of results allows me to recover the market elasticity of demand for electric vehicles. Denote by ε the market elasticity to

Table C.2: Effect of rebates on prices

	Net price		Log of net price	
	(1) OLS	(2) IV	(3) OLS	(4) IV
Avg. rebate	-0.945	-1.032	-0.025	-0.026
	(0.341)	(0.583)	(0.006)	(0.007)
First stage:				
Eligible × Post1		3.265		3.265
		(0.569)		(0.569)
Eligible × Post2		-2.729		-2.729
		(0.680)		(0.680)
Covariates	N	N	N	N
Product FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Observations	362	362	362	362

Notes: The dependent variables are the list price after rebates, in thousand 2018 CAD, and the log of the list price after rebates. The average rebate is computed at the product-year level, also in thousand 2018 CAD. The sample includes all eligible electric vehicles, and their closest non-eligible substitutes obtained using Mahalanobis distance matching, as described in Section 3. All regressions are unweighted. Standard errors in parenthesis are clustered at the product level.

price of electric vehicles. We have that

$$\varepsilon = \frac{d\ln(q)}{d\ln(p-\tau)} = \frac{d\ln(q)}{d\tau} \cdot \frac{d\tau}{d\ln(p-\tau)} = \frac{\alpha}{\psi}, \tag{A3}$$

where α is the semi-elasticity to the rebate, estimated in Table C.1, column (2), and ψ is the rebate passthrough to the log of price, estimated in Table C.2, column (4).

The resulting elasticity and the associated passthrough are reported in Table C.3. I recover a market elasticity for electric vehicles of -2.999 which is comparable to the estimated elasticity in Muehlegger and Rapson (2022). They estimate a market elasticity of -2.1 and an incomplete passthrough of the rebates to consumers. The difference between the two estimates arises partly from the fact that I am using list prices, whereas they have access to transaction prices which are typically lower after bargaining.

My results are also close to other works that estimate demand using a structural estimation. Xing et al. (2021), Remmy (2025), and Li (2023) find an average own-price elasticity of -2.75, -3.54, and -3.70 respectively. Springel (2021) on the other hand finds an own-price

Table C.3: Implied elasticity of demand and passthrough of rebates

	(1) Implied elasticity	(2) Implied Passthrough
Estimate	-3.000	1.032
Std. error	(0.814)	(0.583)

Notes: Standard errors are computed using the Delta method.

elasticity between -1.49 and 1.07. Pavan (2017) estimate the own-price elasticity of alternative fuel vehicles to be between -4.42 and -2.85. All of these studies use list prices to perform their respective analyses.

D Proofs

I prove formally the statements in equations (23) and (28) using Lemmas 1 and 2 below. In what follows, I omit the subscript m and the superscript ev to avoid cluttering the notation, such that $q_t(\cdot)$ and $Q_t(\cdot)$ represent the flow and the stock of electric vehicles in period t, for a given county.

Assumptions. I impose the following three assumptions on the local planner's expectations, which are sufficient conditions for Lemma 1:

A1. $0 \leq \mathbb{E}_t F_{t+k} - \rho \mathbb{E}_t F_{t+k+1} < K_t(\rho), \quad \forall k \geq 1;$

A2. $\mathbb{E}_t q_{t+k}(n) = (1+g_{t+1}) \dots (1+g_{t+k})\, q_t(n), \quad \forall n \in \mathbb{N}, \quad \forall k \geq 1;$

A3. $q_t(n) > q_t(n-1), \quad \forall n \in \mathbb{N}.$

Assumption **A1** imposes some restrictions on the local planner's expectations about the evolution of the installation costs. The fixed cost of installation could go up if the technology improves over time (i.e., faster, more powerful chargers become available), or decrease with economies of scale or increased competition. For reasonable values of ρ, assumption **A1** imposes bounds on the evolution of the expected fixed costs of installation. They can increase over time, at a rate no larger than $(1-\rho)/\rho$, or decrease at a rate defined by the constant $K_t(\rho)$ which I define below. Assumption **A2** states that the local planner's expectation about future sales is the same as current sales multiplied by some exogenous growth rates $(1+g_{t+1}), ..., (1+g_{t+k})$. In some sense, the local planner is uncertain about future advances

in electric vehicle technology, the expansion of the choice set, and consumers' evolving preferences. In this context, his best guess about future sales is based on the current market conditions. Assumption **A3** holds trivially by strict monotonicity of consumer preferences, as long as $\theta_i > 0, \forall i$.

LEMMA 1. *Let* ***A1*** *–* ***A3*** *hold, and* $\rho \in [0, 1)$*. Then*

$$\max_{k \geq 1} \left\{ \rho^k \mathbb{E}_t MV_{t+k}(n, \mathcal{I}_{t+k}) \right\} = \rho \mathbb{E}_t MV_{t+1}(n, \mathcal{I}_{t+1}). \tag{A4}$$

Proof. First, notice that Lemma 1 holds trivially for $\rho = 0$. For $\rho \in (0, 1)$, it is sufficient to show that

$$\rho \mathbb{E}_t MV_{t+1}(n, \mathcal{I}_{t+1}) - \rho^k \mathbb{E}_t MV_{t+k}(n, \mathcal{I}_{t+k}) > 0, \quad \forall k > 1, \tag{A5}$$

or

$$\rho \mathbb{E}_t \overline{MV}_{t+1}(n, \mathcal{I}_{t+1}) - \rho^k \mathbb{E}_t \overline{MV}_{t+k}(n, \mathcal{I}_{t+k}) > \rho \mathbb{E}_t FC_{t+1} - \rho^k \mathbb{E}_t FC_{t+k}, \ \forall k > 1. \tag{A6}$$

By Assumption **A1**, we can rewrite the right-hand side of (A6) as

$$\rho \mathbb{E}_t FC_{t+1} - \rho^k \mathbb{E}_t FC_{t+k} = \sum_{s=1}^{k-1} \rho^s \left(\mathbb{E}_t FC_{t+s} - \rho \mathbb{E}_t FC_{t+s+1} \right) < a_k K_t(\rho), \tag{A7}$$

where the constant a_k is equal to $(\rho - \rho^k)/(1 - \rho)$. Essentially, the expression to the right of the equality in (A7) is a telescoping series that equals the left-hand side after you cancel out all terms except the first and last. The expression to the right of the inequality follows from Assumption **A1**. The proof of Lemma 1 revolves around showing the existence of a constant $K_t(\rho)$ such that

$$\rho \mathbb{E}_t \overline{MV}_{t+1}(n, \mathcal{I}_{t+1}) - \rho^k \mathbb{E}_t \overline{MV}_{t+k}(n, \mathcal{I}_{t+k}) \geq a_k K_t(\rho) > 0, \quad \forall k > 1. \tag{A8}$$

The first term in the left-hand side of equation (A8) can be rewritten as

$$\begin{aligned} \rho \mathbb{E}_t \overline{MV}_{t+1}(n, \mathcal{I}_{t+1}) = \Psi_{mt} \sum_{s=t+1}^{t+k-1} \rho^{s-t} \mathbb{E}_t Q_s(n, \mathcal{I}_{t+1}) \Delta v(n)^\gamma \\ + \Psi_{mt} \sum_{s=t+k}^{\infty} \rho^{s-t} \mathbb{E}_t Q_s(n, \mathcal{I}_{t+1}) \Delta v(n)^\gamma, \end{aligned} \tag{A9}$$

where I have made explicit the dependence of the stock of electric vehicles on the installation date of station n. The second term in the left-hand side of equation (A8) can be rewritten as

$$\rho^k \mathbb{E}_t \overline{MV}_{t+k}(n, \mathcal{I}_{t+k}) = \Psi_{mt} \sum_{s=t+k}^{\infty} \rho^{s-t} \mathbb{E}_t Q_s(n, \mathcal{I}_{t+k}) \Delta v(n)^{\gamma}. \tag{A10}$$

The installed base of electric vehicles in both equations accumulates differently between period $t+1$ and $t+k-1$, since station n is installed in period $t+1$ in the first case and in $t+k$ in the second case. That is, we have that

$$\begin{aligned}
\mathbb{E}_t Q_s(n, \mathcal{I}_{t+1}) &= (1-d)^{s-t} Q_t(n-1) + \sum_{\tau=t+1}^{t+k-1} (1-d)^{s-\tau} \mathbb{E}_t q_\tau(n) \\
&\quad + \sum_{\tau=t+k}^{s} (1-d)^{s-\tau} \mathbb{E}_t q_\tau(n) \\
&= (1-d)^{s-t} Q_t(n-1) + \sum_{\tau=t+1}^{t+k-1} (1-d)^{s-\tau} \Big[\prod_{\kappa=t+1}^{\tau} (1+g_\kappa) \Big] q_t(n) \\
&\quad + \sum_{\tau=t+k}^{s} (1-d)^{s-\tau} \Big[\prod_{\kappa=t+1}^{\tau} (1+g_\kappa) \Big] q_t(n)
\end{aligned} \tag{A11}$$

and

$$\begin{aligned}
\mathbb{E}_t Q_s(n, \mathcal{I}_{t+k}) &= (1-d)^{s-t} Q_t(n-1) + \sum_{\tau=t+1}^{t+k-1} (1-d)^{s-\tau} \mathbb{E}_t q_\tau(n-1) \\
&\quad + \sum_{\tau=t+k}^{s} (1-d)^{s-\tau} \mathbb{E}_t q_\tau(n) \\
&= (1-d)^{s-t} Q_t(n-1) + \sum_{\tau=t+1}^{t+k-1} (1-d)^{s-\tau} \Big[\prod_{\kappa=t+1}^{\tau} (1+g_\kappa) \Big] q_t(n-1) \\
&\quad + \sum_{\tau=t+k}^{s} (1-d)^{s-\tau} \Big[\prod_{\kappa=t+1}^{\tau} (1+g_\kappa) \Big] q_t(n),
\end{aligned} \tag{A12}$$

where the second equality in each equation holds by Assumption **A2**. Combining these results, equation (A8) can be rewritten as

$$\rho \mathbb{E}_t \overline{MV}_{t+1}(n, \mathcal{I}_{t+1}) - \rho^k \mathbb{E}_t \overline{MV}_{t+k}(n, \mathcal{I}_{t+k})$$

$$= \Psi_{mt} \sum_{s=t+1}^{t+k-1} \rho^{s-t} \mathbb{E}_t Q_s(n, \mathcal{I}_{t+1}) \Delta v(n)^{\gamma}$$

$$+ \Psi_{mt} \sum_{s=t+k}^{\infty} \rho^{s-t} \big(\mathbb{E}_t Q_s(n, \mathcal{I}_{t+1}) - \mathbb{E}_t Q_s(n, \mathcal{I}_{t+k}) \big) \Delta v(n)^{\gamma}. \tag{A13}$$

Notice that the first term in the right-hand side of (A13) is greater or equal to zero since $\Psi_{mt} > 0$, and both $\mathbb{E}_t Q_s(\mathcal{I}_{t+1}) \geq 0$ and $\Delta v(n)^{\gamma} \geq 0$ by construction. The terms inside the sum in the second term of equation (A13) can be rewritten as

$$\mathbb{E}_t Q_s(n, \mathcal{I}_{t+1}) - \mathbb{E}_t Q_s(n, \mathcal{I}_{t+k}) = \sum_{\tau=t+1}^{t+k-1} (1-d)^{s-\tau} \Big[\prod_{\kappa=t+1}^{\tau} (1+g_{\kappa}) \Big] \times \big(q_t(n) - q_t(n-1) \big)$$

$$> 0, \tag{A14}$$

since $q_t(n) > q_t(n-1)$ by Assumption **A3**. Combining these results, we have that

$$\rho \mathbb{E}_t \overline{MV}_{t+1}(n, \mathcal{I}_{t+1}) - \rho^k \mathbb{E}_t \overline{MV}_{t+k}(n, \mathcal{I}_{t+k}) > 0, \quad \forall k > 1, \tag{A15}$$

and we can choose $K_t(\rho, n)$ as

$$K_t(\rho, n) = \min_{k>1} \left\{ \frac{\rho \mathbb{E}_t \overline{MV}_{t+1}(n, \mathcal{I}_{t+1}) - \rho^k \mathbb{E}_t \overline{MV}_{t+k}(n, \mathcal{I}_{t+k})}{a_k} \right\} \tag{A16}$$

and

$$K_t(\rho) = \min_{n \in \mathbb{N}} \big\{ K_t(\rho, n) \big\} \tag{A17}$$

which completes the proof.[43]

□

LEMMA 2. *Let $\rho \in [0, 1)$ and define $\widetilde{\rho} = \rho(1-d)$. Then*

$$\rho \mathbb{E}_t \overline{MV}_{t+1}(n, \mathcal{I}_t) - \rho \mathbb{E}_t \overline{MV}_{t+1}(n, \mathcal{I}_{t+1}) = \Psi_{mt} \frac{\widetilde{\rho}}{1-\widetilde{\rho}} \Delta v(n)^{\gamma} \big(q_t(n) - q_t(n-1) \big), \tag{A18}$$

[43] Notice than in the limit, we have $K(0) = \mathbb{E}_t \overline{V}_{t+1}(n, \mathcal{I}_{t+1}) \geq 0$.

Proof. First, notice that Lemma 2 holds trivially for $\rho = 0$. For $\rho \in (0,1)$, we have that

$$\begin{aligned}\rho\mathbb{E}_t\overline{MV}_{t+1}(n,\mathcal{I}_t) - \rho\mathbb{E}_t\overline{MV}_{t+1}(n,\mathcal{I}_{t+1}) \\ = \Psi_{mt}\sum_{s=t+1}^{\infty}\rho^{s-t}\big(\mathbb{E}_tQ_s(n,\mathcal{I}_t) - \mathbb{E}_tQ_s(n,\mathcal{I}_{t+1})\big)\,\Delta v(n)^{\gamma},\end{aligned} \tag{A19}$$

where

$$\mathbb{E}_tQ_s(n,\mathcal{I}_t) = (1-d)^{s-t+1}Q_{t-1} + (1-d)^{s-t}q_t(n) + \sum_{\tau=t+1}^{s}(1-d)^{s-\tau}\mathbb{E}_tq_\tau(n), \tag{A20}$$

and

$$\mathbb{E}_tQ_s(n,\mathcal{I}_{t+1}) = (1-d)^{s-t+1}Q_{t-1} + (1-d)^{s-t}q_t(n-1) + \sum_{\tau=t+1}^{s}(1-d)^{s-\tau}\mathbb{E}_tq_\tau(n). \tag{A21}$$

Combining both equations, we have that

$$\mathbb{E}_tQ_s(n,\mathcal{I}_t) - \mathbb{E}_tQ_s(n,\mathcal{I}_{t+1}) = (1-d)^{s-t}\big(q_t(n) - q_t(n-1)\big), \quad \forall s > t. \tag{A22}$$

Therefore,

$$\begin{aligned}\rho\mathbb{E}_tMV_{t+1}(n,\mathcal{I}_t) - \rho\mathbb{E}_tMV_{t+1}(n,\mathcal{I}_{t+1}) &= \Psi_{mt}\sum_{s=t+1}^{\infty}\rho^{s-t}(1-d)^{s-t} \\ &\quad\times\big(q_t(n) - q_t(n-1)\big)\,\Delta v(n)^{\gamma} \\ &= \Psi_{mt}\sum_{s=t+1}^{\infty}\widetilde{\rho}^{s-t}\big(q_t(n) - q_t(n-1)\big)\Delta v(n)^{\gamma} \\ &= \Psi_{mt}\frac{\widetilde{\rho}}{1-\widetilde{\rho}}\,\Delta v(n)^{\gamma}\big(q_t(n) - q_t(n-1)\big).\end{aligned} \tag{A23}$$

□

E Computational Details

E.1 Details on the demand estimation

The estimation of the demand side parameters follows the best practices described in Conlon and Gortmaker (2020). I include two random coefficients to capture consumers' heterogene-

ity. The random coefficient on prices captures differences in price sensitivity, while the random coefficients on the constant controls for the substitution between the inside good and the outside good. I find that these are important to make sure I do not overestimate the environmental gains due to rebate programs.

Estimation is done in two stages using the Nested Fixed Point algorithm. I set a tight tolerance threshold on the objective function of `1e-5`, as suggested in Conlon and Gortmaker (2020). I concentrate out the linear parameters and focus the estimation on the random coefficients. Fixed effects are differentiated out using Frisch-Waugh-Lovell Theorem. The integration of the market shares is performed using 1,000 independent Halton draws. Finally, I perform the inversion of the market shares using the `squarem` algorithm[44] and a tight convergence threshold of `1e-12`. Reynaert and Verboven (2014) and Conlon and Gortmaker (2020) both show that the `squarem` algorithm is significantly faster than the contraction mapping described in Berry et al. (1995).

I do not use the optimal instruments described in Reynaert and Verboven (2014) and Conlon and Gortmaker (2020). I find that they did not work well in this particular application. Since network size is determined jointly with electric vehicle sales, it is not clear how to deal with the endogenous network size while computing the optimal instruments since it is a stock variable. I also do not include a supply side for cars. As pointed out by Conlon and Gortmaker (2020), including a supply side helps identifying the random coefficients, but can lead to misleading results in case it is misspecified. List prices for cars are set at the Canadian level, hence assuming that manufacturers change prices in response to a local policy would lead to one such misspecification. I instead assume that prices do not respond to the policy, although they are still endogenous since they are correlated to unobserved car attributes.

E.2 Details on the network supply estimation

My preferred specification for the indirect utility for charging $v(N, \theta_i)$ allows for some simplifications of the network supply problem. Recall that

$$v(N, \theta_i) = \theta_i \frac{(1+N)^{\zeta} - 1}{\zeta}, \tag{A24}$$

and

$$\Delta v(N) = \int \frac{\widetilde{r}v(N, \theta_i) - \widetilde{r}v(N-1, \theta_i)}{-\beta_i^{\mathbf{P}}} \, dF(\nu_i | EV). \tag{A25}$$

[44] See Varadhan and Roland (2008).

The curvature parameter does not depend on i, hence it can be taken outside of the integral. I rewrite

$$\begin{aligned}\Delta v(N) &= \frac{(1+N)^{\zeta} - N^{\zeta}}{\zeta} \cdot \widetilde{r} \cdot \int \frac{\theta_i}{-\beta_i^{\mathbf{p}}} \, dF(\nu_i|EV), \\ &= \widetilde{\gamma}_{mt} f(N) \, \widetilde{r}, \end{aligned} \tag{A26}$$

where

$$\widetilde{\gamma}_{mt} = -\int \frac{\theta + \mathbf{D}_{mt}\Gamma^{\mathbf{N}} + \sigma^{\mathbf{N}}\nu_i^{\mathbf{N}}}{\beta^{\mathbf{p}} + \mathbf{D}_{mt}\Gamma^{\mathbf{p}} + \sigma^{\mathbf{p}}\nu_i^{\mathbf{p}}} \, dF(\nu_i|EV) > 0 \tag{A27}$$

and

$$f(N) = \frac{(1+N)^{\zeta} - N^{\zeta}}{\zeta} > 0. \tag{A28}$$

There are a few things to note. First, notice that I do not need to calibrate $\widetilde{r}$ as it is absorbed by the constant included in $\mathbf{D}_{mt}$ (after I take the logarithm). Accordingly, I can omit the discount rate $\widetilde{r}$ completely, as it has no impact on the estimates.[45]

Second, the integral in $\widetilde{\gamma}_{mt}$ is computed over consumers that have purchased an electric vehicle. These consumers could differ from the population in terms of their preferences, therefore, in general, we will have $F(\nu_i|EV) \neq F(\nu_i)$. Define $\overline{\gamma}_{mt}$ to be the counterpart of $\widetilde{\gamma}_{mt}$, where the integral is computed over the unconditional distribution of preference in the population rather than the distribution of preferences of electric vehicle owners, that is,

$$\overline{\gamma}_{mt} = -\int \frac{\theta + \mathbf{D}_{mt}\Gamma^{\mathbf{N}} + \sigma^{\mathbf{N}}\nu_i^{\mathbf{N}}}{\beta^{\mathbf{p}} + \mathbf{D}_{mt}\Gamma^{\mathbf{p}} + \sigma^{\mathbf{p}}\nu_i^{\mathbf{p}}} \, dF(\nu_i) \tag{A29}$$

Estimating the model with $\overline{\gamma}_{mt}$ instead of $\widetilde{\gamma}_{mt}$ can lead to biased estimates whenever $\overline{\gamma}_{mt}$ is not proportional to $\widetilde{\gamma}_{mt}$.[46] In the estimation, we approximate the (unconditional) probability distribution of consumers preferences, $dF(\nu_i)$, using R simulation draws with equal weights $w = 1/R$. Therefore, the integral in (A29) can be computed as

$$\overline{\gamma}_{mt} = -\frac{1}{R}\sum_{i=1}^{R} \frac{\theta + \mathbf{D}_{mt}\Gamma^{\mathbf{N}} + \sigma^{\mathbf{N}}\nu_i^{\mathbf{N}}}{\beta^{\mathbf{p}} + \mathbf{D}_{mt}\Gamma^{\mathbf{p}} + \sigma^{\mathbf{p}}\nu_i^{\mathbf{p}}}. \tag{A30}$$

To approximate the distribution of preferences of electric vehicle owners, I proceed by reweighting consumers in (A30) to account for the propensity they own an electric vehicle. Weights are computed recursively, in each market, for periods $t = 1, ..., T$. These weights

[45] Alternatively, this means that $\ln \Psi_{mt} - \eta_{mt} + \ln \widetilde{r} = \mathbf{D}_{mt}\lambda^{\mathbf{D}}$.

[46] If $\overline{\gamma}_{mt}$ is proportional to $\widetilde{\gamma}_{mt}$, for example, if $\overline{\gamma}_{mt}/\widetilde{\gamma}_{mt} = \kappa_t$, then integrating over the wrong distribution of consumer preferences would only bias the coefficients of the year fixed effects included in $\mathbf{D}_{mt}$.

are of the form

$$w_{imt}(n) = \underbrace{w_{im,t-1} \cdot \frac{\delta Q^{ev}_{m,t-1}}{Q^{ev}_{mt}(n)}}_{\text{Contribution from past purchases}} + \underbrace{\frac{\sum_{j \in EV} s_{ijmt}(n)}{\sum_{i=1}^{R} \sum_{j \in EV} s_{ijmt}(n)} \cdot \frac{q^{ev}_{mt}(n)}{Q^{ev}_{mt}(n)}}_{\text{Contribution from new purchases}}, \tag{A31}$$

with $w_{im0} = 1/R$ is the initial condition in each market. By construction, these weights sum to 1, since they are a linear combination of weights that sum to 1, and the recursive construction ensures that we keep track of past and current electric vehicle purchases. We can then approximate the integral over electric vehicle owners by

$$\widetilde{\gamma}_{mt}(n) = -\sum_{i=1}^{R} \frac{\theta + \mathbf{D}_{mt}\Gamma^{\mathbf{N}} + \sigma^{\mathbf{N}}\nu_i^{\mathbf{N}}}{\beta^{\mathbf{p}} + \mathbf{D}_{mt}\Gamma^{\mathbf{p}} + \sigma^{\mathbf{p}}\nu_i^{\mathbf{p}}} \cdot w_{imt}(n). \tag{A32}$$

Notice that $\widetilde{\gamma}_{mt}$ has to be recomputed at each n during the estimation routine, and for counterfactual experiments.

I now rewrite the estimating equation and the other structural functions in terms of the specific functional form for $\Delta v(N)$. Notice that these structural functions involve an infinite sum over potential network sizes. In practice, I truncate these infinite sums to avoid needless computation. In economic terms, the idea is that there exists a saturation point that local planners will not exceed in terms of network size. A natural saturation point is the county-level population, L_{mt}. In practice, the analysis is very robust to the choice of this cutoff, as long as it is well above realistic network sizes. The current saturation point is set to $S_{mt} = L_{mt}/200$ in each county.

1. Conditional log-likelihood. The conditional log-likelihood becomes:

$$\begin{aligned}
\ell(\lambda \mid Q^{ev}_{m,t-1}, \mathbf{D}_{mt}) = \sum_m \sum_t \ln \Bigg[& \Phi\Bigg(\mathbf{D}_{mt}\lambda^{\mathbf{D}} + \lambda^{\mathbf{N}} \ln\Big(\widetilde{\gamma}_{mt}(N_{mt}) f(N_{mt})\Big) \\
& \quad + \lambda^{\mathbf{Q}} \ln\Big(Q^{ev}_{mt}(N_{mt}) + \frac{\widetilde{\rho}}{1-\widetilde{\rho}} \Delta q^{ev}_{mt}(N_{mt})\Big)\Bigg) \\
& - \Phi\Bigg(\mathbf{D}_{mt}\lambda^{\mathbf{D}} + \lambda^{\mathbf{N}} \ln\Big(\widetilde{\gamma}_{mt}(N_{mt}+1) f(N_{mt}+1)\Big) \\
& \quad + \lambda^{\mathbf{Q}} \ln\Big(Q^{ev}_{mt}(N_{mt}+1) + \frac{\widetilde{\rho}}{1-\widetilde{\rho}} \Delta q^{ev}_{mt}(N_{mt}+1)\Big)\Bigg)\Bigg].
\end{aligned} \tag{A33}$$

2. Network supply. The supply function becomes:

$$N_{mt} = \sum_{n=1}^{+\infty} n \cdot \mathbb{1}\Big\{ \mathbf{D}_{mt}\lambda^{\mathbf{D}} + \lambda^{\mathbf{N}} \ln\Big(\widetilde{\gamma}_{mt}(n) f(n)\Big) + \lambda^{\mathbf{Q}} \ln\Big(Q^{ev}_{mt}(n) + \frac{\widetilde{\rho}}{1-\widetilde{\rho}} \Delta q^{ev}_{mt}(n)\Big) > \epsilon^{\mathbf{n}}_{mt} \geq \mathbf{D}_{mt}\lambda^{\mathbf{D}} + \lambda^{\mathbf{N}} \ln\Big(\widetilde{\gamma}_{mt}(n+1) f(n+1)\Big) + \lambda^{\mathbf{Q}} \ln\Big(Q^{ev}_{mt}(n+1) + \frac{\widetilde{\rho}}{1-\widetilde{\rho}} \Delta q^{ev}_{mt}(n+1)\Big)\Big\}. \tag{A34}$$

3. Conditional expectation. The conditional expectation becomes:

$$\mathbb{E}_{\epsilon^{\mathbf{n}}}(N_{mt} \mid Q^{ev}_{m,t-1}, \mathbf{D}_{mt}) = \sum_{n=1}^{+\infty} \Phi\Big(\mathbf{D}_{mt}\lambda^{D} + \lambda^{\mathbf{N}} \ln\Big(\widetilde{\gamma}_{mt}(n) f(n)\Big) + \lambda^{\mathbf{Q}} \ln\Big(Q^{ev}_{mt}(n) + \frac{\widetilde{\rho}}{1-\widetilde{\rho}} \Delta q^{ev}_{mt}(n)\Big)\Big). \tag{A35}$$

4. Marginal effect. Finally, the marginal effect becomes:

$$\frac{\partial N_{mt}}{\partial Q^{ev}_{mt}} = \sum_{n=1}^{+\infty} \phi\Big(\mathbf{D}_{mt}\lambda^{D} + \lambda^{\mathbf{N}} \ln\Big(\widetilde{\gamma}_{mt}(n) f(n)\Big) + \lambda^{\mathbf{Q}} \ln\Big(Q^{ev}_{mt}(n) + \frac{\widetilde{\rho}}{1-\widetilde{\rho}} \Delta q^{ev}_{mt}(n)\Big)\Big) \cdot \frac{\lambda^{\mathbf{Q}}}{Q^{ev}_{mt}(n) + \frac{\widetilde{\rho}}{1-\widetilde{\rho}} \Delta q^{ev}_{mt}(n)}. \tag{A36}$$

The derivation of the marginal effect follows in the next section.

E.3 Elasticities

Elasticity of network supply and marginal effect. I first derive an expression for the elasticity of network supply,

$$\eta_{mt} = \frac{\partial N_{mt}}{\partial Q^{ev}_{mt}} \cdot \frac{Q^{ev}_{mt}}{N_{mt}}. \tag{A37}$$

The supply equation is a step function, hence its derivative $\partial N_{mt}/\partial Q^{ev}_{mt}$ is either zero or it is not differentiable. Following Blundell and Powell (2004), rewrite network supply as $N_{mt} = N(Q^{ev}_{m,t-1}, \epsilon^{\mathbf{n}}_{mt})$ to make the dependence on $Q^{ev}_{m,t-1}$ and $\epsilon^{\mathbf{n}}_{mt}$ explicit, and consider the average structural function,

$$\mathbb{E}_{\epsilon^{\mathbf{n}}}(N_{mt} \mid Q^{ev}_{m,t-1}, \mathbf{D}_{mt}) = \int N(Q^{ev}_{m,t-1}, \mathbf{D}_{mt}, \epsilon^{\mathbf{n}})\, dF(\epsilon^{\mathbf{n}}). \tag{A38}$$

I can show that for $\epsilon^{\mathbf{n}}_{mt}$ distributed as standard normal, the average structural function can be written as

$$\mathbb{E}_{\epsilon^{\mathbf{n}}}(N_{mt} \mid Q^{ev}_{m,t-1}, \mathbf{D}_{mt}) = \sum_{n=1}^{+\infty} \Phi\Bigg(\mathbf{D}_{mt}\lambda^{D} + \lambda^{\mathbf{N}} \ln\big(\Delta v(n)\big) + \lambda^{\mathbf{Q}} \ln\left(Q^{ev}_{mt}(n) + \frac{\widetilde{\rho}}{1-\widetilde{\rho}}\,\Delta q^{ev}_{mt}(n)\right)\Bigg). \tag{A39}$$

The partial effect can be recovered as the derivative of the average structural function, that is,

$$\frac{\partial N_{mt}}{\partial Q^{ev}_{mt}} = \sum_{n=1}^{+\infty} \phi\Bigg(\mathbf{D}_{mt}\lambda^{D} + \lambda^{\mathbf{N}} \ln\big(\Delta v(n)\big) + \lambda^{\mathbf{Q}} \ln\left(Q^{ev}_{mt}(n) + \frac{\widetilde{\rho}}{1-\widetilde{\rho}}\,\Delta q^{ev}_{mt}(n)\right)\Bigg) \cdot \frac{\lambda^{\mathbf{Q}}}{Q^{ev}_{mt}(n) + \frac{\widetilde{\rho}}{1-\widetilde{\rho}}\,\Delta q^{ev}_{mt}(n)}. \tag{A40}$$

where $\Phi(\cdot)$ and $\phi(\cdot)$ represent the cumulative and probability distribution functions of the standard normal distribution.

Elasticity of demand. The elasticity to price can be computed using chain rule. We have that

$$\varepsilon^{j,k}_{mt} = \frac{\partial s_{jmt}}{\partial p_{kt}} \cdot \frac{(p_{kt} - \tau_{kt})}{s_{jmt}}, \tag{A41}$$

where

$$\frac{\partial s_{jmt}}{\partial p_{kt}} = \frac{\partial s_{jmt}(\mathbf{p}_t, N_{mt})}{\partial p_{kt}} + \frac{\partial s_{jmt}(\mathbf{p}_t, N_{mt})}{\partial N_{mt}} \cdot \frac{\partial N_{mt}}{\partial Q^{ev}_{mt}} \cdot \frac{\partial Q^{ev}_{mt}}{\partial p_{kt}}. \tag{A42}$$

It can be shown that the terms in (A42) are

$$\frac{\partial s_{jmt}(\mathbf{p}_t, N_{mt})}{\partial p_{kt}} = \begin{cases} \int \beta^{\mathbf{p}}_i \, s_{ijmt}\,(1 - s_{ijmt})\, dF(\nu_i) & \text{if } j = k \\ -\int \beta^{\mathbf{p}}_i \, s_{ijmt}\, s_{ikmt}\, dF(\nu_i) & \text{if } j \neq k \end{cases}, \tag{A43}$$

$$\frac{\partial s_{jmt}(\mathbf{p}_t, N_{mt})}{\partial N_{mt}} = \begin{cases} \int \theta_i (1 + N_{mt})^{\zeta - 1}\, s_{ijmt} \left(1 - \sum_{\ell \in EV} s_{i\ell mt}\right) dF(\nu_i) & \text{if } j \in EV \\ -\int \theta_i (1 + N_{mt})^{\zeta - 1}\, s_{ijmt} \sum_{\ell \in EV} s_{i\ell mt}\, dF(\nu_i) & \text{if } j \notin EV \end{cases}, \tag{A44}$$

$$\frac{\partial N_{mt}}{\partial Q^{ev}_{mt}} = \sum_{n=1}^{S_{mt}} \phi\Bigg(\mathbf{D}_{mt}\lambda^{D} + \lambda^{\mathbf{N}} \ln\left(\Delta v(n)\right) + \lambda^{\mathbf{Q}} \ln\left(Q^{ev}_{mt}(n) + \frac{\widetilde{\rho}}{1-\widetilde{\rho}}\Delta q^{ev}_{mt}(n)\right)\Bigg) \cdot \frac{\lambda^{\mathbf{Q}}}{Q^{ev}_{mt}(n) + \frac{\widetilde{\rho}}{1-\widetilde{\rho}}\Delta q^{ev}_{mt}(n)}, \tag{A45}$$

$$\frac{\partial Q^{ev}_{mt}}{\partial p_{kt}} = \frac{L_{mt} \cdot \sum_{\ell \in EV} \frac{\partial s_{\ell mt}}{\partial p_{kt}}}{1 - L_{mt} \cdot \frac{\partial N_{mt}}{\partial Q^{ev}_{mt}} \cdot \sum_{\ell \in EV} \frac{\partial s_{\ell mt}}{\partial N_{mt}}}. \tag{A46}$$

where L_{mt} is the market size.

Demand elasticities are useful to assess the quality of the estimation. Figure A.10 depicts the distribution of own price elasticities and Table E.1 reports the full elasticity matrix for selected battery electric and plug-in hybrid vehicles, in 2018. The average over all vehicles is -3.27, which is comparable to other studies on the car market. Previous works find that the cross-price elasticities between electric vehicles are negative, suggesting that these products become complements once network effects are accounted for, see Springel (2021). I find the opposite: when network operators are forward-looking, all cross-price elasticities are positive, meaning that electric vehicles remain substitutes when I account for network effects. This follows from the fact that the marginal effects are very small on the network supply side, leading to weak network effects. Several cross-price elasticities become negative once we ignore the forward-looking behavior of local planners in favor of static network supply.

Figure A.10: Distribution of own-price elasticities

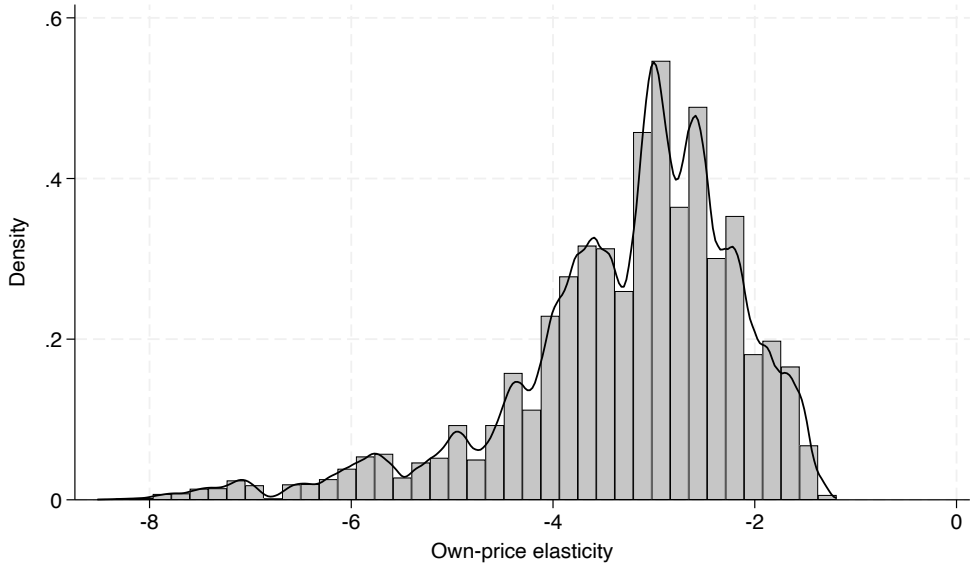


Table E.1: Average elasticities of electric vehicles, in 2018

	Bolt EV	Volt	Pacifica	C-Max	Fusion	Soul EV	Outlander	Leaf	Model 3	e-Golf
Panel A. Forward-looking Network Supply										
Chevrolet Bolt EV	**-2.953**	0.00186	0.00026	0.00009	0.00056	0.00030	0.00092	0.00203	0.00277	0.00046
Chevrolet Volt	0.00141	**-2.573**	0.00026	0.00011	0.00053	0.00036	0.00091	0.00218	0.00270	0.00052
Chrysler Pacifica	0.00107	0.00123	**-3.800**	0.00006	0.00051	0.00026	0.00132	0.00169	0.00343	0.00033
Ford C-Max	0.00156	0.00240	0.00035	**-2.205**	0.00073	0.00052	0.00172	0.00255	0.00328	0.00070
Ford Fusion	0.00129	0.00170	0.00027	0.00009	**-3.276**	0.00028	0.00098	0.00186	0.00294	0.00040
Kia Soul EV	0.00146	0.00227	0.00031	0.00012	0.00065	**-2.358**	0.00116	0.00248	0.00290	0.00060
Mitsubishi Outlander	0.00113	0.00143	0.00026	0.00007	0.00044	0.00022	**-3.581**	0.00167	0.00292	0.00034
Nissan Leaf	0.00143	0.00201	0.00025	0.00010	0.00052	0.00034	0.00101	**-2.803**	0.00282	0.00048
Tesla Model 3	0.00115	0.00126	0.00033	0.00006	0.00050	0.00022	0.00113	0.00149	**-4.208**	0.00025
Volkswagen e-Golf	0.00160	0.00243	0.00028	0.00012	0.00066	0.00044	0.00131	0.00249	0.00283	**-2.381**
Panel B. Static Network Supply										
Chevrolet Bolt EV	**-2.954**	-0.00035	-0.00005	0.00003	-0.00007	-0.00002	-0.00132	-0.00003	0.00090	-0.00003
Chevrolet Volt	0.00019	**-2.575**	-0.00005	0.00004	-0.00012	0.00004	-0.00140	0.00013	0.00083	0.00003
Chrysler Pacifica	-0.00006	-0.00077	**-3.800**	0.00001	-0.00006	-0.00003	-0.00070	-0.00032	0.00154	-0.00015
Ford C-Max	0.00057	0.00077	0.00010	**-2.205**	0.00022	0.00029	-0.00003	0.00083	0.00153	0.00029
Ford Fusion	0.00009	-0.00045	-0.00004	0.00002	**-3.277**	-0.00004	-0.00125	-0.00021	0.00107	-0.00009
Kia Soul EV	0.00032	0.00019	0.00004	0.00006	0.00004	**-2.358**	-0.00093	0.00038	0.00101	0.00012
Mitsubishi Outlander	-0.00009	-0.00078	-0.00005	0.00001	-0.00021	-0.00011	**-3.583**	-0.00039	0.00105	-0.00015
Nissan Leaf	0.00022	-0.00018	-0.00005	0.00004	-0.00012	0.00001	-0.00125	**-2.805**	0.00094	-0.00001
Tesla Model 3	-0.00002	-0.00089	0.00007	-0.00001	-0.00010	-0.00008	-0.00101	-0.00056	**-4.210**	-0.00023
Volkswagen e-Golf	0.00043	0.00036	-0.00002	0.00006	0.00007	0.00014	-0.00080	0.00042	0.00094	**-2.381**

E.4 Computing counterfactuals

Computing counterfactuals relies on a simple fixed point iteration to determine jointly network size and electric vehicle sales. Recall that the structural model can be written as $N_{mt} = N(Q^{ev}_{m,t-1}, \mathbf{D}_{mt}, \epsilon^{\mathbf{n}})$, with conditional expectation

$$\begin{aligned}\mathbb{E}_{\epsilon^{\mathbf{n}}}(N_{mt} \mid Q^{ev}_{m,t-1}, \mathbf{D}_{mt}) &= \int N(Q^{ev}_{m,t-1}, \mathbf{D}_{mt}, \epsilon^{\mathbf{n}})\, dF(\epsilon^{\mathbf{n}}), \\ &= \sum_{n=1}^{+\infty} \Phi\Bigg(\mathbf{D}_{mt}\lambda^{\mathbf{D}} + \lambda^{\mathbf{N}} \ln\big(\Delta v(n)\big) \\ &\qquad + \lambda^{\mathbf{Q}} \ln\Big(Q^{ev}_{mt}(n) + \frac{\widetilde{\rho}}{1-\widetilde{\rho}}\, \Delta q^{ev}_{mt}(n)\Big)\Bigg). \end{aligned} \tag{A47}$$

Any structural function N_{mt} can be decomposed into its conditional expectation and a disturbance, that is,

$$N_{mt} = \mathbb{E}_{\epsilon^{\mathbf{n}}}(N_{mt} \mid Q^{ev}_{m,t-1}, \mathbf{D}_{mt}) + \epsilon_{mt}. \tag{A48}$$

Notice that I can estimate $\widehat{\epsilon}_{mt}$ using parameter estimates $\widehat{\lambda}$ and the data, that is,

$$\widehat{\epsilon}_{mt} = N_{mt} - \mathbb{E}_{\epsilon^{\mathbf{n}}}(N_{mt} \mid Q^{ev}_{m,t-1}, \mathbf{D}_{mt}, \widehat{\lambda}). \tag{A49}$$

With these estimates in hand, I can then compute counterfactual networks as

$$\widetilde{N}_{mt} = \mathbb{E}_{\epsilon^{\mathbf{n}}}(N_{mt} \mid \widetilde{Q}^{ev}_{m,t-1}, \mathbf{D}_{mt}, \widehat{\lambda}) + \widehat{\epsilon}_{mt}, \tag{A50}$$

for any sequence of $\widetilde{Q}^{ev}_{m,t-1}$. Since the structural model takes as inputs the stock of electric vehicles and the stock of available charging stations, I need to solve counterfactuals recursively starting from $t = 1$. Let $\widetilde{Q}^{ev}_{m0} = Q^{ev}_{m0}$, $\widetilde{N}_{m0} = 0$, and consider counterfactual policy $\widetilde{\tau}$. The algorithm is as follows:

1. Start from $t = 1$;
2. The initial price vector is $\widetilde{\mathbf{p}}^0_t = (p_{1t}, ..., p_{Jt})$;
3. For each county, the initial fleet of electric vehicles is $\widetilde{Q}^{ev}_{m,t-1}$;
4. Set initial network size $\widetilde{N}^0_{mt} = \widetilde{N}_{m,t-1}$;
5. Compute market shares $s_{jmt}(\widetilde{\mathbf{p}}^0_t, \widetilde{\tau}, \widetilde{N}^0_{mt})$;
6. Compute electric vehicle sales $\widetilde{q}^{ev}_{mt}(\mathbf{p}^0_t, \widetilde{N}^0_{mt}) = L_{mt} \cdot \sum_{j \in EV} s_{jmt}(\widetilde{\mathbf{p}}^0_t, \widetilde{\tau}, \widetilde{N}^0_{mt})$;
7. Compute electric vehicle fleet $\widetilde{Q}^{ev}_{mt}(\widetilde{\mathbf{p}}^0_t, \widetilde{N}^0_{mt}) = (1 - d)\, \widetilde{Q}^{ev}_{m,t-1} + \widetilde{q}^{ev}_{mt}(\mathbf{p}^0_t, \widetilde{N}^0_{mt})$;
8. Update network size $\widetilde{N}^1_{mt} = \mathbb{E}_{\epsilon^{\mathbf{n}}}(N_{mt} \mid \widetilde{Q}^{ev}_{m,t-1}, \mathbf{D}_{mt}, \widehat{\lambda}) + \widehat{\epsilon}_{mt}$;
9. Repeat steps 4-7 until convergence in $\widetilde{N}_{mt}$;
10. Update prices $\widetilde{\mathbf{p}}^1_t(\widetilde{N}_{1t}, ..., \widetilde{N}_{Mt})$ (using the firm's first-order conditions);
11. Repeat 3-10 until convergence in $\widetilde{\mathbf{p}}_t$;
12. Update fleet variable $\widetilde{Q}^{ev}_{mt} = \widetilde{Q}^{ev}_{mt}(\widetilde{\mathbf{p}}, \widetilde{N}_{mt})$;
13. Repeat steps 2-12 recursively for $t = 2, 3, ..., T$.

F Alternative local planner objective

F.1 Setup

The main specification for the network supply model assumes that local planners install stations optimally to maximize consumers' utility of charging. In this section, I consider alternative objectives for the local planners. I omit the county index m and time index t for simplicity in what follows.

The starting point for studying alternative problems is the specification of the aggregate marginal benefits function. Consider the marginal benefits of the local planner, from consumer i, denoted by

$$b_i(n, j(i), \nu_i) = \begin{cases} b_i(n, \nu_i), & \text{if } j(i) \in EV \\ 0, & \text{otherwise} \end{cases}, \tag{A51}$$

where n is the network size, $j(i)$ denotes the vehicle owned by i and ν_i is an individual-specific taste parameter. The local planner derives some marginal benefits from installing station n, provided that consumer i owns an electric vehicle.

For the moment, I am deliberately silent about the nature of the marginal benefits contained in $b_i(n, \nu_i)$. They could represent the utility of charging from electric vehicle owners, the revenues from charging, the value of one more electric vehicle adoption, or the value of emission abatement, to name a few. Aggregating over consumers, we recover the aggregate marginal benefits from installing station n, denoted by

$$\begin{aligned} MB(n) &= \sum_i w_i \cdot b_i(n, j(i), \nu_i), \\ &= \sum_i w_i \cdot \mathbb{1}\{j(i) \in EV \mid n, \nu_i\} \cdot b_i(n, \nu_i), \end{aligned} \tag{A52}$$

where w_i is a consumer-specific weight (e.g., the share of consumers with preference ν_i).

Notice that we can always multiply and divide the right-hand side of (A52) by $\sum_\ell w_\ell \cdot \mathbb{1}\{j(\ell) \in EV \mid n, \nu_\ell\}$, that is,

$$\begin{aligned} MB(n) &= \left(\sum_\ell w_\ell \cdot \mathbb{1}\{j(\ell) \in EV \mid n, \nu_\ell\} \right) \\ &\quad \cdot \left(\sum_i \frac{w_i \cdot \mathbb{1}\{j(i) \in EV \mid n, \nu_i\} \cdot b_i(n, \nu_i)}{\sum_\ell w_\ell \cdot \mathbb{1}\{j(\ell) \in EV \mid n, \nu_\ell\}} \right), \\ &= Q(n) \cdot \int b_i(n, \nu_i) \, dF(\nu_i | EV), \\ &= Q(n) \cdot b(n) \end{aligned} \tag{A53}$$

where $F(\nu_i|EV)$ is, as previously, the distribution of electric vehicle owner preferences, $Q(n)$ is the installed base of electric vehicles, and $b(n)$ represents the average marginal benefits accrued to the local planner from electric vehicle owners.

With this in mind, studying alternative local planner objectives revolves around spec-

Table F.1: Alternative specifications for the local planner objective

Objective	$b(n)$	$Q(n)$	Fixed cost	Forward-looking
(1) Utility of charging	$\Psi_{mt}\Delta v(n)^{\gamma}$	$Q^{ev}_{mt}(n)$	FC_{mt}	Yes
(2) Revenues from charging	$\psi n^{\gamma-1}$	$Q^{ev}_{mt}(n)$	FC_{mt}	Yes
(3) EV adoption	ψ	$\Delta q^{ev}_{mt}(n)$	$\kappa n^{\gamma}\widetilde{FC}_{mt}$	No
(4) CO_2 emissions	ψ_t	$-\sum_j \Delta q_{jmt}(n)\, \boldsymbol{m}_{jmt}\, e_{jt}$	$\kappa n^{\gamma}\widetilde{FC}_{mt}$	No

ifying an adequate market size function $Q(n)$ and an adequate average marginal benefit function $b(n)$. For example, if $Q(n)$ is the emissions abated from installing station n and $b(n)$ is the average value of these emissions, we can change the objective of the local planner to reducing carbon emissions, given the fixed cost of expanding the network.

F.2 Alternative objectives

I consider several alternative local planner objectives to test the robustness of my results. First, I consider the case where the local planner installs chargers to maximize the revenues for charging, following closely Springel (2021). Second, I consider the objective to maximize local electric vehicle adoption. While this doesn't correlate perfectly with emission reductions, this is a recurring objective stated by governments to justify subsidies. Finally, I consider the objective to minimize local CO_2 emissions.

For some of these alternative objective, the benefit function is not decreasing in n, which is a necessary condition to obtain an equilibrium. In this case, the benefits of adding more stations grow to surpass any constant fixed costs and local planners install additional stations without bounds. This is inconsistent with what I observe in the data. One way to circumvent this issue and make the model consistent with the data is to specify the fixed costs of installation to grow with n, at a faster rate than the growth in benefits. It is not clear whether or not this assumption is reasonable in practice, however, it allows us to study these alternative objectives.

Table F.1 describe the average marginal benefit function, the market size function, and the fixed costs associated with each local planner objective described above. I abstract from more complex problems which combines two or more objectives. I describe the solution to each of these new objectives in what follows and provide additional details in the following sections.

Note that the objectives that involve maximizing electric vehicle adoption or reducing fleet emissions are, by design, static objectives. The reason is that installing new stations doesn't generate additional benefits beyond current adopters (e.g., new stations cannot

change past electric vehicle adoption). Therefore, there is no dynamic tradeoff in the installation decision: the local planner considers either current electric vehicle adoption (as opposed to the full fleet of electric vehicles) or the lifetime emissions associated with current sales, and there are no additional benefits that carry into the future.

F.3 Solution to the revenue maximizing problem

The solution to the local planner's problem under revenue maximization follows closely the methodology developed in Section 4.

The specification for revenues follows closely Springel (2021). She suggests that average revenues per station can be approximated by a nonlinear function of network size, $b(n) = \psi n^{\gamma-1}$, where $\psi > 0$ is a markup term (operators are price takers) and $0 < \gamma < 1$ captures decreasing returns from additional stations, which is consistent with the story that more profitable locations are opened first. This ensures that $b(n)$ is a decreasing function of n as the average revenues per station decreases mechanically with n (hence the $\gamma - 1$). It is worth mentioning that the linear parameter ψ, is not separately identified from fixed effects once we take the logarithm of $b(n)$.

F.4 Solution to the electric vehicle adoption problem

I derive the network supply for the electric vehicle adoption objective. In this case, the benefit function is

$$MB_{mt}(n) = \psi \Delta q^{ev}_{mt}(n), \tag{A54}$$

where $\Delta q^{ev}_{mt}(n) = q^{ev}_{mt}(n) - q^{ev}_{mt}(n-1)$ are the marginal electric vehicle sales in county m at time t, from opening station n, and ψ is a shadow price that converts these sales to dollars.

Note that this benefit function is not guaranteed to be decreasing in n. By strict monotonicity of consumers' preferences, Δq^{ev}_{mt} is guaranteed to be positive, but its derivative could be positive or negative over some range of n depending on the curvature of demand. Therefore, a model based on this specification of $MB(n)$ is not guaranteed to have an interior solution (the problem described above). To solve this issue, I allow the fixed installation costs to be increasing in network size. This would occur if the local planner installs the most accessible sites first, which have low installation costs, then move on to more complex installations which are more costly.

Consider the following parametrization for fixed costs,

$$FC_{mt}(n) = \kappa n^{\gamma} \widetilde{FC}_{mt}, \tag{A55}$$

where $\kappa > 0$ and $\gamma > 0$ are parameters controling the complexity of the installation which scale up the fixed costs $\overline{F}_{mt}$. The lifetime value of installing station n is simply

$$MV_{mt}(n) = -FC_{mt}(n) + MB_{mt}(n), \tag{A56}$$

which does not depend on a continuation value. The model is static by construction and the local planner installs station as long as $MV_{mt}(n) \geq 0$.

The optimal network size, N, has to satisfy the following two inequality conditions,

$$MB_{mt}(N) \geq FC_{mt}(N) \tag{A57}$$

$$MB_{mt}(N+1) < FC_{mt}(N+1), \tag{A58}$$

or

$$\frac{\psi}{\kappa} N^{-\gamma} \Delta q_{mt}^{ev}(N) \geq \widetilde{FC}_{mt} \tag{A59}$$

$$\frac{\psi}{\kappa} (N+1)^{-\gamma} \Delta q_{mt}^{ev}(N+1) < \widetilde{FC}_{mt}. \tag{A60}$$

Taking logs and reorganizing, the network supply equation is

$$N_{mt} = \sum_{n=1}^{+\infty} n \cdot \mathbb{1}\Big\{ \lambda^{\mathbf{1}} + \lambda^{\mathbf{N}} \ln(n) + \lambda^{\mathbf{Q}} \ln\left(\Delta q_{mt}^{ev}(n)\right) \geq \epsilon_{mt}^{\mathbf{n}} > \lambda^{\mathbf{1}} + \lambda^{\mathbf{N}} \ln(n+1) + \lambda^{\mathbf{Q}} \ln\left(\Delta q_{jmt}^{ev}(n+1)\right) \Big\}, \tag{A61}$$

where $\epsilon_{mt}^{\mathbf{n}} = (1/\omega) \ln(\widetilde{FC}_{mt})$ is distributed as standard normal, $\lambda^{\mathbf{1}} = (1/\omega) \ln(\psi/\kappa)$, $\lambda^{\mathbf{N}} = -\gamma/\omega$, and $\lambda^{\mathbf{Q}} = 1/\omega$.

F.5 Solution to the emission reduction problem

I derive the network supply for the emission reduction objective. In this case, the benefit function is simply the current emissions of the new electric vehicle sales multiplied by the value of these emissions, namely,

$$MB_{mt}(n) = -\psi_t \sum_j \Delta q_{jmt}(n)\, m_{jmt}\, e_{jt}, \tag{A62}$$

where $\Delta q_{jmt}(n)$ are the marginal sales (including fuel and hybrid vehicles) of model j in county m at time t, due to the installation of station n, m_{jmt} is the average mileage of a driver of model j in county m at time t, e_{jt} are the emissions per kilometer of model j, and

ψ_t is the value of emissions in period t (e.g., a carbon price).

Note that this specification suffers from two important problems. First, $MB_{mt}(n)$ is not guaranteed to be positive. For example, increasing n can lead to substitution from the outside good to a plug-in hybrid with positive carbon emissions, increasing total emissions. Second, $MB_{mt}(n)$ is not guaranteed to be decreasing in n. To solve these issues, I consider the same parametrization for the fixed costs as above and I exclude markets where $MB_{mt}(n)$ is negative.

The lifetime value of installing station n is

$$\begin{aligned} MV_{mt}(n) &= -FC_{mt}(n) + \overline{MV}_{mt}(n), \\ &= -\kappa n^{\gamma} \widetilde{FC}_{mt} - \psi_t \sum_{s=t}^{t+T_j} \rho^{s-t} \sum_j \Delta q_{jmt}(n)\, \overline{m}\, e_{jt}, \end{aligned} \tag{A63}$$

where T_j is the vehicle's expected lifetime, ρ is the discount factor, and I have assumed that $\mathbb{E}_t(m_{jms}) = \overline{m}$ for all j, m, and t.[47] Notice that there are no dynamic considerations in this problem as installing stations do not change the environmental impact of past sales, hence the model is fully static, and local planners install stations as long as $MV_{mt}(n) \geq 0$.

The optimal network size, N, has to satisfy the following two inequality conditions,

$$\overline{MV}_{mt}(N) \geq FC_{mt}(N) \tag{A64}$$

$$\overline{MV}_{mt}(N+1) < FC_{mt}(N), \tag{A65}$$

or

$$-\frac{\psi_t}{\kappa} N^{-\gamma} \sum_{s=t}^{t+T_j} \rho^{s-t} \sum_j \Delta q_{jmt}(N)\, \overline{m}\, e_{jt} \geq \widetilde{FC}_{mt} \tag{A66}$$

$$-\frac{\psi_t}{\kappa} (N+1)^{-\gamma} \sum_{s=t}^{t+T_j} \rho^{s-t} \sum_j \Delta q_{jmt}(N+1)\, \overline{m}\, e_{jt} < \widetilde{FC}_{mt}. \tag{A67}$$

Taking logs and reorganizing, the network supply equation is

$$\begin{aligned} N_{mt} = \sum_{n=1}^{+\infty} n \cdot \mathbb{1} \Big\{ & \lambda_t^{\mathbf{1}} + \lambda^{\mathbf{N}} \ln(n) \\ & + \lambda^{\mathbf{Q}} \ln\left(- \sum_j \Delta q_{jmt}(n)\, \overline{m}\, e_{jt} \right) \geq \epsilon_{mt}^{\mathbf{n}} \end{aligned}$$

[47] I set $\rho = 0.95$, $\overline{m} = 22{,}083$km, and $T_j = 12.02$ years, see Section 6.

$$> \lambda_t^{\mathbf{1}} + \lambda^{\mathbf{N}} \ln(n+1)$$
$$+\lambda^{\mathbf{Q}} \ln\left(-\sum_j \Delta q_{jmt}(n+1)\, \overline{m}\, e_{jt} \right) \Bigg\}, \tag{A68}$$

where $\epsilon_{mt}^{\mathbf{n}} = (1/\omega)\ln(\widetilde{FC}_{mt})$ is distributed as standard normal, $\lambda^{\mathbf{N}} = -\gamma/\omega$, $\lambda^{\mathbf{Q}} = 1/\omega$, and $\lambda_t^{\mathbf{1}} = (1/\omega)\ln\big((\psi_t/\kappa)\cdot(1-\rho^{T_j+1})/(1-\rho)\big)$.

F.6 Estimation and counterfactuals

I estimate the specification described in Table F.1. The results are presented in Table F.2 below. I augment all specifications with county level demographics to preserve comparability with the baseline results (e.g., $\psi = \Psi_{mt}$ and $\psi_t = \Psi_{mt}$). There are several interesting findings. First, maximizing the network utility of consumers or the revenues from charging leads to network effects of similar magnitudes: the elasticity of network supply to the stock of electric vehicles is slightly higher under revenue maximization, 0.114 versus 0.108.

In both cases, the local planners install more stations in counties with a higher share of graduates (which captures environmental awareness) and in urban counties with fewer homeowners (which captures the potential for home-charging). One interesting difference emerges. Under utility maximization, local planners install stations more in low-income counties, targeting vulnerable populations which may benefit more from additional local stations. The reverse is true when local planners maximize revenues: they install more chargers in counties where income is high, and the potential for revenues is high.

Second, maximizing electric vehicle adoption or minimizing carbon emissions do not lead to network effects on the supply side (i.e., the elasticity of supply is zero). This happens mechanically as the objective function does not take into account changes to the stock of electric vehicles (through $Q_{mt}^{ev}(n)$ or $q^{ev}(n)$). Instead, the model depends on marginal adopters, $\Delta q_{mt}^{ev}(n)$, with $\partial \Delta q^{ev}/\partial q^{ev} = 0$, and the equilibrium is driven entirely by the increasing fixed costs. In other words, we could have that $\Delta q^{ev}(n)$ is constant in n, and we would still obtain an equilibrium from the increasing fixed costs where network effects do not play a role.

It is difficult to compare the parameter values of the four specifications (beyond the sign), as they arise from different models. To assess whether or not the choice of the objective drives my results, I replicate the basic counterfactual simulations under these additional local planner objectives. The results are presented in Table F.3.

I find that the revenues maximizing objective predicts 2,305 additional electric vehicle sales and 210 additional charging stations over the utility maximizing objective. These larger

Table F.2: Network supply estimation for alternative local planner objectives

	(1) Network utility	(2) Revenues from charging	(3) EV adoption	(4) CO_2 emissions
$\lambda^{\mathbf{N}}$	2.383	-1.703	-1.539	-1.539
	(0.132)	(0.099)	(0.097)	(0.097)
$\lambda^{\mathbf{Q}}$	0.375	0.391	0.227	0.234
	(0.110)	(0.103)	(0.104)	(0.100)
Avg. income	-0.580	0.274	0.222	0.217
	(0.240)	(0.225)	(0.226)	(0.227)
Avg. Age	1.824	0.725	0.654	0.670
	(0.556)	(0.482)	(0.500)	(0.495)
Avg. household size	-0.092	-0.990	-0.810	-0.823
	(1.497)	(1.458)	(1.441)	(1.445)
Share graduates	8.405	6.849	7.805	7.734
	(2.461)	(2.254)	(2.315)	(2.317)
Share homeowners	-6.024	-6.397	-6.825	-6.887
	(1.989)	(1.809)	(1.879)	(1.864)
Urban	0.435	0.362	0.502	0.487
	(0.328)	(0.309)	(0.311)	(0.311)
Elasticity to EV	0.108	0.114	0	0
EV for one additional station	75.3	70.52	n/a	n/a
Observations	830	830	814	813
Log-likelihood	-2,123.0	-2,136.6	-2,134.5	-2,131.4

Notes: Includes year fixed effects. Markets without electric vehicles in circulation are excluded from specifications (1) and (2). Markets without electric vehicles in the choice set are excluded from specification (3). Markets in which increasing network size increases emissions are excluded from specification (4). In this case, the local planner foregoes the installation of additional stations, and that observation does not contribute to the likelihood. Specifications (1) and (2) are estimated using a discount factor of $\widetilde{\rho} = 0.871$ and allow the local planners to be forward-looking. By construction, specifications (3) and (4) are static and the elasticity to the stock of electric vehicles is zero. Standard errors in parenthesis are clustered at the county level and are computed using 500 bootstrap replications.

effects arise from the fact that network effects are stronger on the supply side under that specification. These additional electric vehicle sales are associated to 794 fewer internal combustion and hybrid vehicles, which decrease emissions by an additional 0.044 million metric ton over the lifetime of vehicles. Overall, the difference in environmental outcomes between the two specifications is not too large since electric vehicles do not substitute internal combustion engines one-to-one, and the difference in adoption is not large.

The results are different for the other two specifications, which focus on electric vehicle adoption and emission reductions. As stated above, there are no network effects on the network supply side for these specifications, hence the predicted growth in networks is sig-

Table F.3: Counterfactual simulations for alternative local planner objectives

		Counterfactuals: No Subsidies			
	(1) Baseline	(3) Network utility	(4) Revenues from charging	(5) EV adoption	(6) CO_2 emissions
Key outcomes					
Δ Total sales	3,248,085	-14,567	-15,361	-14,272	-14,096
Δ Sales (fuel)	3,119,123	+28,140	+29,618	+27,596	+27,303
Δ Sales (battery electric)	47,305	-25,664	-26,911	-25,198	-24,899
Δ Sales (plug-in hybrid)	36,869	-16,302	-17,360	-15,918	-15,740
Δ Sales (hybrid)	44,788	-741	-709	-753	-761
Δ Charging stations	2,811	-267	-477	-191	-126
Δ CO_2 emissions	141.46	+1.080	+1.136	+1.060	+1.048
Δ Consumer surplus	0	-589.4	-623.7	-576.7	-569.4
Δ Total cost	737.8	0	0	0	0
Implied abatement costs					
Avg. cost per ton CO_2	–	-683	-649	-696	-704
Avg. cost per electric vehicle	–	-17,581	-16,666	-17,944	-18,155

nificantly lower: 191 additional stations under the electric vehicle adoption objective and 126 additional stations under the carbon emission objective. This translates to fewer electric vehicle sales, and worse environmental performance. This means that, interestingly, focusing on consumers' utility or revenues from charging leads to more adoption and fewer emissions than focusing on adoption or emissions directly. This is a direct consequence of the positive network effects on the supply side that give rise to a positive feedback loop that increases adoption and station deployment.

I could not find evidence that the local planners in Quebec had emission targets or electric vehicle adoption targets. They are for the most part local county governments that invest in charging infrastructures to benefit their local population, in the same way as they develop public transports or other crucial infrastructures. This seems aligned with the network utility maximization objective.

I cannot rule out completely that some of these planners intend to raise public funds using stations, to cover for example the cost of installing future networks. Therefore, the true objective could include a mix of the network utility and revenues maximizing objectives. In this case, we should expect slightly larger network effects than I estimate, slightly more adoption, slightly lower emissions overall, and slightly lower abatement costs. However,

since the two specifications yield very similar results, I am not concerned that the chosen specification produces misleading results.

G Cost-benefit analysis with future environmental gains

In this section, I describe how I incorporate future environmental gains to the cost-benefit analysis framework described in Section 6.

Setup. I consider the following policy implementation:

1. The policymaker offers subsidy τ, starting in period 1.
2. Electric vehicle adoption and investments in charging infrastructure occur in periods $t = 1, ..., T_1$, leading to $Q^{ev}_{T_1}(\tau)$ electric vehicles and $N_{T_1}(\tau)$ stations in period T_1.
3. Subsidies are phased out in period $T_1 + 1$.
4. Electric vehicle adoption and investment in charging infrastructures occur for an additional T_2 periods, leading to $Q^{ev}_{T_1+T_2}(N_{T_1}(\tau))$ electric vehicles and $N_{T_1+T_2}(Q^{ev}_{T_1}(\tau))$ stations.

To study the long term benefits of the policy, I then compare the environmental outcomes in period $T_1 + T_2$ between the policy implementation described above and a counterfactual where electric vehicles are never subsidized.

Forecasting demand. Solving for the counterfactual described above is only possible at the cost of additional assumptions, hence the results must be carefully interpreted with these caveats in mind. One difficulty arises from the fact that we need to forecast the demand for electric vehicles, yet we do not observe key features of the market in periods T_1+1 to T_1+T_2. For example, I do not observe the evolution of the choice set (e.g., new electric vehicles being introduced) or the evolution of the characteristics of existing models. As shown previously, new electric vehicles are being introduced every year and battery capacity is improving over time.

To forecast the demand for electric vehicles, I repeat the choice set that is available to consumers in 2020 to cover the period between 2021 and 2030. As shown in Section 6.4, electric vehicle adoption is essentially the same between 2021 and 2030 whether or not electric vehicles were initially subsidized (see panel (a) of Figure 4) and most of the environmental gains accrue from the period when subsidies were offered. In this context, I expect that the

marginal abatement cost estimated from repeating the 2020 choice set is not far from the estimate I would get if I knew the true evolution of the choice set and the characteristics of vehicles between 2021 and 2030.

Network supply. I forecast network supply in the same way, by repeating the county-level demographics from 2020 to cover the period between 2021 and 2030. There is very little variation over time in these average demographics at the county level.

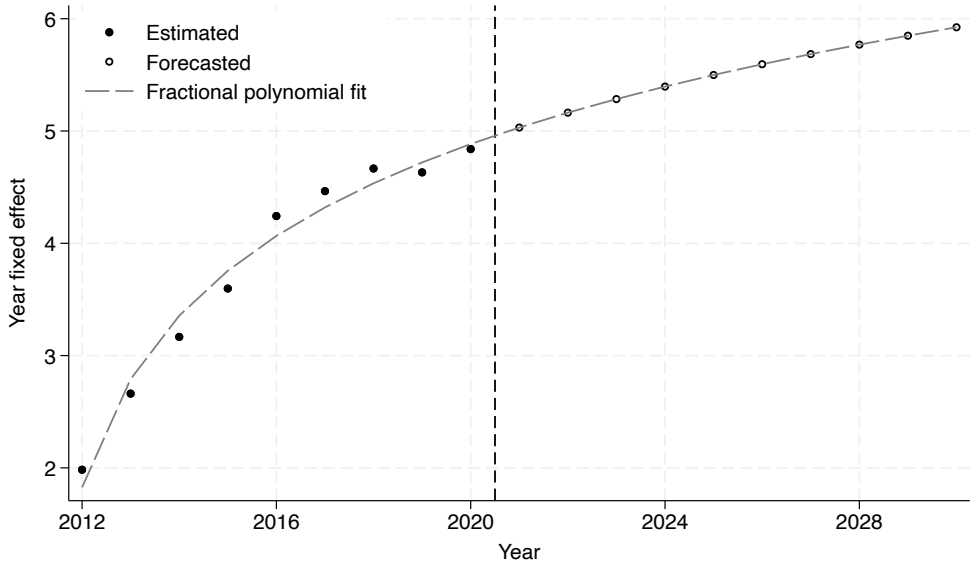
The network supply model includes time fixed effects. I observe a clear increasing pattern, partly driving the increase in network size. To avoid underestimating the growth in local network, I forecast these fixed effects in period 2021 to 2030 using a fractional polynomial fit on the estimated fixed effects from 2012 to 2020. The results of the regression are presented in Table F.4 and the forecasted values are presented in Figure A.11. The regression includes only 9 data points, so I use a polynomial of order 1 to estimate the model and produce the forecasted values.

Table F.4: Fractional polynomial regression for year fixed effects

	(1) Year fixed effects
Log of time trend	1.392
	(0.076)
Constant	1.826
	(0.132)
Observations	9
R-squared	0.978

Notes: This table presents the result from a fractional polynomial regression of the year fixed effects from the network supply model on a time trend. I restrict the polynomial to be of order 1. The functional form resulting from the estimation is the log of the time trend (i.e., power p = 0). Robust standard errors in parenthesis.

Figure A.11: Estimated and forecasted year fixed effects



Notes: This figure presents the forecasted values for the time fixed effects from the network supply model, using the results from the fractional polynomial fit in Table F.4.